SHARP OPENING RATES OF SINGLE-STRIP SUBSOLUTIONS FOR THE MUSKAT PROBLEM WITH VISCOSITY JUMP

DRAFT MANUSCRIPT

Date: 9 November 2026.

Credit and provenance. Mixing solutions for fully unstable Muskat data with viscosity jump are due to P. Luna-Velasco and F. Mengual [LM26], whose analysis we follow. From [LM26] come the Saffman–Taylor formulation, i.e. the inversion in terms of σ rather than the vorticity (Proposition 2.10 is [LM26, left relation in (3.7b), and Sec. 3.1.3] in our normalisation; the $A = 0$ formulas of Example 2.12 are in [LM26, Secs. 1.2, 3.1.3]); the single-layer construction with a second order expansion in time (after [FS18, CFM22]), with its first-order matching (Lemma 5.1; cf. [LM26, Lemmas 5.1–5.2]) and its second-order step, of which Lemmas 5.5–5.6 are the periodic version with a nonzero target [LM26, Lemma 5.3, (5.21), (8.41)], solvable by [LM26, Prop. 6.1] (periodic case: [CCG11]); the components of $a^{(0)}$ in Lemma 5.2 (normal: [LM26, (7.11), Prop. 7.4]; tangential: from [LM26, (3.7b)]); and the flat single strip with arbitrary intermediate density [LM26, Sec. 3.2]. The periodic operator theory (smoothing of $\mathcal{K}$, invertibility of $I - 2\xi\mathcal{K}$ for $|\xi| \leq 1$) is due to Córdoba, Córdoba and Gancedo [CCG11] (we use Hölder versions); the linearisation of Darcy's law, the relaxation and the h-principle to Mengual [Men22] (Székelyhidi [Sze12] for $A = 0$). Our $\mathcal{U}_A$ is the set O of [Men22], written (O) in [LM26], whose closure is the Λ-convex hull [LM26, Thm. 2.4]; the function g characterising it is [LM26, (A.1)], see [Men22, Lemmas 4.1–4.2]. *Provenance.* Sections 2–4 were first written (commit `e8efc0f`, 6 November 2026) from the seminar abstract announcing [LM26] only, before the paper was read; that abstract already announced the key idea of inverting in terms of σ, so the independence concerns details, not the idea. The subsection of Section 2 on the vortex-sheet strength (the operator $\mathcal{T}$, Proposition 2.10, γ_0 and $\Pi(z_0)$), the constant C_Ψ and the tilt hypothesis of Lemma 4.5, and Remark 4.6 were added after reading [LM26] (commit `a1c40e3`, 7 November 2026), as were the results (i)–(iii) below; (iv) was in the first version. The identification of $\mathcal{U}_A$ with O in Section 3 (eq. (3.8), whose last identity was first written in Section 8 of `a1c40e3`), Remark 3.3(b)–(c) in its present form, and all comparisons with [LM26] in Sections 2–4 were also written after reading [LM26], in the revisions of 7 and 9 November 2026; the revision of 9 November also made smaller corrections in these sections. *New relative to* [LM26]*:* (i) Theorem 1.2, necessity within single-strip subsolutions of constant intermediate density, a short argument built from the ingredients of [LM26] (flat case: [LM26, Sec. 3.2]); (ii) an explicit vortex-sheet-dependent single-layer range for $A \neq 0$ (Theorem 1.1, Corollary 1.3), including openings on pinch-free, partially Saffman–Taylor stable data; it is latent in the construction of [LM26], whose single layer is already tilted and satisfies the exact hull condition [LM26, (4.14)] at $t = 0$ on a γ_0-dependent range, so that a few-line modification of [LM26, Sec. 7.4], with an opening c such that $c - c_\infty \in H^{k-1}$ and $\inf c > 0$ [LM26, Sec. 4.1 and (5.1)], yields such a range in their setting, but none is stated there; (iii) the optimal tilt δ_* for $A \neq 0$ and general ρ_{m} (Lemma 5.2), sharp within single strips, which for $\rho_{\mathrm{m}} = A$ multiplies the γ_0^2 term of the range of the single layer of [LM26] by $(2 + A^2)^2/(4(1 + A^2)) \in [1, 9/8)$, with no gain for $A = 0$; (iv) curved strips with $\rho_{\mathrm{m}} \neq A$ and the horizontally periodic Hölder setting, both routine. The multilayers of [LM26, Sec. 7.5] with several intermediate strips open faster than single strips for fully unstable data wherever $|\gamma_0|$ is small. Corollary 1.5 is a change of frame and of time scale that applies equally to [LM26, Thm. 1.1]. For $A = 0$ our existence statements follow from the construction of [LM26], and our $A = 0$ statements are very likely covered by earlier work: see [FS18] for stable non-horizontal data and [CFM22] for turned interfaces; we have not compared our $A = 0$ statements with their precise admissible ranges. Details: `NOVELTY.md` and `COMPARISON.md` in the source repository.

Abstract. We study mixing solutions of the incompressible porous media equation for two fluids with different densities and different viscosities, for horizontally periodic interfaces. For asymptotically flat interfaces, Luna-Velasco and Mengual recently constructed such solutions for *fully unstable* Muskat data, i.e. when the Saffman–Taylor function σ_0 of the initial interface is negative everywhere, from subsolutions with one strip of constant intermediate density (a single layer) or with several such strips (multilayers). We work in the class of single-strip subsolutions and follow their Saffman–Taylor formulation. We first observe that their single layer, with its exact hull condition evaluated in full instead of through its normal component, already admits openings that depend on the vortex-sheet strength γ_0 of the interface; this is not stated in their paper. Optimising the target of the second order compatibility condition, which fixes the tangential component (the tilt) of the relaxed flux in the strip, we then obtain the sharp single-strip range. With A the viscosity Atwood number, ρ_{m} the intermediate density and

$$\Phi(\kappa) := \kappa\big(2(1 - A\rho_{\mathrm{m}})\kappa + \sigma_0\big) - \frac{(1 - A^2)\,\gamma_0^2}{8(1 + A\rho_{\mathrm{m}})},$$

we show that for interfaces of class $C^{6,\gamma}$ every opening function $\kappa \in C^{4,\gamma}$ with $\kappa > 0$ and $\Phi(\kappa) < 0$ is realised by a single strip, which yields infinitely many mixing solutions, and that every single-strip subsolution has a centre of mass that starts with the Muskat velocity and an opening that satisfies $\Phi(\kappa) \le 0$. For $\rho_{\mathrm{m}} = A$ the optimal tilt improves the γ_0^2 term of the range of their single layer by a factor in $[1, 9/8)$, and not at all when $A = 0$. Consequently, single strips with everywhere positive opening exist if and only if $\gamma_0 \neq 0$ wherever $\sigma_0 \ge 0$. This condition allows partially Saffman–Taylor stable data: an explicit overturned interface satisfies it for $A = 0$ and, numerically, for several values of $A \neq 0$. Mixing solutions for such data are latent in the work of Luna-Velasco and Mengual (for $A = 0$ with exactly our range) but not stated there; for $A = 0$ they are very likely also covered by earlier work of Förster–Székelyhidi and of Castro–Faraco–Mengual, whose precise admissible ranges we have not compared with ours. The multilayers of Luna-Velasco and Mengual with several intermediate strips open faster for fully unstable data wherever $|\gamma_0|$ is small. As in their work, the Darcy field $u + Am = -\nabla p - \rho e_2$ of a subsolution, where m is its momentum, is written as a sum of Cauchy integrals whose densities are affine in the Saffman–Taylor densities, and the resulting equations are solved by inverting $I \mp 2A\mathcal{K}$, where $\mathcal{K}$ is the adjoint Neumann–Poincaré operator of the interface. A change of frame and of time scale, which applies equally to their theorem, transfers the existence results to injection-driven viscous fingering.

Contents

1. Introduction 2
2. Cauchy integrals and the Saffman–Taylor formulation of the Muskat problem 8
3. Relaxation of IPM with viscosity jump 13
4. Piecewise constant subsolutions 16
5. Proof of Theorem 1.1 20
6. Necessary conditions: proof of Theorem 1.2 25
7. Consequences: partially stable data, equal viscosities and viscous fingering 28
8. Remarks and open problems 31
Appendix A. Periodic Cauchy integrals 34
References 36

1. Introduction

1.1. **The equations.** We study two incompressible fluids in a two-dimensional, horizontally periodic porous medium (or Hele-Shaw cell) under gravity. The fluids have different densities and different viscosities, and their evolution is governed by the *incompressible porous media equation with viscosity jump*

$$\partial_t \rho + \operatorname{div}(\rho u) = 0, \qquad \operatorname{div} u = 0, \qquad \mu(\rho)\, u = -\nabla p - \rho\, e_2, \qquad (x,t) \in \mathbb{T} \times \mathbb{R} \times (0,T), \tag{1.1}$$

where $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, $e_2 = (0,1)$, u is the velocity, p the pressure, ρ the density and $\mu(\rho)$ the viscosity (permeability and gravity are normalised to one). After subtracting a hydrostatic pressure and rescaling (as in [LM26, Sec. 1.1]),

$$\rho \in \{-1,1\}, \qquad \mu(\rho) = 1 + A\rho, \qquad A = \frac{\mu^+ - \mu^-}{\mu^+ + \mu^-} \in (-1,1), \tag{1.2}$$

where $\mu^\pm = 1 \pm A$ are the viscosities of the heavy ($\rho = 1$) and light ($\rho = -1$) fluids and A is the viscosity Atwood number. Since μ is affine in ρ and $\rho^2 = 1$, Darcy's law can be written linearly as $u + Am = -\nabla p - \rho e_2$ with $m = \rho u$ [Men22].

When $\rho = 1$ above a curve $\Gamma(t)$ and $\rho = -1$ below it, (1.1) is the *Muskat problem* [Mus34]. Two scalar quantities on the interface govern it: the *Saffman–Taylor function* and the *vortex-sheet strength*

$$\sigma := \partial_n p^+ - \partial_n p^-, \qquad \gamma := u^+ \cdot \tau - u^- \cdot \tau, \tag{1.3}$$

where n and τ are the upward normal and the tangent, and $\pm$ denote traces from above and below. If $\sigma > 0$ the problem is locally well-posed [Amb04, CG07, CCG11, Mat19, NP20] (see the surveys [Gan17, GBL20]); if $\sigma < 0$ it is ill-posed [SCH04, CG07]: this is the Saffman–Taylor instability [ST58, Hom87]. In our normalisation $\sigma = -2(n_2 + AU)$, where $U = u \cdot n$ (cf. [LM26, (3.8)]), so the instability has a gravitational part ($n_2 > 0$: heavy fluid on top) and a viscous part ($AU > 0$: the less viscous fluid pushes into the more viscous one).

1.2. **Mixing solutions.** In the unstable regime one looks for weak solutions in which the fluids mix at small scales inside a growing *mixing zone*. Otto [Ott99] described the flat case by a relaxation based on the gradient-flow structure; Székelyhidi [Sze12], building on [DLS09, CFG11, Shv11], computed the relaxation of IPM for $A = 0$ and constructed mixing solutions by convex integration. For curved interfaces and $A = 0$, mixing solutions were constructed for fully unstable data in [CCF21] (see also [ACF21]), by piecewise constant subsolutions in [FS18, NS21] (according to its abstract, [FS18] also treats non-horizontal data in the Rayleigh–Taylor stable regime), with degraded profiles in [CFM19], and with localised mixing zones for bubbles and turned interfaces, reportedly opened slightly inside the stable regime, in [CFM22]; we have not checked these statements on the stable regime against the full texts of [FS18, CFM22]. For $|A| < 1$, Mengual [Men22] computed the lamination convex hull of IPM, proved the h-principle and treated the flat interface, showing how a *pinch singularity* in the relaxation prevents the two fluids from mixing wherever there is neither Rayleigh–Taylor nor vorticity at the interface; the identification of the relaxation was completed in [LM26, Thm. 2.4].

Very recently, Luna-Velasco and Mengual [LM26] constructed mixing solutions for curved interfaces with viscosity jump, namely asymptotically flat chord-arc curves with $z_0 - (\alpha, 0) \in H^5(\mathbb{R})$, under the assumption that the data are *fully unstable*, $\sigma_0 < 0$ everywhere. Their key idea is to formulate the inversion of the Birkhoff–Rott operator in terms of σ rather than the vorticity, which also gives a Saffman–Taylor formulation of the Muskat problem [LM26, Sec. 3.1]. Their subsolutions consist of one strip of intermediate density A, or of finitely many strips of different densities, around a central curve, defined through a second order expansion in time as in [FS18, CFM22]; the opening speeds are proportional to σ_0: in their notation $c = -\bar{c}\,\sigma^{(0)}/|\partial_\alpha z^\circ|$ [LM26, (7.13)], with $0 < \bar{c} < \frac{1}{2}$ for one strip and any $0 < \bar{c} < 1$ for several strips [LM26, Sec. 7.5] (see Section 1.5 for the comparison with our notation).

1.3. **Main results.** Let $z_0 \in \mathcal{Z}^{k,\gamma}$, $k \geq 3$, be a horizontally periodic interface (Definition 2.1) with upward normal n_0 and tangent τ_0. By Propositions 2.7 and 2.10, the Saffman–Taylor function, the normal velocity and the vortex-sheet strength of the Muskat problem at $t = 0$ are

$$\sigma_0 = -2\big(I - 2A\mathcal{K}_0\big)^{-1}[n_{0,2}], \qquad U_0 = -\mathcal{K}_0\sigma_0, \qquad \gamma_0 = -2\big(I + 2A\mathcal{K}_0\big)^{-1}[\tau_{0,2}], \tag{1.4}$$

where $\mathcal{K}_0$ is the adjoint Neumann–Poincaré operator of z_0. These are the horizontally periodic counterparts of [LM26, (5.12)–(5.13)] and of the relations between σ and the vorticity in [LM26,

(3.7b) and Sec. 3.1.3]. For an intermediate density $\rho_{\mathrm{m}} \in (-1,1)$ let $\mu_{\mathrm{m}} = 1 + A\rho_{\mathrm{m}}$ and define the *admissibility function* and the *maximal opening function*

(1.5)
$$\Phi(\kappa) := \kappa\big(2(1-A\rho_{\mathrm{m}})\kappa+\sigma_0\big) - \frac{(1-A^2)\gamma_0^2}{8\mu_{\mathrm{m}}}, \quad \kappa_{\max} := \frac{-\sigma_0 + \sqrt{\sigma_0^2 + (1-A\rho_{\mathrm{m}})(1-A^2)\gamma_0^2/\mu_{\mathrm{m}}}}{4(1-A\rho_{\mathrm{m}})}.$$

For $\kappa > 0$, $\Phi(\kappa) < 0$ if and only if $\kappa < \kappa_{\max}$. Finally, the *pinch set* of z_0 is

$$\Pi(z_0) := \{\alpha \in \mathbb{T} : \ \sigma_0(\alpha) \geq 0 \ \text{ and } \ \gamma_0(\alpha) = 0\} = \{\kappa_{\max} = 0\}.$$

Note that $\kappa_{\max} \geq -\sigma_0/(2(1-A\rho_{\mathrm{m}}))$, with equality if and only if $\gamma_0 = 0$ and $\sigma_0 \leq 0$.

Theorem 1.1 (Existence of single-strip mixing zones)**.** *Let $A \in (-1,1)$, $\gamma \in (0,1)$, $z_0 \in \mathcal{Z}^{6,\gamma}$, $\rho_{\mathrm{m}} \in (-1,1)$, and let $\kappa \in C^{4,\gamma}(\mathbb{T})$ satisfy*

$$0 < \kappa(\alpha) < \kappa_{\max}(\alpha) \qquad \text{for every } \alpha \in \mathbb{T}. \tag{1.6}$$

Then there exist $T > 0$, $a \in C^{3,\gamma}(\mathbb{T})$ and a polynomial h with $|h(t)| \leq Ct^3$ such that, setting

(1.7)
$$Z(\alpha,t) = z_0(\alpha) + \Big(tU_0(\alpha) + \tfrac{t^2}{2}a(\alpha)\Big)n_0(\alpha) + h(t)e_2, \qquad z^{\pm}(\alpha,t) = Z(\alpha,t) \pm t(1\pm\rho_{\mathrm{m}})\kappa(\alpha)n_0(\alpha),$$

and denoting by $\Omega_{\mathrm{mix}}(t)$ the open strip between the curves $\Gamma^{\pm}(t) = z^{\pm}(\mathbb{R},t)$, there are infinitely many weak solutions (ρ,u) of (1.1) in $\mathbb{R}^2 \times (0,T)$, that is, with $x \in \mathbb{R}^2$ in place of $x \in \mathbb{T}\times\mathbb{R}$, with the Muskat initial datum $\rho_0 = \mathbf{1}_{\Omega^+(z_0)} - \mathbf{1}_{\Omega^-(z_0)}$ such that

- (i) *$\rho = 1$ above $\Gamma^+(t)$ and $\rho = -1$ below $\Gamma^-(t)$;*
- (ii) *ρ mixes in Ω_{mix}: both phases have positive measure in every open subset of $\{(x,t) : 0 < t < T,\ x \in \Omega_{\mathrm{mix}}(t)\}$;*
- (iii) *a sequence of such solutions converges weakly-$*$ to the density equal to 1, ρ_{m}, -1 above, inside and below the strip.*

The subsolution behind Theorem 1.1 is 2π-periodic in x_1; the convex integration is applied to its periodic lift to $\mathbb{R}^2$, so the solutions need not be periodic. The condition (1.6) is sharp within the class of single-strip subsolutions, whose mixing zone is one strip of constant intermediate density (Definition 6.1):

Theorem 1.2 (Necessary conditions)**.** *Let $z_0 \in \mathcal{Z}^{3,\gamma}$ and $\rho_{\mathrm{m}} \in (-1,1)$, and let $(\bar\rho, \bar u, \bar m)$ be a single-strip subsolution with intermediate density ρ_{m} emanating from z_0, whose boundary curves have initial normal velocities $W^{\pm}$. Then*

- (a) *the centre of mass starts with the Muskat velocity: $\frac{1}{2}[(1-\rho_{\mathrm{m}})W^+ + (1+\rho_{\mathrm{m}})W^-] = U_0$; equivalently $W^{\pm} = U_0 \pm (1\pm\rho_{\mathrm{m}})\kappa$ with $\kappa = \frac{1}{2}(W^+ - W^-) \geq 0$;*
- (b) *$\Phi(\kappa) \leq 0$ on $\mathbb{T}$; that is, $\kappa \leq \kappa_{\max}$, and in particular $\kappa = 0$ on $\Pi(z_0)$.*

Theorem 1.2 constrains the type of subsolution rather than the width of the mixing zone: the multilayer subsolutions of [LM26, Sec. 7.5], whose mixing zone contains several intermediate densities, are not single-strip subsolutions and can open faster (Section 1.5). The proof is short and is built from ingredients of [LM26]: the boundary motion determines the Darcy field and, through the Rankine–Hugoniot conditions, the normal component of the relaxed flux on the boundary of the strip [LM26, Secs. 3.3.1–3.3.2 and (4.12)]; part (a) is the necessity counterpart of the first order compatibility condition [LM26, Lemma 5.2]; and the term in γ_0 comes from [LM26, (3.7b)]. For the flat interface both parts are already contained in [LM26, Sec. 3.2], within its explicit ansatz with an arbitrary intermediate density: the centre of mass is at rest, and the hull condition reads $\kappa < 1/(1-A\rho_{\mathrm{m}}) = \kappa_{\max}$. We expect Theorem 1.2 to hold also for the analogous class of subsolutions in the asymptotically flat setting of [LM26], which contains their single-layer subsolutions (with $\rho_{\mathrm{m}} = A$); Steps 1 and 4 of the proof, which use the periodic far field and the invertibility of $I - 2A\mathcal{K}_0$ on periodic Hölder spaces, are the ones that need adaptation. For the single-layer subsolutions actually constructed in [LM26] the conclusions are immediate: (a) holds by construction [LM26, Lemma 5.2], and letting $t \to 0$ in their hull condition [LM26, (4.14)], with [LM26, (4.10), (7.9)–(7.11) and (3.7b)], gives (1.8) below with $\leq$ in place of $<$, which implies $\Phi(\kappa) \leq 0$. Combining the two theorems:

Corollary 1.3 (Pinch-free data)**.** *Let $z_0 \in \mathcal{Z}^{6,\gamma}$. A single-strip subsolution emanating from z_0 with everywhere positive opening exists if and only if $\Pi(z_0) = \emptyset$. In that case the conclusions of Theorem 1.1 hold with $\kappa = \theta\kappa_{\max}$ for every $\theta \in (0,1)$.*

Pinch-free data need not be fully unstable. For $A = 0$, Corollary 1.4 and the overturned interface of Section 7 give pinch-free data that are Rayleigh–Taylor stable on part of the interface; for $A \in \{0.3, -0.6, 0.9\}$ the same interface is pinch-free and partially Saffman–Taylor stable according to a numerical computation, without validated error bounds. On the part of the interface where $\sigma_0 \geq 0$ the admissible opening comes entirely from the vortex-sheet term $(1-A^2)\gamma_0^2/(8\mu_{\mathrm{m}})$ in Φ. For $\rho_{\mathrm{m}} = A$ the existence part of Corollary 1.3 is latent in the asymptotically flat setting of [LM26], with the range (1.8), which for $\kappa > 0$ is contained in (1.6): it would follow from the unmodified single layer of [LM26] by keeping the full modulus $|Au + e_2|$ in their final estimate [LM26, Sec. 7.4] and choosing an opening with $c - c_\infty \in H^{k-1}$ and $\inf c > 0$ (Sections 1.4 and 1.5); [LM26] do not state it, and we have not written out this modification (Section 8, open problem (3)).

Corollary 1.4 (Equal viscosities)**.** *Let $A = 0$ and $z_0 \in \mathcal{Z}^{6,\gamma}$. Then $\kappa_{\max} = \frac{1}{2}(1 + n_{0,2})$ and $\Pi(z_0) = \{\alpha : \tau_0(\alpha) = -e_1\}$. Hence, if the unit tangent of z_0 never equals $-e_1$, mixing solutions exist with any opening $\kappa \in C^{4,\gamma}(\mathbb{T})$ with $0 < \kappa < \frac{1}{2}(1+n_{0,2})$; this includes overturned interfaces, which are Rayleigh–Taylor stable on the overturned part.*

The formulas $\sigma_0 = -2n_{0,2}$ and $\gamma_0 = -2\tau_{0,2}$ for $A = 0$ (Example 2.12), on which Corollary 1.4 rests, are those of [LM26, Secs. 1.2 and 3.1.3]. Moreover, for $A = 0$ we have $|Au + e_2| = 1$, and by [LM26, (4.14) and (7.9)] the hull condition of the single layer of [LM26] at $t = 0$ reads $|2\kappa - n_{0,2}| < 1$; this is the range of Corollary 1.4, i.e. (1.8) with $A = 0$. For $A = 0$, see also [FS18] for stable non-horizontal data and [CFM22] for turned interfaces; we have not compared Corollary 1.4 with their precise admissible ranges.

Corollary 1.5 (Viscous fingering)**.** *The existence statements above (Theorem 1.1, the 'if' part of Corollary 1.3, and Corollary 1.4) apply, after a change of frame and of time scale, to the problem with a uniform injection velocity Ve_2 at infinity and a gravity constant $G \geq 0$, provided $G_{\mathrm{eff}} := G + AV > 0$; the opening functions, in particular $\kappa_{\max}$, then refer to the rescaled time $G_{\mathrm{eff}}t$. In particular, for fluids of equal density ($G = 0$), mixing solutions exist whenever the less viscous fluid is injected into the more viscous one ($AV > 0$) and the data are pinch-free; see Corollary 7.1.*

The same change of frame applies to [LM26, Thm. 1.1], which therefore already gives injection-driven mixing solutions for fully unstable data, also without density contrast. The condition $G + AV > 0$ is the Saffman–Taylor criterion for the flat interface [ST58] (see also [Hom87]); for an optimal bound on the growth rate of the mixing zone in viscous fingering see [Ott97].

Figure 1 illustrates Corollary 1.3 on an overturned interface.

1.4. **The mechanism: the tangential component of the relaxed flux.** The relaxation of (1.1) is described in Section 3 by a vector D (a normalised relaxed flux) lying in the open disc $|D - a| < |a|$, where $a = e_2 + \frac{A}{\mu}v$ is computed from the density and the Darcy field $v = u + Am$ of the state; this is the relaxed set of [Men22], written (O) in [LM26], described at fixed Darcy field. In a thin mixing zone, mass conservation and the Rankine–Hugoniot conditions fix the *normal* component of D, namely $D \cdot n_0 = -\sigma_0 - 2(1 - A\rho_{\mathrm{m}})\kappa$ at leading order. The *tangential* component of D is the transverse derivative of a stream function across a strip of width $O(t)$; it is therefore controlled by t^{-1} times the $O(t)$ part of the primitive of the flux mismatch M between the two boundaries (Lemma 4.5), and that part is governed by the normal acceleration of the central curve, which is the unknown of the second order compatibility condition. This is the structure of the construction of [LM26]: the boundary motion determines the tangential derivatives of their potential on the boundaries [LM26, Sec. 3.3.1 and (4.12)]; the transverse derivative involves $t^{-1}\partial_\lambda G$ [LM26, Sec. 3.3.4]; and the second order condition is affine in the acceleration, with linear part $I - AP$, our $I - 2A\mathcal{K}_0$ [LM26, Lemma 5.3, (5.21) and (8.41)],

and solvable for every right-hand side [LM26, Prop. 6.1] (in the periodic setting, by [CCG11]). Lemmas 5.5 and 5.6 are this second order step with a prescribed nonzero target. Writing $a^{(0)}$ for the limit of a as $t \to 0$, and δ for the limit of $D \cdot \tau_0$,

$$a^{(0)} \cdot n_0 = -\frac{\sigma_0}{2} - \frac{A^2(1-\rho_{\mathrm{m}}^2)\kappa}{\mu_{\mathrm{m}}}, \qquad a^{(0)} \cdot \tau_0 = -\frac{(1-A^2)\gamma_0}{2\mu_{\mathrm{m}}} =: \delta_*,$$

(for $\rho_{\mathrm{m}} = A$ the normal component is computed in [LM26, (7.11) and the proof of Prop. 7.4], and the tangential one follows from [LM26, (3.7b)]), and the limit $D^{(0)}$ of D satisfies

$$D^{(0)} \cdot \big(D^{(0)} - 2a^{(0)}\big) = \frac{2(1-A^2)}{\mu_{\mathrm{m}}}\,\Phi(\kappa) + (\delta - \delta_*)^2$$

(Lemma 5.2). The admissible range is thus governed by $|a|$, which combines the Saffman–Taylor function and the vortex-sheet strength (in the variables of [LM26] the radius of the disc is $|Au + e_2|$, and a positive lower bound for it in the mixing zone is what [LM26] single out as the additional ingredient required by the viscosity jump [LM26, Secs. 3.3.4 and 7.3]), and the best choice is $\delta = \delta_*$, which places D on the line through the centre of the disc parallel to n_0. Prescribing the target of the second order condition accordingly gives Theorem 1.1. As $\kappa \to 0$ the disc degenerates only where $\sigma_0 = \gamma_0 = 0$; on the rest of $\Pi(z_0)$, where $\sigma_0 > 0$ and $\gamma_0 = 0$, it is the forced normal component of D that lies outside the disc. (The name of $\Pi(z_0)$ refers to the pinch singularity of [Men22].)

The single layer of [LM26] is already tilted. For $\rho_{\mathrm{m}} = A$, their momentum ansatz [LM26, (4.6)] and Darcy's law [LM26, (4.10)] give, in our variables, $D = (Au + e_2) - 2\nabla^\perp g$ and $a = (e_2 - A\nabla p)/(1+A^2)$, so that $D \cdot (D - 2a) = (4|\nabla g|^2 - |Au + e_2|^2)/(1+A^2)$ and their exact hull condition $2|\nabla g| < |Au + e_2|$ [LM26, (4.14)] is our disc. They choose the acceleration so as to cancel the $O(t)$ part of $\partial_\alpha G_+ - \partial_\alpha G_-$ [LM26, Sec. 3.3.4 and Lemma 5.3], a condition on their potential g (with $G = g \circ Z$) rather than on our mismatch M. Then $\partial_\lambda G = O(t^2)$, ∇g is tangential at leading order [LM26, Props. 7.2–7.3], and, by [LM26, (4.10), (7.11) and (3.7b)],

$$D \cdot \tau_0 = (Au + e_2) \cdot \tau_0 + O(t) = \delta_{\mathrm{LM}} + O(t), \qquad \delta_{\mathrm{LM}} := -\frac{(1-A^2)\gamma_0}{2+A^2} = \frac{2(1+A^2)}{2+A^2}\,\delta_*.$$

Thus their layer is tilted towards the centre of the disc, and exactly onto it when $A = 0$; in terms of our mismatch, $M = -2t\,\partial_\alpha(\kappa\delta_{\mathrm{LM}}) + O(t^2)$ on their subsolution. (The choice $\delta = 0$, i.e. $M = O(t^2)$, gives a different subsolution, not used in [LM26], whose range $0 < 2(1 - A\rho_{\mathrm{m}})\kappa < -\sigma_0$ coincides with the range stated in [LM26] for $\rho_{\mathrm{m}} = A$.) The stated range $\bar{c} < \frac{1}{2}$ of [LM26] comes only from the lower bound $|Au + e_2| \geq (Au + e_2) \cdot n_0 = -\frac{1}{2}\sigma_0 + O(t)$ in their final estimate [LM26, Sec. 7.4], which discards the tangential part; up to that point their opening c is free [LM26, (5.1)]. What we add is to keep the full modulus and to choose the normalisation optimally, $\delta = \delta_*$; compared with δ_{LM} this gains little for $A \neq 0$ and nothing for $A = 0$ (Section 1.5). Theorem 1.2 shows that nothing better can be achieved with a single strip of constant intermediate density: the normal component of D on the boundary of the strip is forced, and the hull condition at $t = 0$, minimised over the free tangential component (that is, for $\delta = \delta_*$), is exactly $\Phi(\kappa) \leq 0$. Theorem 1.2 is proved only in our periodic setting; for the single layer of [LM26] the bound $\Phi(\kappa) \leq 0$ follows directly from their hull condition [LM26, (4.14)] (see the discussion after Theorem 1.2).

1.5. **Relation to** [LM26]**.** Sections 2–4 give a self-contained account, in a horizontally periodic Hölder setting, of the Saffman–Taylor formulation and of the single-strip ansatz of [LM26], and of the relaxation of [Men22]. The periodic operator theory used in Section 2 (the smoothing property of $\mathcal{K}$ and the invertibility of $I - 2\xi\mathcal{K}$ for $|\xi| \leq 1$) is due to [CCG11], in Sobolev spaces; we use Hölder versions. In the periodic setting the mass balance of the strip (Lemma 4.3) requires the vertical correction h in (1.7), which has no counterpart in [LM26].

In the opening speed $c = -\bar{c}\,\sigma^{(0)}/|\partial_\alpha z^\circ|$ of [LM26], the quantity $\sigma^{(0)}/|\partial_\alpha z^\circ|$ is our σ_0; the boundaries of their mixing zone move with the normal speeds $c/(1 \mp A)$ relative to the central curve, so that $c = (1 - A^2)\kappa$ in our notation. The main result [LM26, Thm. 1.1] allows any $0 < \bar{c} < 1$, which is reached with finitely many strips of different densities [LM26, Sec. 7.5].

These multilayer subsolutions are not single-strip subsolutions. The half-width of their mixing zone grows like $t\kappa$ with $\kappa = c/(1-A^2) = -\bar{c}\,\sigma_0/(1-A^2)$, which for $\bar{c}$ close to 1 exceeds $\kappa_{\max}$ (with $\rho_{\mathrm{m}} = A$) wherever $(1-A^2)^2\gamma_0^2 < 8(1+A^2)\sigma_0^2$; where $\gamma_0 = 0$ the supremum $-\sigma_0/(1-A^2)$ equals $2\kappa_{\max}$. Our results are therefore to be compared with the single layer of [LM26], whose stated range is $c < -\sigma^{(0)}/(2|\partial_\alpha z^\circ|)$, i.e. $\bar{c} < \frac{1}{2}$, which in our notation is $0 < \kappa < -\sigma_0/(2(1-A^2))$. Relative to [LM26], what is new is the following.

(i) Theorem 1.2, that is, necessity within single-strip subsolutions of constant intermediate density. It is short and built from the ingredients of [LM26], and its flat case is contained in [LM26, Sec. 3.2].

(ii) An explicit single-layer range depending on the vortex sheet for $A \neq 0$, including openings on pinch-free data that are partially Saffman–Taylor stable (Corollary 1.3). Such a range is latent in the construction of [LM26], but it is not stated there (nor, as far as we know, elsewhere for $A \neq 0$): keeping the full modulus $|Au + e_2|$, the leading order as $t \to 0$ of the exact hull condition [LM26, (4.14)] for the unchanged single layer of [LM26], with $c = (1-A^2)\kappa$, is

$$\kappa\big(2(1-A^2)\kappa + \sigma_0\big) < \frac{(1-A^2)\gamma_0^2}{2(2+A^2)^2}, \tag{1.8}$$

equivalently $\frac{2(1-A^2)}{\mu_{\mathrm{m}}}\Phi(\kappa) + (\delta_{\mathrm{LM}} - \delta_*)^2 < 0$ with $\rho_{\mathrm{m}} = A$. Turning (1.8) into an existence theorem in their setting requires only a modification of a few lines of [LM26, Sec. 7.4], together with the conditions $c - c_\infty \in H^{k-1}$ and $\inf c > 0$ of [LM26, Secs. 4.1 and 5]; [LM26] state that their aim is to construct mixing solutions rather than to optimise the opening speed [LM26, Sec. 7.5].

(iii) The optimal tilt δ_* for $A \neq 0$ and general ρ_{m}, which makes the range sharp within single strips. For $\rho_{\mathrm{m}} = A$ it improves the coefficient of γ_0^2 in (1.8) by the factor $(2+A^2)^2/(4(1+A^2)) \in [1, \frac{9}{8})$, and it gives nothing new for $A = 0$. On the overturned interface of Section 7, with $\rho_{\mathrm{m}} = A$, the minimum of $\kappa_{\max}$ over the interface is 0.2350, 0.2297, 0.2265, 0.2959 for $A = 0$, 0.3, -0.6, 0.9, against 0.2350, 0.2294, 0.2224, 0.2778 for the range (1.8) (computed numerically), while the range stated in [LM26] is empty on the stable part of that interface.

(iv) Curved strips with $\rho_{\mathrm{m}} \neq A$, i.e. mixing zones in which the opening speeds of the two boundaries have the ratio $(1+\rho_{\mathrm{m}})/(1-\rho_{\mathrm{m}})$, and the horizontally periodic Hölder setting; both extensions are routine. For the flat interface, [LM26, Sec. 3.2] already treat an arbitrary intermediate density θ_{mix}, with the same hull condition and mass balance, and choose $\theta_{\mathrm{mix}} = A$ because it maximises the product h_+h_- of the opening speeds. Their curved single layer has the intermediate density A; their momentum ansatz and Darcy law [LM26, (4.6), (4.10)] are written for a general intermediate phase, and their multilayers use them [LM26, (7.17)] in strips with the levels θ_j of [LM26, (7.14)], which include $\theta_N = A$ and, for $N \geq 2$, other values.

Our existence statements for $A = 0$ are very likely covered by [FS18, CFM22], although we have not compared them with the precise admissible ranges there, and they also follow from the construction of [LM26]: for $A = 0$, (1.8) is $\Phi(\kappa) < 0$. Corollary 1.5 is a change of frame, which applies equally to [LM26, Thm. 1.1]. Finally, Theorem 1.1(ii) asserts mixing in space–time, whereas [LM26] prove mixing in every time slice, a property going back to [CFM19].

Organisation. Section 2 recalls periodic Cauchy integrals, the Saffman–Taylor formulation of [LM26] and, following [LM26, Sec. 3.1.3], the relation between the vortex-sheet strength and σ. Section 3 contains the relaxation, following [Men22], and the h-principle of [Men22]. Section 4 introduces the single-strip ansatz. Section 5 proves Theorem 1.1, Section 6 proves Theorem 1.2, and Section 7 proves the corollaries and discusses an example. Section 8 contains remarks and open problems, and the appendix collects facts on Cauchy integrals.

2. Cauchy integrals and the Saffman–Taylor formulation of the Muskat problem

2.1. **Curves and notation.** We identify $\mathbb{R}^2$ with $\mathbb{C}$, $x = x_1 + ix_2$. For a vector $w = (w_1, w_2)$ we write $w^\flat := w_1 - iw_2$ (the complex conjugate of w seen as a complex number); a vector field w is divergence-free and curl-free in an open set if and only if $w^\flat$ is holomorphic there. We write $w^\perp = (-w_2, w_1)$.

Definition 2.1. For $k \geq 1$ and $\gamma \in (0,1)$ let $\mathcal{Z}^{k,\gamma}$ be the set of maps $z \in C^{k,\gamma}_{\mathrm{loc}}(\mathbb{R};\mathbb{R}^2)$ such that $z(\alpha) - (\alpha, 0)$ is 2π-periodic and

$$\delta(z) := \inf\Big\{ \frac{|\sin\frac{z(\alpha)-z(\beta)}{2}|}{|\sin\frac{\alpha-\beta}{2}|} : \ \alpha,\beta \in \mathbb{R},\ \alpha - \beta \notin 2\pi\mathbb{Z} \Big\} > 0,$$

where sin is the complex sine. We norm $\mathcal{Z}^{k,\gamma}$ by $\|z - (\cdot, 0)\|_{C^{k,\gamma}(\mathbb{T})}$.

Since $|\sin\frac{w}{2}|^2 = \sin^2\frac{\operatorname{Re} w}{2} + \sinh^2\frac{\operatorname{Im} w}{2}$, the condition $\delta(z) > 0$ says that z induces an embedding of $\mathbb{T}$ into the cylinder $\mathbb{T}\times\mathbb{R}$, quantitatively; in particular $|\partial_\alpha z| \geq \delta(z)$. The curve $\Gamma = z(\mathbb{R})$ is invariant under $x \mapsto x + 2\pi e_1$ and $\mathbb{R}^2 \setminus \Gamma$ has two connected components $\Omega^+(z)$ and $\Omega^-(z)$, containing $\{x_2 > \max z_2\}$ and $\{x_2 < \min z_2\}$ respectively. We write

$$\tau = \frac{\partial_\alpha z}{|\partial_\alpha z|}, \qquad n = \tau^\perp \ \ (\text{so } n = i\tau \text{ as complex numbers}), \qquad \mathrm{d}s = |\partial_\alpha z|\,\mathrm{d}\alpha,$$

so that n points into $\Omega^+(z)$, and $n_2 = \partial_\alpha z_1/|\partial_\alpha z|$. For a function w defined near Γ we denote by $w^\pm(\alpha)$ its limits at $z(\alpha)$ from $\Omega^\pm(z)$, by $[\![w]\!] = w^+ - w^-$ its jump and by $\langle w\rangle = \frac12(w^+ + w^-)$ its average. The constant C may change from line to line.

2.2. **Periodic Cauchy integrals.**

Definition 2.2. Let $z \in \mathcal{Z}^{1,\gamma}$ and $f \in C^{0,\gamma}(\mathbb{T};\mathbb{R}^2)$. The *Cauchy field* of f along z is the vector field $\mathcal{C}[z;f]$ on $\mathbb{R}^2\setminus\Gamma$ defined by

$$\mathcal{C}[z;f]^\flat(x) = \frac{1}{4\pi i}\int_{\mathbb{T}} f^\flat(\beta)\,\partial_\beta z(\beta)\,\cot\frac{z(\beta)-x}{2}\,\mathrm{d}\beta. \tag{2.1}$$

Since $\frac12\cot\frac{w}{2} = \frac1w + O(w)$, formula (2.1) is the periodisation of the classical Cauchy integral

$$\frac{1}{2\pi i}\int \frac{f^\flat(\zeta)\,\mathrm{d}\zeta}{\zeta - x}.$$

Lemma 2.3. *Let $z \in \mathcal{Z}^{1,\gamma}$, $f \in C^{0,\gamma}(\mathbb{T};\mathbb{R}^2)$ and $\mathcal{C} = \mathcal{C}[z;f]$.*

(a) *$\mathcal{C}$ is real analytic and 2π-periodic in x_1 on $\mathbb{R}^2\setminus\Gamma$, with $\operatorname{div}\mathcal{C} = \operatorname{curl}\mathcal{C} = 0$ there.*

(b) (Plemelj–Privalov) *The restrictions of $\mathcal{C}$ to $\Omega^\pm(z)$ extend to $C^{0,\gamma}(\overline{\Omega^\pm(z)})$ with*

$$\|\mathcal{C}\|_{C^{0,\gamma}(\overline{\Omega^\pm})} \leq C\|f\|_{C^{0,\gamma}},$$

where C depends on γ, $\delta(z)$ and $\|z\|_{\mathcal{Z}^{1,\gamma}}$. Moreover $[\![\mathcal{C}]\!] = f$, and $\langle\mathcal{C}\rangle$ is given by (2.1) with $x = z(\alpha)$ and the integral understood in the principal value sense.

(c) *As $x_2 \to \pm\infty$, $\mathcal{C}(x) \to \mathcal{C}_{\pm\infty}$ uniformly in x_1, where $\mathcal{C}^\flat_{\pm\infty} = \pm\frac{1}{4\pi}\int_{\mathbb{T}} f^\flat\partial_\beta z\,\mathrm{d}\beta$. In particular, for a constant vector c we have $\mathcal{C}[z;c] = \pm\frac{c}{2}$ in $\Omega^\pm(z)$, and for a scalar ϕ*

$$\mathcal{C}[z;\phi n]_{\pm\infty} = \pm\frac{1}{4\pi}\Big(\int_{\mathbb{T}}\phi\,|\partial_\beta z|\,\mathrm{d}\beta\Big)e_2. \tag{2.2}$$

(d) *If $j \geq 0$, $z \in \mathcal{Z}^{j+1,\gamma}$ and $f \in C^{j,\gamma}$, then $\mathcal{C} \in C^{j,\gamma}(\overline{\Omega^\pm(z)})$ and $\mathcal{C}^\pm \in C^{j,\gamma}(\mathbb{T})$, with bounds.*

(e) (Representation) *Let w be a bounded vector field on $\mathbb{R}^2\setminus\Gamma$, 2π-periodic in x_1, divergence-free and curl-free in $\mathbb{R}^2\setminus\Gamma$, whose restrictions to $\Omega^\pm(z)$ extend continuously to the closures with $[\![w]\!] \in C^{0,\gamma}(\mathbb{T})$, and with $w(x) \to 0$ as $x_2 \to \pm\infty$. Then $w = \mathcal{C}[z;[\![w]\!]]$.*

Proof. (a) is immediate. (b) and (d) are the periodic versions of the Plemelj–Privalov theorem and are discussed in Appendix A. (c) When $x_2 \to +\infty$ (resp. $-\infty$), $\cot\frac{z(\beta)-x}{2} \to i$ (resp. $-i$) uniformly in β, which gives $\mathcal{C}_{\pm\infty}$. If $f = c$ is constant, then $\partial_x \int_{\mathbb{T}} c^\flat \partial_\beta z \cot\frac{z(\beta)-x}{2}\,\mathrm{d}\beta = -c^\flat \int_{\mathbb{T}} \partial_\beta\big[\cot\frac{z(\beta)-x}{2}\big]\,\mathrm{d}\beta = 0$ by periodicity, so $\mathcal{C}[z;c]$ is constant in each component and equals its limit $\pm\frac{1}{4\pi}c^\flat \cdot 2\pi = \pm\frac{c^\flat}{2}$; recall $\int_{\mathbb{T}} \partial_\beta z\,\mathrm{d}\beta = 2\pi$. For $f = \phi n$ we have $n^\flat \partial_\beta z = -i|\partial_\beta z|$, so $\mathcal{C}^\flat_{+\infty} = -\frac{i}{4\pi}\int \phi|\partial_\beta z|$, i.e. (2.2). (e) The field $W = w - \mathcal{C}[z;[\![w]\!]]$ is bounded, periodic, continuous across Γ, and $W^\flat$ is holomorphic off Γ; hence $W^\flat$ is entire (Morera), and by Liouville's theorem on the cylinder it is constant. By (c), $W \to -\mathcal{C}_{+\infty}$ as $x_2 \to +\infty$ and $W \to -\mathcal{C}_{-\infty} = \mathcal{C}_{+\infty}$ as $x_2 \to -\infty$. Therefore $\mathcal{C}_{+\infty} = 0$ and $W \equiv 0$. □

2.3. **The adjoint Neumann–Poincaré operator.**

Definition 2.4. For $z \in \mathcal{Z}^{2,\gamma}$ and a real function $\phi \in C^{0,\gamma}(\mathbb{T})$ we set

$$\mathcal{K}(z)\phi := \langle \mathcal{C}[z;\phi n]\rangle \cdot n.$$

Since the jump of $\mathcal{C}[z;\phi n]$ is normal, Lemma 2.3(b) gives

$$\mathcal{C}[z;\phi n]^\pm \cdot n = \mathcal{K}(z)\phi \pm \tfrac{1}{2}\phi, \qquad \mathcal{C}[z;\phi n]^+ \cdot \tau = \mathcal{C}[z;\phi n]^- \cdot \tau. \tag{2.3}$$

In potential-theoretic terms, $\mathcal{C}[z;\phi n]$ is the gradient of the single layer potential of ϕ and $\mathcal{K}(z)$ is the (periodic) adjoint Neumann–Poincaré operator.

Lemma 2.5(a) and Proposition 2.6 below are Hölder-space versions of classical results of Córdoba, Córdoba and Gancedo [CCG11] in the horizontally periodic setting: the smoothing property of $\mathcal{K}(z)$ (cf. [CCG11, Lemmas 3.1 and 5.2]) and, in Sobolev spaces, the invertibility of $I - 2\xi\mathcal{K}(z)$ for $|\xi| \le 1$, by a harmonic-energy argument (cf. [CCG11, Sec. 4]). They are recalled in [LM26, Secs. 3.1.3, 8.2 and 8.3], whose operator P corresponds to our $2\mathcal{K}$, and the versions for asymptotically flat curves are [LM26, Props. 6.1 and 8.3].

Lemma 2.5. *Let $k \ge 2$ and $z \in \mathcal{Z}^{k,\gamma}$.*

(a) $\mathcal{K}(z)\phi(\alpha) = \int_{\mathbb{T}} k_z(\alpha,\beta)\phi(\beta)\,\mathrm{d}\beta$, *with*

$$k_z(\alpha,\beta) = \frac{|\partial_\beta z(\beta)|}{4\pi|\partial_\alpha z(\alpha)|}\ \mathrm{Im}\Big[\partial_\alpha z(\alpha)\cot\frac{z(\beta)-z(\alpha)}{2}\Big],$$

and k_z extends from $\{\alpha \ne \beta\}$ to a function in $C^{k-2,\gamma}(\mathbb{T}\times\mathbb{T})$. Consequently $\mathcal{K}(z)$ maps $C^0(\mathbb{T})$ boundedly into $C^{k-2,\gamma}(\mathbb{T})$, and for $k \ge 3$ it is compact on $C^{j,\gamma}(\mathbb{T})$ for $0 \le j \le k-3$.

(b) $\int_{\mathbb{T}} \big(\mathcal{K}(z)\phi\big)\,|\partial_\alpha z|\,\mathrm{d}\alpha = 0$ *for every $\phi \in C^{0,\gamma}(\mathbb{T})$.*

Proof. (a) The complex density of ϕn is $\phi\, n^\flat \partial_\beta z = -i\phi|\partial_\beta z|$. Using $w \cdot n = \mathrm{Re}(w^\flat n)$ and $n = i\partial_\alpha z/|\partial_\alpha z|$, the principal value formula of Lemma 2.3(b) gives the stated kernel, once we show it is bounded (then no principal value is needed). Away from the diagonal $\{\alpha - \beta \in 2\pi\mathbb{Z}\}$ the kernel is $C^{k-1,\gamma}$, because $\delta(z) > 0$. Near the diagonal write $\cot\frac{w}{2} = \frac{2}{w} + g(w)$ with g analytic for $|w| < 2\pi$, and

$$z(\beta) - z(\alpha) = (\beta-\alpha)Z_1(\alpha,\beta), \qquad \partial_\alpha z(\alpha) - Z_1(\alpha,\beta) = -(\beta-\alpha)Z_2(\alpha,\beta),$$

with $Z_1 = \int_0^1 \partial_\alpha z(\alpha + s(\beta-\alpha))\,\mathrm{d}s \in C^{k-1,\gamma}$, $Z_2 = \int_0^1\int_0^1 s\,\partial_\alpha^2 z(\alpha + rs(\beta-\alpha))\,\mathrm{d}r\,\mathrm{d}s \in C^{k-2,\gamma}$ and $|Z_1| \ge c > 0$. Then

$$\mathrm{Im}\Big[\frac{2\partial_\alpha z(\alpha)}{z(\beta)-z(\alpha)}\Big] = \frac{2}{\beta-\alpha}\,\mathrm{Im}\Big[\frac{\partial_\alpha z(\alpha) - Z_1}{Z_1}\Big] = -2\,\mathrm{Im}\Big[\frac{Z_2}{Z_1}\Big] \in C^{k-2,\gamma},$$

while $\partial_\alpha z(\alpha) g(z(\beta)-z(\alpha)) \in C^{k-1,\gamma}$. Differentiating under the integral sign gives $\mathcal{K}(z) : C^0 \to C^{k-2,\gamma}$, and compactness follows from the compact embedding $C^{k-2,\gamma}(\mathbb{T}) \subset C^{j,\gamma}(\mathbb{T})$ for $j \le k-3$.

(b) Let $w = \mathcal{C}[z;\phi n]$. Since w is divergence-free and periodic, its flux through Γ from above equals its flux through $\{x_2 = L\}$ for every large L, and letting $L \to \infty$ we get from (2.2)

$\int_{\mathbb{T}} w^+ \cdot n \, \mathrm{d}s = 2\pi(\mathcal{C}_{+\infty})_2 = \frac{1}{2}\int \phi \, \mathrm{d}s$. Similarly $\int_{\mathbb{T}} w^- \cdot n \, \mathrm{d}s = 2\pi(\mathcal{C}_{-\infty})_2 = -\frac{1}{2}\int \phi \, \mathrm{d}s$. Adding, $\int \langle w \rangle \cdot n \, \mathrm{d}s = 0$. □

Proposition 2.6. *Let $|A| < 1$, $k \geq 3$ and $z \in \mathcal{Z}^{k,\gamma}$. Then $I - 2A\mathcal{K}(z)$ is an isomorphism of $C^{j,\gamma}(\mathbb{T})$ for $0 \leq j \leq k-3$. Moreover, if $f \in C^{j,\gamma}(\mathbb{T})$ with $0 \leq j \leq k-2$, the solution $\sigma \in C^{0,\gamma}(\mathbb{T})$ of $(I - 2A\mathcal{K}(z))\sigma = f$ belongs to $C^{j,\gamma}(\mathbb{T})$, with $\|\sigma\|_{C^{j,\gamma}} \leq C\|f\|_{C^{j,\gamma}}$, and $\int \sigma \, \mathrm{d}s = \int f \, \mathrm{d}s$.*

Proof. By Lemma 2.5(a) and the Riesz–Schauder theory, $I - 2A\mathcal{K}(z)$ is a Fredholm operator of index zero on $C^{j,\gamma}(\mathbb{T})$, $0 \leq j \leq k-3$; so it suffices to prove injectivity on $C^{0,\gamma}$. Let $\phi \in C^{0,\gamma}$ with $\phi = 2A\mathcal{K}(z)\phi$. If $A = 0$ there is nothing to prove, so let $A \neq 0$. By Lemma 2.5(b), $\int \phi \, \mathrm{d}s = 0$, so $w := \mathcal{C}[z; \phi n]$ tends to 0 as $x_2 \to \pm\infty$, by (2.2); being the conjugate of a periodic holomorphic function, it does so exponentially fast. Since the jump of w is normal, w is curl-free in $\mathcal{D}'(\mathbb{R}^2)$, and since its circulation along horizontal lines tends to zero, $w = \nabla\Phi$ for a periodic, Lipschitz function Φ which is $C^{1,\gamma}$ up to Γ from both sides and bounded. By (2.3) and $\mathcal{K}(z)\phi = \phi/(2A)$,

$$\frac{w^+ \cdot n}{1+A} = \frac{w^- \cdot n}{1-A} = \frac{\phi}{2A} =: g.$$

Integrating by parts on one period of $\Omega^\pm \cap \{|x_2| < L\}$ and letting $L \to \infty$ (the contributions of the horizontal boundaries vanish by the exponential decay, the lateral ones cancel by periodicity),

$$\frac{1}{1+A}\int_{\Omega^+} |w|^2 + \frac{1}{1-A}\int_{\Omega^-} |w|^2 = -\int_{\mathbb{T}} \Phi \, \frac{w^+ \cdot n}{1+A} \, \mathrm{d}s + \int_{\mathbb{T}} \Phi \, \frac{w^- \cdot n}{1-A} \, \mathrm{d}s = 0,$$

where the integrals over $\Omega^\pm$ are over one period. Hence $w \equiv 0$ and $\phi = [\![w]\!] \cdot n = 0$.

For the regularity statement with $j = k-2$: the solution $\sigma \in C^{k-3,\gamma}$ satisfies $\sigma = f + 2A\mathcal{K}(z)\sigma$, and $\mathcal{K}(z)\sigma \in C^{k-2,\gamma}$ by Lemma 2.5(a). Finally $\int \sigma \, \mathrm{d}s = \int f \, \mathrm{d}s$ by Lemma 2.5(b). □

2.4. **The Muskat problem with viscosity jump.** Let $z \in \mathcal{Z}^{k,\gamma}$, $k \geq 3$, and $\rho = \pm 1$ in $\Omega^\pm(z)$. The *instantaneous Darcy problem* consists in finding u and p, $C^{0,\gamma}$ and $C^{1,\gamma}$ up to Γ from both sides, such that

$$\operatorname{div} u = 0, \quad (1 \pm A)\, u = -\nabla p \mp e_2 \ \text{ in } \Omega^\pm(z), \qquad [\![p]\!] = 0, \quad [\![u \cdot n]\!] = 0 \ \text{ on } \Gamma, \tag{2.4}$$

with $u \to 0$ and $\nabla p + \rho e_2 \to 0$ as $x_2 \to \pm\infty$ (the fluids are at rest at infinity). We introduce the *Darcy field* and the *Saffman–Taylor function*

$$v := \mu(\rho)u = -\nabla p - \rho e_2, \qquad \sigma := [\![\partial_n p]\!] = \partial_n p^+ - \partial_n p^- \ \text{ on } \Gamma.$$

Proposition 2.7 and Corollary 2.8 below are the Saffman–Taylor formulation $(M\sigma)$ of Luna-Velasco and Mengual [LM26, Sec. 3.1], transcribed to the horizontally periodic setting: (2.6) says that $\nabla p = \mathcal{C}[z; \sigma n]$, which is the representation [LM26, (3.2)] of the pressure gradient; (2.7) is [LM26, (3.6)]; the identity $\sigma = -2(n_2 + AU)$ is [LM26, (3.7a) and (3.8)]; and (2.5) is [LM26, (5.12)]. The dictionary is $[\![\theta]\!] = -2$ for the phase jump of [LM26], whose Saffman–Taylor density is $|\partial_\alpha z|\sigma$ and whose operator P corresponds to $2\mathcal{K}(z)$. The proof below follows the derivation of $(M\sigma)$ in [LM26, Sec. 3.1.5].

Proposition 2.7. *Let $|A| < 1$, $k \geq 3$, $z \in \mathcal{Z}^{k,\gamma}$, and let $\sigma \in C^{k-2,\gamma}(\mathbb{T})$ be the solution of*

$$\big(I - 2A\mathcal{K}(z)\big)\sigma = -2n_2. \tag{2.5}$$

Then (2.4) has a solution, unique up to an additive constant in p, given by

$$v = -\mathcal{C}[z; \sigma n] \mp e_2 \quad \text{and} \quad u = \frac{v}{1 \pm A} \qquad \text{in } \Omega^\pm(z). \tag{2.6}$$

Its Saffman–Taylor function is σ, the normal velocity $U := u \cdot n$ on Γ is

$$U = -\mathcal{K}(z)\sigma, \tag{2.7}$$

and moreover $\sigma = -2(n_2 + AU)$ and $\int_{\mathbb{T}} \sigma \, \mathrm{d}s = -4\pi$.

Proof. Define v and u by (2.6). By (2.3), $v^\pm\cdot n=-\mathcal{K}\sigma\mp\frac12\sigma\mp n_2$, and (2.5) says $\frac12\sigma+n_2=A\mathcal{K}\sigma$. Therefore $v^\pm\cdot n=-(1\pm A)\mathcal{K}\sigma$, i.e. $u^+\cdot n=u^-\cdot n=-\mathcal{K}\sigma$, which gives $[\![u\cdot n]\!]=0$ and (2.7); then $\sigma=-2n_2+2A\mathcal{K}\sigma=-2(n_2+AU)$. Integrating (2.5) and using Lemma 2.5(b) and $\int n_2\,\mathrm{d}s=\int\partial_\alpha z_1\,\mathrm{d}\alpha=2\pi$, we get $\int\sigma\,\mathrm{d}s=-4\pi$; by (2.2), $v\to\mp\frac{1}{4\pi}(\int\sigma\,\mathrm{d}s)e_2\mp e_2=0$ as $x_2\to\pm\infty$. Next, $v+\rho e_2=-\mathcal{C}[z;\sigma n]$ in $\mathbb{R}^2\setminus\Gamma$; this field is curl-free in $\mathcal{D}'(\mathbb{R}^2)$, because its jump is normal, and its circulation along horizontal lines vanishes (its limits at $\pm\infty$ are vertical). Hence $v+\rho e_2=-\nabla p$ for a continuous function p, periodic in x_1, and $\partial_n p^\pm=\mathcal{C}[z;\sigma n]^\pm\cdot n=\mathcal{K}\sigma\pm\frac12\sigma$, so that $[\![\partial_n p]\!]=\sigma$. The field u is divergence-free in each $\Omega^\pm$ and has no normal jump, so $\operatorname{div}u=0$ in $\mathcal{D}'(\mathbb{R}^2)$. Regularity follows from Lemma 2.3(d).

For uniqueness, let (u',p') be the difference of two solutions. Then $\operatorname{div}u'=0$, $(1\pm A)u'=-\nabla p'$ in $\Omega^\pm$, p' is continuous, $u'\cdot n$ has no jump, and $u'\to0$ at infinity, exponentially fast since u' is the gradient of a periodic harmonic function in $\{|x_2|>L\}$. Integrating by parts over one period,

$$\int(1+A\rho)|u'|^2=-\int\nabla p'\cdot u'=\int_{\mathbb{T}}p'\left(u'^+-u'^-\right)\cdot n\,\mathrm{d}s=0,$$

so $u'=0$ and $\nabla p'=0$. □

The identity $\int_{\mathbb{T}}\sigma\,\mathrm{d}s=-4\pi$ is the periodic counterpart of the normalisation $\sigma^{(0)}+2\in H^{k-1}$ of [LM26, Prop. 6.3]. It also follows by integrating $\sigma=-2(n_2+AU)$ over a period, since $\int_{\mathbb{T}}U\,\mathrm{d}s=0$ by Lemma 2.5(b) (conservation of area).

Since the interface moves with the normal velocity of the fluid, Proposition 2.7 gives the following formulation of the Muskat problem with viscosity jump, in which the only operator to be inverted acts on the Saffman–Taylor function.

Corollary 2.8. *A family of curves $z(\cdot,t)\in\mathcal{Z}^{k,\gamma}$, $k\geq3$, solves the Muskat problem with viscosity jump if and only if*

$$\partial_t z\cdot n=-\mathcal{K}(z)\sigma,\qquad\big(I-2A\mathcal{K}(z)\big)\sigma=-2n_2.$$

Proof. By definition, z solves the Muskat problem if, for every t, $\partial_t z\cdot n=u\cdot n$ on $\Gamma(t)$ (the kinematic condition, cf. [LM26, (3.1)]), where u is the velocity of the solution of (2.4) for the curve $z(\cdot,t)$. By Proposition 2.7, $u\cdot n=-\mathcal{K}(z)\sigma$, where σ is the unique solution of (2.5) (Proposition 2.6). □

Remark 2.9 (Duality with the vorticity formulation). For a real function γ, the complex densities of $\gamma\tau$ and γn are $\gamma|\partial_\beta z|$ and $-i\gamma|\partial_\beta z|$, so $\mathcal{C}[z;\gamma\tau]^\flat=i\,\mathcal{C}[z;\gamma n]^\flat$, i.e. $\mathcal{C}[z;\gamma\tau]$ is obtained by rotating $\mathcal{C}[z;\gamma n]$ by $-\pi/2$; this is the relation between the representations of the velocity and of the pressure gradient in [LM26, (3.2)]. In particular $\langle\mathcal{C}[z;\gamma\tau]\rangle\cdot\tau=\langle\mathcal{C}[z;\gamma n]\rangle\cdot n=\mathcal{K}(z)\gamma$. The velocity is curl-free off Γ with jump $\gamma\tau$, $\gamma=[\![u\cdot\tau]\!]$, hence $u=\mathcal{C}[z;\gamma\tau]$ by Lemma 2.3(e). Hence $\langle u\rangle=\mathrm{BR}(z,\varpi)$, where $\mathrm{BR}(z,\varpi)^\flat(\alpha):=\frac{1}{4\pi i}\,\mathrm{p.v.}\int_{\mathbb{T}}\varpi(\beta)\cot\frac{z(\alpha)-z(\beta)}{2}\,\mathrm{d}\beta$ is the (periodised) Birkhoff–Rott integral and $\varpi:=-\gamma|\partial_\alpha z|$ is the vorticity density of the sheet per unit α (so that $\operatorname{curl}u=-\gamma\,\mathcal{H}^1\llcorner\Gamma$; this is the density ϖ of [LM26]). The tangential part of Darcy's law, $[\![v\cdot\tau]\!]=-[\![\rho]\!]\tau_2=-2\tau_2$, reads $\gamma+2A\langle u\rangle\cdot\tau=-2\tau_2$, i.e. $(I+2A\mathcal{K}(z))\gamma=-2\tau_2$; in terms of ϖ this is $\varpi-2A\,\mathrm{BR}(z,\varpi)\cdot\partial_\alpha z=2\partial_\alpha z_2$, which is $(I+AP)\varpi=-[\![\theta]\!]\partial_\alpha z_2$ of [LM26, Sec. 3.1.3] with $[\![\theta]\!]=-2$, and the equation for the vorticity of [CCG11] up to the sign conventions used there for the jumps and the Atwood number (cf. [LM26, Sec. 3.1.3]). Thus the vorticity is obtained by inverting $I+2A\mathcal{K}(z)$ and the Saffman–Taylor function by inverting $I-2A\mathcal{K}(z)$: as observed in [LM26, Sec. 3.1.3], both problems reduce to inverting operators of the form $I\pm AP$, and both are covered by Proposition 2.6. In the vorticity formulation the normal velocity $U=\langle u\rangle\cdot n=-\langle\mathcal{C}[z;\gamma n]\rangle\cdot\tau$ is a singular integral of γ, whereas in (2.7) it is the smoothing operator $\mathcal{K}(z)$ applied to σ. This concerns only the map from the density to the normal velocity at a fixed interface (cf. [LM26, Prop. 8.3]) and is not a gain of regularity for the evolution: the loss of derivatives behind the Saffman–Taylor instability sits in the dependence of $\mathcal{K}(z)$ on z, and the singular operator $\mathcal{T}(z)$ of Section 2.5 reappears when γ is recovered from σ (Proposition 2.10). As explained in [LM26, Secs. 3.1.4 and 3.3.5], the advantage of the Saffman–Taylor formulation appears for subsolutions with a mixing zone: the pressure remains

harmonic in each region, whereas the velocity carries vorticity in the interior of the mixing zone, so that a vorticity formulation would couple the reconstruction of the velocity to the interior extension of the stream function of the relaxed flux (Ψ in Section 4, the potential g in [LM26]); in the strips of Section 4, $\operatorname{curl}\bar{u} = -A\operatorname{curl}\bar{m} = -\frac{A(1-\rho_{\mathrm{m}}^2)}{2(1-A^2)}\Delta\Psi$ in S_t.

2.5. **The vortex-sheet strength and the pinch set.** The letter γ denotes both the Hölder exponent and the vortex-sheet strength; the meaning is always clear from the context. For $z \in \mathcal{Z}^{2,\gamma}$ and a real function $\phi \in C^{0,\gamma}(\mathbb{T})$ we also set

$$\mathcal{T}(z)\phi := \langle \mathcal{C}[z;\phi n]\rangle \cdot \tau.$$

Unlike $\mathcal{K}(z)$, the operator $\mathcal{T}(z)$ is a singular integral operator: the argument in the proof of Lemma 2.5(a) shows that, for $z \in \mathcal{Z}^{k,\gamma}$ with $k \geq 2$, $\mathcal{T}(z)\phi - \frac{1}{2|\partial_\alpha z|}H\big[|\partial_\alpha z|\phi\big]$ is given by a kernel in $C^{k-2,\gamma}(\mathbb{T}\times\mathbb{T})$, where $H\phi(\alpha) := \frac{1}{2\pi}\,\mathrm{p.v.}\int_{\mathbb{T}}\phi(\beta)\cot\frac{\alpha-\beta}{2}\,\mathrm{d}\beta$ is the periodic Hilbert transform; for the flat interface $\mathcal{T} = \frac{1}{2}H$ (cf. [LM26, (8.14)] on $\mathbb{R}$). The operator $\mathcal{T}(z)$ is bounded on $C^{j,\gamma}(\mathbb{T})$ for $z \in \mathcal{Z}^{j+1,\gamma}$ by Lemma 2.3(d). As in [LM26, Sec. 3.1.3], the vortex-sheet strength of the Muskat problem can be recovered from σ by a local formula in σ and $\mathcal{T}(z)\sigma$:

Proposition 2.10. *In the setting of Proposition 2.7, the vortex-sheet strength $\gamma = [\![u\cdot\tau]\!]$ is*

$$\gamma = -\frac{2\big(\tau_2 - A\,\mathcal{T}(z)\sigma\big)}{1-A^2}, \tag{2.8}$$

and it is the unique solution in $C^{0,\gamma}(\mathbb{T})$ of $(I+2A\mathcal{K}(z))\gamma = -2\tau_2$. In particular $\gamma \in C^{k-2,\gamma}(\mathbb{T})$.

Proof. By (2.6), $v^{\pm}\cdot\tau = -\mathcal{T}(z)\sigma \mp \tau_2$ (the tangential traces of $\mathcal{C}[z;\sigma n]$ coincide). Since $u^{\pm} = v^{\pm}/(1\pm A)$,

$$\gamma = \frac{-\mathcal{T}\sigma-\tau_2}{1+A} - \frac{-\mathcal{T}\sigma+\tau_2}{1-A} = \frac{2A\,\mathcal{T}\sigma - 2\tau_2}{1-A^2}.$$

The velocity is curl-free and divergence-free off Γ, has no normal jump, tends to 0 at infinity and has tangential jump γ, so $u = \mathcal{C}[z;\gamma\tau]$ by Lemma 2.3(e); Remark 2.9 then gives $(I+2A\mathcal{K}(z))\gamma = -2\tau_2$, and uniqueness and regularity follow from Proposition 2.6 applied with $-A$ in place of A. □

Identity (2.8) is the left relation of [LM26, (3.7b)], $(1-A^2)\varpi = [\![\theta]\!](AQ-e_2)\cdot\partial_\alpha z$, in our normalisation: $[\![\theta]\!] = -2$, $\varpi = -|\partial_\alpha z|\gamma$ (Remark 2.9), the Saffman–Taylor density of [LM26] is $|\partial_\alpha z|\sigma$, and their mean pressure-gradient trace is $Q = \langle\mathcal{C}[z;\sigma n]\rangle$, so that $Q\cdot\partial_\alpha z = |\partial_\alpha z|\,\mathcal{T}(z)\sigma$ and $e_2\cdot\partial_\alpha z = |\partial_\alpha z|\tau_2$. Luna-Velasco and Mengual [LM26, Sec. 3.1.3] recover ϖ from σ in exactly this way, and the equation $(I+2A\mathcal{K}(z))\gamma = -2\tau_2$ is their $(I+AP)\varpi = -[\![\theta]\!]\partial_\alpha z_2$; only the uniqueness and regularity statements in periodic Hölder spaces are added here.

Definition 2.11. Let $z_0 \in \mathcal{Z}^{k,\gamma}$, $k \geq 3$, and $\mathcal{K}_0 := \mathcal{K}(z_0)$, $\mathcal{T}_0 := \mathcal{T}(z_0)$. The Saffman–Taylor function, the normal velocity and the vortex-sheet strength of the Muskat data $\rho_0 = \mathbf{1}_{\Omega^+(z_0)} - \mathbf{1}_{\Omega^-(z_0)}$ are

$$\sigma_0 := -2\big(I-2A\mathcal{K}_0\big)^{-1}[n_{0,2}], \qquad U_0 := -\mathcal{K}_0\sigma_0, \qquad \gamma_0 := -2\big(I+2A\mathcal{K}_0\big)^{-1}[\tau_{0,2}].$$

The data are *fully unstable* if $\sigma_0 < 0$ on $\mathbb{T}$. The *pinch set* is $\Pi(z_0) := \{\alpha :\ \sigma_0(\alpha)\geq 0,\ \gamma_0(\alpha) = 0\}$.

The functions σ_0 and U_0 are the periodic counterparts of $\sigma^{(0)}$ and of the normal component of $z^{(1)}$ in [LM26, (5.12)–(5.13)], and full instability is the hypothesis of [LM26, Thm. 1.1]. For the relation of $\Pi(z_0)$ with the pinch singularity of [Men22] see Remark 3.3(b).

Example 2.12. For the flat interface $z_0(\alpha) = (\alpha, 0)$ the kernel k_{z_0} vanishes (the cotangent of a real number is real), so $\mathcal{K}_0 = 0$, $\sigma_0 \equiv -2$, $U_0 \equiv 0$ and $\gamma_0 \equiv 0$. If $A = 0$, then $\sigma_0 = -2n_{0,2}$ and $\gamma_0 = -2\tau_{0,2}$ for every z_0: the data are fully unstable if and only if z_0 is a graph, and $\Pi(z_0) = \{n_0 = -e_2\} = \{\tau_0 = -e_1\}$. These formulas, and the resulting characterisation of fully unstable data for $A = 0$, are those of [LM26, Secs. 1.2 and 3.1.3]: there $\sigma = [\![\theta]\!]$ for the flat interface, and $\sigma = [\![\theta]\!]\,\partial_\alpha z_1$, $\varpi = -[\![\theta]\!]\,\partial_\alpha z_2$ for $A = 0$, with $[\![\theta]\!] = -2$.

For $z_0 \in \mathcal{Z}^{6,\gamma}$, Propositions 2.6 (with $k=6$, $j=4$) and 2.10, and Lemma 2.5(a), give $\sigma_0, U_0, \gamma_0 \in C^{4,\gamma}(\mathbb{T})$. The quantities Φ and $\kappa_{\max}$ of (1.5) are then well defined, $\kappa_{\max}=0$ exactly on $\Pi(z_0)$, and since the expression under the square root can only vanish where $\sigma_0=\gamma_0=0$, we have $\kappa_{\max}\in C^{4,\gamma}(\mathbb{T})$ whenever $\Pi(z_0)=\emptyset$.

3. Relaxation of IPM with viscosity jump

3.1. **IPM as a differential inclusion.** A pair $(\rho,u)\in L^\infty(\mathbb{T}\times\mathbb{R}\times(0,T))^{1+2}$ with $\rho^2=1$ a.e. is a *weak solution* of (1.1) with initial datum ρ_0 if for every $\varphi\in C_c^\infty(\mathbb{T}\times\mathbb{R}\times[0,T))$

$$\begin{aligned}&\int_0^T\!\!\int \big(\rho\,\partial_t\varphi+\rho u\cdot\nabla\varphi\big)\,\mathrm{d}x\,\mathrm{d}t+\int\rho_0\varphi(\cdot,0)\,\mathrm{d}x=0, \qquad \int_0^T\!\!\int u\cdot\nabla\varphi\,\mathrm{d}x\,\mathrm{d}t=0,\\ &\int_0^T\!\!\int \big((1+A\rho)\,u\cdot\nabla^\perp\varphi+\rho\,\partial_1\varphi\big)\,\mathrm{d}x\,\mathrm{d}t=0,\end{aligned} \tag{3.1}$$

where $\nabla^\perp=(-\partial_2,\partial_1)$; the last identity is the curl of Darcy's law, $\operatorname{curl}(\mu(\rho)u)=-\partial_1\rho$. Consider the linear system for $(\rho,u,m)\in\mathbb{R}\times\mathbb{R}^2\times\mathbb{R}^2$

$$\partial_t\rho+\operatorname{div}m=0, \qquad \operatorname{div}u=0, \qquad \operatorname{curl}(u+Am)=-\partial_1\rho, \tag{3.2}$$

and the constitutive set

$$\mathsf{K}:=\{(\rho,u,m):\ \rho\in\{-1,1\},\ m=\rho u\}.$$

Since $(1+A\rho)u=u+Am$ on K, a pair (ρ,u) is a weak solution if and only if $(\rho,u,\rho u)$ solves (3.2) in the sense of distributions, with initial datum ρ_0, and takes values in K a.e. This linearisation of Darcy's law is due to [Men22].

For a state (ρ,u,m) we define the *Darcy field* $v:=u+Am$. In terms of (ρ,v,m) the system (3.2) becomes

$$\partial_t\rho+\operatorname{div}m=0, \qquad \operatorname{curl}v=-\partial_1\rho, \qquad \operatorname{div}v=-A\,\partial_t\rho, \qquad u=v-Am. \tag{3.3}$$

The point of (3.3) is that v *is determined by* ρ *and* $\partial_t\rho$ *alone* (modulo harmonic fields), whereas the vorticity of $u=v-Am$ depends on $\operatorname{curl}m$. This is the observation behind the pressure formulation of [LM26]: writing $v=-\nabla p-\rho e_2$, (3.3) gives $\Delta p=A\partial_t\rho-\partial_2\rho$ [LM26, Sec. 3.1.5], so the pressure is harmonic in every region where ρ is constant, for instance in each region of a strip subsolution [LM26, Secs. 3.3.2 and 3.3.5]. On K we have $v=(1+A\rho)u=\mu(\rho)u=-\nabla p-\rho e_2$.

3.2. **Laminates.** Following [Men22] (see also [LM26, Sec. 2.1]), the wave cone of (3.2) is the set Λ of states $\bar W=(\bar\rho,\bar u,\bar m)$ for which there exist $\xi\in\mathbb{R}^2\setminus\{0\}$ and $c\in\mathbb{R}$ such that $\bar W h(\xi\cdot x+ct)$ solves (3.2) for every $h\in C^1(\mathbb{R})$, that is,

$$c\bar\rho+\xi\cdot\bar m=0, \qquad \xi\cdot\bar u=0, \qquad \xi^\perp\cdot(\bar u+A\bar m)+\xi_1\bar\rho=0. \tag{3.4}$$

For a state (ρ,u,m) with $|\rho|<1$ we introduce

$$\mu:=1+A\rho, \quad v:=u+Am, \quad d:=\frac{2\big[(1-A^2)m-(\rho-A)v\big]}{1-\rho^2}, \quad D:=d+2e_2, \quad a:=e_2+\frac{A}{\mu}v. \tag{3.5}$$

For fixed ρ the map $(u,m)\mapsto(v,d)$ is a linear isomorphism of $\mathbb{R}^4$, with inverse

$$m=\frac{(\rho-A)v+\frac12(1-\rho^2)d}{1-A^2}, \qquad u=v-Am.$$

Lemma 3.1 and Proposition 3.2 re-derive, in this chart at fixed Darcy field v, the inclusion of the relaxed set O of Mengual [Men22] in the lamination convex hull of K [Men22] (Székelyhidi [Sze12] for $A=0$); the identification of $\mathcal{U}_A$ with O is given after Proposition 3.2.

Lemma 3.1 (First order laminates)**.** *Let $\rho\in(-1,1)$, $\lambda_\pm=\frac{1\pm\rho}{2}$, and $P_+=(1,u_+,u_+)$, $P_-=(-1,u_-,-u_-)\in\mathsf{K}$. Then $P_+-P_-\in\Lambda$ if and only if there is a unit vector ν with*

$$u_+\cdot\nu=u_-\cdot\nu, \qquad \big((1+A)u_+-(1-A)u_-\big)\cdot\nu^\perp=-2\nu_1. \tag{3.6}$$

In that case the barycentre $P = \lambda_+ P_+ + \lambda_- P_- = (\rho, u, m)$ *satisfies*

$$D = 2(a\cdot\nu)\nu, \qquad \text{in particular} \qquad |D-a| = |a|. \tag{3.7}$$

Conversely, for every $\rho \in (-1,1)$, $v \in \mathbb{R}^2$ *and unit vector* ν, *the state with these values of* ρ, v *and with* $D = 2(a\cdot\nu)\nu$ *is such a barycentre.*

Proof. We have $P_+ - P_- = (2, u_+ - u_-, u_+ + u_-)$. Taking $\xi = \nu$ in (3.4), the first condition only fixes c, the second is $\nu\cdot(u_+ - u_-) = 0$ and the third is $\nu^\perp\cdot(u_+ - u_- + A(u_+ + u_-)) + 2\nu_1 = 0$, which is (3.6). Let $v_\pm := (1\pm A)u_\pm$ be the Darcy fields of the two phases and $\delta := v_+ - v_-$. Then $v = u + Am = \lambda_+ v_+ + \lambda_- v_-$, so $v_+ = v + \lambda_-\delta$, $v_- = v - \lambda_+\delta$, and

$$(1-A^2)m = (1-A)\lambda_+ v_+ - (1+A)\lambda_- v_- = (\rho - A)v + 2\lambda_+\lambda_-\delta = (\rho - A)v + \tfrac{1}{2}(1-\rho^2)\delta,$$

that is, $d = \delta$. The first condition in (3.6) reads $(1-A)v_+\cdot\nu = (1+A)v_-\cdot\nu$, i.e. $\delta\cdot\nu\,[(1-A)\lambda_- + (1+A)\lambda_+] = 2A\,v\cdot\nu$; since $(1-A)\lambda_- + (1+A)\lambda_+ = \mu$, this is $\delta\cdot\nu = \frac{2A}{\mu}v\cdot\nu$. The second condition reads $\delta\cdot\nu^\perp = -2\nu_1 = -2e_2\cdot\nu^\perp$. Hence

$$\delta = \frac{2A}{\mu}(v\cdot\nu)\nu - 2(e_2\cdot\nu^\perp)\nu^\perp = -2e_2 + 2\Big(\big(e_2 + \tfrac{A}{\mu}v\big)\cdot\nu\Big)\nu,$$

where we used $\nu^\perp\otimes\nu^\perp = I - \nu\otimes\nu$. This is (3.7), and $|2(a\cdot\nu)\nu - a|^2 = |a|^2$. The computation can be reversed: given (ρ, v, ν) put $\delta = -2e_2 + 2(a\cdot\nu)\nu$, $v_+ = v + \lambda_-\delta$, $v_- = v - \lambda_+\delta$ and $u_\pm = v_\pm/(1\pm A)$. □

Proposition 3.2 (The relaxed set)**.** *The set*

$$\mathcal{U}_A := \big\{(\rho,u,m)\in\mathbb{R}\times\mathbb{R}^2\times\mathbb{R}^2 :\ |\rho|<1,\ |D-a|<|a|\big\} = \big\{|\rho|<1,\ D\cdot(D-2a)<0\big\},$$

with D, a *given by* (3.5)*, is open, and every point of* $\mathcal{U}_A$ *is the barycentre of a* Λ*-segment whose endpoints are first order laminates of* K *as in Lemma 3.1. In particular* $\mathcal{U}_A$ *is contained in the lamination convex hull* K^{lc}*, hence in the interior of the* Λ*-convex hull* K^Λ*.*

Proof. Openness is clear. For $\bar m \neq 0$ the state $\bar W = (0, -A\bar m, \bar m)$ belongs to Λ: take $\xi = \bar m^\perp$ and any c in (3.4). Moving along $\bar W$ leaves ρ and $v = u + Am$ (hence a) unchanged and moves d, hence D, in the direction $\bar m$. Let $(\rho, u, m)\in\mathcal{U}_A$; then D lies in the open disc $\{|D'-a|<|a|\}$, and any line through D meets the circle $\{|D'-a| = |a|\}$ at two points D_1, D_2 with $D\in(D_1, D_2)$. The circle is exactly $\{2(a\cdot\nu)\nu :\ |\nu| = 1\}$ (the circle through the origin with centre a), so by the converse part of Lemma 3.1 the states with values (ρ, v, D_j) are first order laminates. The equivalence of the two descriptions of $\mathcal{U}_A$ follows from $|D-a|^2 - |a|^2 = D\cdot(D-2a)$. □

Identification with the relaxed set of [Men22]. The set $\mathcal{U}_A$ is the open set O of [Men22], written (O) in [LM26], which consists of the states with

$$\big|2(1-A\rho)(m-\rho u) + (1-\rho^2)(Au+e_2)\big| < (1-\rho^2)\,|Au+e_2|;$$

this inequality forces $|\rho|<1$. Let $g := (1-A\rho)|m-\rho u|^2 + (1-\rho^2)(m-\rho u)\cdot(Au+e_2)$ be the function of [LM26, (A.1)], which characterises $\overline{O}$ as $\{|\rho|\le 1,\ g\le 0\}$ ([Men22, Lemmas 4.1–4.2]; [Sze12, Prop. 2.3] for $A = 0$). For $|\rho|<1$, a direct computation from (3.5) gives

$$\begin{aligned}(1-\rho^2)\big(D-(Au+e_2)\big) &= 2(1-A\rho)(m-\rho u) + (1-\rho^2)(Au+e_2),\\ D\cdot\big(D-2(Au+e_2)\big) = \frac{4(1-A\rho)\,g}{(1-\rho^2)^2}, &\qquad D\cdot(D-2a) = \frac{4(1-A^2)\,g}{(1-\rho^2)^2(1+A\rho)}.\end{aligned} \tag{3.8}$$

Hence (O) reads $|D-(Au+e_2)|<|Au+e_2|$, and $O = \{|\rho|<1,\ g<0\} = \mathcal{U}_A$. Thus our disc $|D-a|<|a|$ is the disc $|D-(Au+e_2)|<|Au+e_2|$ of [Men22, LM26], re-centred by slicing at fixed Darcy field v instead of fixed velocity u; the two centres differ by $a-(Au+e_2) = \frac{A^2}{\mu}(m-\rho u)$. Moreover $\mathsf{K}^{lc} = \overline{O}$ by [Men22], and $\overline{O}$ is also the Λ-convex hull K^Λ and the weak relaxation in the sense of [LM26, Sec. 2.1] by [LM26, Thm. 2.4]; for $A = 0$ all these identities are due to [Sze12].

Remark 3.3. (a) For $A = 0$ we have $v = u$, $a = e_2$ and $D - a = \frac{2}{1-\rho^2}\big(m - \rho u + \frac{1}{2}(1-\rho^2)e_2\big)$, so $\mathcal{U}_0$ is the interior of the hull $\{|m - \rho u + \frac{1}{2}(1-\rho^2)e_2| \le \frac{1}{2}(1-\rho^2)\}$ computed by Székelyhidi [Sze12].

(b) The disc degenerates (it is empty) exactly when $a = 0$, i.e. $v = -\frac{\mu}{A}e_2$; for $A = 0$ we have $a = e_2$, and it never degenerates. For a state in a thin strip around an interface z_0, at leading order $a \cdot n_0 = -\frac{1}{2}\sigma_0 - \frac{A^2(1-\rho_{\mathrm{m}}^2)\kappa}{\mu_{\mathrm{m}}}$ and $a \cdot \tau_0 = -\frac{(1-A^2)\gamma_0}{2\mu_{\mathrm{m}}}$ (Lemma 5.2), so as $\kappa \to 0$ the disc degenerates only where $\sigma_0 = \gamma_0 = 0$. On a Muskat interface, $\sigma = \gamma = 0$ holds exactly where $A\langle u\rangle + e_2 = 0$ [LM26, (3.7a)], i.e. where the right-hand side of (O) vanishes for $u = \langle u\rangle$: this is the trace on the interface of the pinch singularity of [Men22]. In the chart of [Men22, LM26], where the disc is $|D - (Au + e_2)| < |Au + e_2|$, the corresponding non-degeneracy is a positive lower bound for $|Au + e_2|$, which [LM26] single out as the additional ingredient required by the viscosity jump and obtain from its normal component [LM26, Sec. 3.3.4 and Prop. 7.4]. On the rest of $\Pi(z_0)$, where $\sigma_0 > 0$ and $\gamma_0 = 0$, the disc does not degenerate (at leading order its radius is at least $\frac{1}{2}\sigma_0$), but the normal component of D forced by the Rankine–Hugoniot conditions, $D \cdot n_0 = -\sigma_0 - 2(1 - A\rho_{\mathrm{m}})\kappa$ at leading order, lies outside it for every $\kappa > 0$, because $a \cdot \tau_0 = 0$ and $D \cdot n_0 - 2a \cdot n_0 = -\frac{2(1-A^2)\kappa}{\mu_{\mathrm{m}}} < 0$ there (proof of Lemma 5.2). Theorem 1.2 makes this precise within single strips: a single-strip subsolution cannot open at linear order in t on $\Pi(z_0)$. Strips opening at rate $o(t)$, localised mixing zones and other classes of subsolutions are not excluded.

(c) By (3.8) and [LM26, Thm. 2.4], $\overline{\mathcal{U}_A} = \overline{O} = \mathsf{K}^\Lambda$; since the points of the boundary circles $|D - a| = |a|$ and the states with $|\rho| = 1$ are not interior points of $\overline{O}$, also $\mathcal{U}_A = O = \operatorname{int}\mathsf{K}^\Lambda$. We only use the inclusion $\mathcal{U}_A \subset \operatorname{int}\mathsf{K}^\Lambda$ of Proposition 3.2.

3.3. **Subsolutions and the h-principle.**

Definition 3.4. Let $T > 0$ and $\rho_0 \in L^\infty(\mathbb{T}\times\mathbb{R})$ with $\rho_0^2 = 1$. A triple $(\bar\rho, \bar u, \bar m) \in L^\infty(\mathbb{T}\times\mathbb{R}\times(0,T))^{1+2+2}$ is a *subsolution* with initial datum ρ_0 and mixing zone Ω_{mix}, an open subset of $\mathbb{T}\times\mathbb{R}\times(0,T)$, if

(i) (3.2) holds in the sense of distributions on $\mathbb{T}\times\mathbb{R}\times[0,T)$ with initial datum ρ_0, i.e. for every $\varphi \in C_c^\infty(\mathbb{T}\times\mathbb{R}\times[0,T))$

$$\iint \big(\bar\rho\partial_t\varphi + \bar m\cdot\nabla\varphi\big) + \int \rho_0\varphi(\cdot,0) = 0, \quad \iint \bar u\cdot\nabla\varphi = 0, \quad \iint \big((\bar u + A\bar m)\cdot\nabla^\perp\varphi + \bar\rho\,\partial_1\varphi\big) = 0;$$

(ii) $(\bar\rho, \bar u, \bar m)$ is continuous on Ω_{mix} and takes values in $\mathcal{U}_A$ there;

(iii) $(\bar\rho, \bar u, \bar m) \in \mathsf{K}$ a.e. in the complement of Ω_{mix}.

The following h-principle is [LM26, Thm. 2.3]; it was proved by Székelyhidi [Sze12] for $A = 0$ (see also [CFG11, CCF21]) and by Mengual [Men22] for $|A| < 1$. In [LM26, Thm. 2.3] it is stated on $(0,T]\times\mathbb{R}^2$, for subsolutions that are continuous with values in the set O of [Men22] in the mixing zone and that are pure phases ($\bar\rho = \pm 1$, $\bar m = \bar\rho\bar u$) in two open sets which, together with the mixing zone, cover every time slice up to a null set. Since $O = \mathcal{U}_A$ by (3.8), the first requirement is condition (ii) of Definition 3.4; in our applications the pure phases are the open regions above and below a strip. We apply the theorem to the 2π-periodic lift to $\mathbb{R}^2$ of a subsolution in the sense of Definition 3.4. The resulting solutions are defined on $\mathbb{R}^2$ and need not be 2π-periodic in x_1; we do not discuss whether the convex integration scheme can be run on $\mathbb{T}\times\mathbb{R}$. We refer to [Men22] and [LM26, Sec. 2] for the precise functional framework, which also requires weak-$*$ continuity in time [LM26, Def. 2.1].

Theorem 3.5 (h-principle [Sze12, Men22], [LM26, Thm. 2.3])**.** *Let $(\bar\rho, \bar u, \bar m)$ be a subsolution with initial datum ρ_0 and mixing zone Ω_{mix}, and denote by the same symbols the 2π-periodic lifts of $(\bar\rho, \bar u, \bar m)$ and Ω_{mix} to $\mathbb{R}^2\times(0,T)$ and of ρ_0 to $\mathbb{R}^2$. Then there exist infinitely many weak solutions (ρ, u) of (1.1) in $\mathbb{R}^2\times(0,T)$, i.e. satisfying (3.1) with $\mathbb{T}\times\mathbb{R}$ replaced by $\mathbb{R}^2$, with initial datum ρ_0, such that $(\rho, u) = (\bar\rho, \bar u)$ a.e. outside Ω_{mix} and ρ mixes in Ω_{mix}, i.e. both $\{\rho = 1\}$ and $\{\rho = -1\}$ have positive measure in every open subset of Ω_{mix}. Moreover there is a sequence of such solutions with $(\rho_k, u_k) \overset{*}{\rightharpoonup} (\bar\rho, \bar u)$ in L^∞.*

The mixing property in [LM26, Thm. 2.3], a theorem that [LM26] recall from [Sze12, Men22], is stronger: it holds in every time slice [LM26, Def. 2.2] (see [CCF21, CFM19]); we only state and use the space–time version above. The last assertion of Theorem 3.5 is not part of [LM26, Thm. 2.3]; it is the approximation property of the h-principle of [Men22, Thm. 2.1], which is also used in [LM26, App. A.1.2].

3.4. **The flat interface.** The following explicit subsolutions illustrate the role of the viscosity jump. They are not new. The flat interface is treated in [Men22], whose macroscopic profile [Men22, Sec. 2] has the flux $\frac{1-\rho^2}{1-A\rho}$ (denoted F_A in [LM26, Sec. 7.5]; it is F_1 below). The ansatz $\bar u = 0$, $\bar m = -F_\theta(\bar\rho)e_2$ below and the reduction of the hull condition to $0<\theta<1$ are those of the flat single strip of [LM26, Sec. 3.2], whose constant γ is our $\theta-\frac12$ and whose intermediate density is a constant. Finally, for the flat interface the multilayer subsolutions of [LM26, Sec. 7.5] with $\bar c = \theta$ have the mixing zones of Example 3.6, and Luna-Velasco and Mengual identify the limit of their profiles, as the number of layers tends to infinity, as the viscosity-dependent Otto profile with time scaled by $\bar c$ [LM26, Sec. 7.5], i.e. the profile of Example 3.6.

Example 3.6. Let $\theta\in(0,1)$ and $F_\theta(\rho) := \theta\frac{1-\rho^2}{1-A\rho}$ for $\rho\in[-1,1]$. Then $F_\theta(\pm1)=0$,

$$F_\theta'(\rho) = \theta\,\frac{A-2\rho+A\rho^2}{(1-A\rho)^2}, \qquad F_\theta''(\rho) = -\frac{2\theta(1-A^2)}{(1-A\rho)^3} < 0,$$

so $-F_\theta'$ increases from $-\frac{2\theta}{1+A}$ at $\rho=-1$ to $\frac{2\theta}{1-A}$ at $\rho=1$. Let $R_\theta:\mathbb{R}\to[-1,1]$ be equal to -1 on $(-\infty,-\frac{2\theta}{1+A}]$, to 1 on $[\frac{2\theta}{1-A},\infty)$, and to the inverse of $-F_\theta'$ in between, and set

$$\bar\rho(x,t) = R_\theta(x_2/t), \qquad \bar u = 0, \qquad \bar m = -F_\theta(\bar\rho)\,e_2.$$

Then $\partial_t\bar\rho+\partial_2\bar m_2 = -\frac{R_\theta'}{t}\big(\frac{x_2}{t}+F_\theta'(R_\theta)\big)=0$ (a rarefaction wave), $\operatorname{div}\bar u=0$, and $\operatorname{curl}(\bar u+A\bar m) = -A\partial_1F_\theta(\bar\rho)=0=-\partial_1\bar\rho$; all fields are continuous and R_θ is Lipschitz (since $-F_\theta''\ge\frac{2\theta(1-A^2)}{(1+|A|)^3}$), and $\bar\rho(\cdot,t)\to\operatorname{sign}(x_2)$ in L^1_{loc} as $t\to0$, so (3.2) holds with initial datum $\operatorname{sign}(x_2)$. Outside the fan $\{-\frac{2\theta t}{1+A}<x_2<\frac{2\theta t}{1-A}\}$ the state is $(\pm1,0,0)\in\mathsf K$. Inside, $v=A\bar m$, $(1-A^2)\bar m-(\bar\rho-A)v = (1-A\bar\rho)\bar m$, hence

$$D = 2(1-\theta)e_2, \qquad a = \Big(1-\frac{A^2\theta(1-\bar\rho^2)}{1-A^2\bar\rho^2}\Big)e_2, \qquad D\cdot(D-2a) = -\frac{4\theta(1-\theta)(1-A^2)}{1-A^2\bar\rho^2}<0.$$

So $(\bar\rho,\bar u,\bar m)$ is a subsolution, and Theorem 3.5 gives mixing solutions whose mixing zone is $\{-\frac{2\theta t}{1+A}<x_2<\frac{2\theta t}{1-A}\}$. The same formulas with $\theta=1$ define the extremal profile, for which $D=0$ lies on the boundary circle $|D-a|=|a|$ of the hull; it is not a subsolution in the sense of Definition 3.4, and the family is obtained from it by scaling time: $\bar\rho_\theta(x,t)=\bar\rho_1(x,\theta t)$ and $\bar m_\theta(x,t)=\theta\,\bar m_1(x,\theta t)$. For $A>0$ (the heavy fluid on top is the more viscous one) the mixing zone grows faster upwards, in line with the Saffman–Taylor picture [ST58] of the less viscous fluid penetrating the more viscous one; this asymmetry, with the factors $1/(1\mp A)$, is already present in the profile of [Men22] and in [LM26, Secs. 1.4 and 3.2]. Moreover $\int(\bar\rho-\operatorname{sign}x_2)\,\mathrm{d}x_2 = t\int_{-1}^1F_\theta'(r)\,\mathrm{d}r=0$, i.e. $x_2=0$ remains the equal-mass level of the mixing zone (the counterpart of the centre of mass (4.3) of a strip); this is conservation of mass, since $F_\theta(\pm1)=0$, and its discrete counterpart for multilayers is the identity $\sum_j(\theta_j-\theta_{j-1})h_j=0$ of [LM26, Sec. 7.5].

4. Piecewise constant subsolutions

In this section $z_0\in\mathcal Z^{6,\gamma}$, $\rho_{\mathrm m}\in(-1,1)$ and $\kappa\in C^{4,\gamma}(\mathbb T)$ with $\kappa>0$ are fixed, $\sigma_0,U_0\in C^{4,\gamma}(\mathbb T)$ are as in Definition 2.11, and

$$\mu_{\mathrm m} := 1+A\rho_{\mathrm m}, \qquad c^\pm := (1\pm\rho_{\mathrm m})\kappa.$$

Moreover $a\in C^{3,\gamma}(\mathbb T)$ and a polynomial h with $h(0)=h'(0)=0$ are given; they will be chosen in Section 5.

4.1. **Geometry.** We define the central curve and the two boundary curves of the mixing zone,

(4.1)
$$Z(\alpha,t) := z_0(\alpha) + \Big(tU_0(\alpha) + \tfrac{t^2}{2}a(\alpha)\Big)n_0(\alpha) + h(t)e_2, \qquad z^{\pm}(\alpha,t) := Z(\alpha,t) \pm t\, c^{\pm}(\alpha)\, n_0(\alpha),$$

and the map

$$X(\alpha,\lambda,t) := \tfrac{1}{2}(1-\lambda)\, z^-(\alpha,t) + \tfrac{1}{2}(1+\lambda)\, z^+(\alpha,t), \qquad \partial_\lambda X = \tfrac{1}{2}(z^+ - z^-) = t\kappa\, n_0. \tag{4.2}$$

Lemma 4.1. *There is $T_1 > 0$ such that for $t \in (0,T_1]$: $z^{\pm}(\cdot,t) \in \mathcal{Z}^{3,\gamma}$ with $\delta(z^{\pm}(\cdot,t)) \geq \frac{1}{2}\delta(z_0)$; $\Gamma^+(t) \subset \Omega^+(z^-(\cdot,t))$; and $X(\cdot,\cdot,t)$ is a $C^{3,\gamma}$ diffeomorphism from $\mathbb{T}\times[-1,1]$ onto the closure of the strip*

$$S_t := \Omega^-(z^+(\cdot,t)) \cap \Omega^+(z^-(\cdot,t)),$$

mapping $\lambda = \pm 1$ onto $\Gamma^{\pm}(t)$. Moreover $z^{\pm}$ are polynomials in t with coefficients in $C^{3,\gamma}$, and $z^{\pm}(\cdot,t) \to z_0$ in $C^{3,\gamma}$ as $t \to 0$.

Proof. All statements follow from $z^{\pm}(\cdot,t) - z_0 = O(t)$ in $C^{3,\gamma}$, the continuity of δ with respect to the C^1 norm, $z^+ - z^- = 2t\kappa n_0$ with $\kappa > 0$, and the fact that $\partial_\alpha X = \partial_\alpha z_0 + O(t)$ and $\partial_\lambda X = t\kappa n_0$ are linearly independent; injectivity of X for small t is the usual tubular neighbourhood argument, as in [CFM22, Lemma B.1] and [LM26, Lemma 4.1]. □

We denote by $n^{\pm}$ the upward unit normals of $\Gamma^{\pm}(t)$ and by

$$V^{\pm} := \partial_t z^{\pm} \cdot n^{\pm}$$

their normal velocities. Note that the central curve Z is the *centre of mass* of the strip: since $(1-\rho_{\mathrm{m}})c^+ = (1+\rho_{\mathrm{m}})c^-$,

$$Z = \tfrac{1}{2}\big[(1-\rho_{\mathrm{m}})z^+ + (1+\rho_{\mathrm{m}})z^-\big]. \tag{4.3}$$

The relation $(1-\rho_{\mathrm{m}})c^+ = (1+\rho_{\mathrm{m}})c^-$ is the mass balance $(1-\theta_{\mathrm{mix}})h_+ = (1+\theta_{\mathrm{mix}})h_-$ of the flat strip of [LM26, Sec. 3.2]; for $\rho_{\mathrm{m}} = A$ it is built into the opening speeds $c/(1\mp A)$ of the boundaries in [LM26, (4.2)], and [LM26, Sec. 7.5] contains its multilayer analogue.

4.2. **The subsolution.** Let $[\![\rho]\!]^+ := 1-\rho_{\mathrm{m}}$ and $[\![\rho]\!]^- := 1+\rho_{\mathrm{m}}$ be the jumps of the density across $\Gamma^{\pm}$, and define

$$f^{\pm} := [\![\rho]\!]^{\pm}\Big(-e_2 + \big(n_2^{\pm} + AV^{\pm}\big)n^{\pm}\Big), \qquad v := \mathcal{C}[z^+;f^+] + \mathcal{C}[z^-;f^-] \quad \text{on } \mathbb{R}^2 \setminus (\Gamma^+\cup\Gamma^-). \tag{4.4}$$

Let $v_S^{\pm}(\alpha,t)$ be the limits of v at $z^{\pm}(\alpha,t)$ from inside the strip:

$$v_S^+ = \mathcal{C}[z^+;f^+]^- + \mathcal{C}[z^-;f^-]\circ z^+, \qquad v_S^- = \mathcal{C}[z^-;f^-]^+ + \mathcal{C}[z^+;f^+]\circ z^-. \tag{4.5}$$

We define the *boundary fluxes* and the *flux mismatch*

$$F^+ := |\partial_\alpha z^+|\Big\{\tfrac{2}{1+\rho_{\mathrm{m}}}\big(v_S^+\cdot n^+ - (1-A)V^+\big) + 2n_2^+\Big\}, \tag{4.6}$$

$$F^- := |\partial_\alpha z^-|\Big\{\tfrac{2}{1-\rho_{\mathrm{m}}}\big((1+A)V^- - v_S^-\cdot n^-\big) + 2n_2^-\Big\}, \tag{4.7}$$

$$M := F^+ - F^-, \tag{4.8}$$

the functions

$$\Psi^-(\alpha,t) := \int_0^\alpha F^-(\beta,t)\,\mathrm{d}\beta, \qquad \Psi^+(\alpha,t) := \int_0^\alpha F^+(\beta,t)\,\mathrm{d}\beta + C_\Psi(t), \tag{4.9}$$

where C_Ψ is a continuous function of time to be chosen (it fixes the relative normalisation of the two boundary values and will produce the tilt in Section 5; $C_\Psi \equiv 0$ is allowed), and the stream function Ψ on $\overline{S_t}$ by

$$\Psi\big(X(\alpha,\lambda,t),t\big) := \tfrac{1}{2}(1-\lambda)\Psi^-(\alpha,t) + \tfrac{1}{2}(1+\lambda)\Psi^+(\alpha,t). \tag{4.10}$$

Finally we set $\bar\rho(\cdot,0)=\rho_0$ and, for $0<t\le T_1$,

(4.11)
$$(\bar\rho,\bar u,\bar m):=\begin{cases}\Big(1,\ \dfrac{v}{1+A},\ \dfrac{v}{1+A}\Big) & \text{in } \Omega^+(z^+),\\[2mm] \Big(\rho_{\mathrm m},\ v-A\bar m,\ \dfrac{(\rho_{\mathrm m}-A)v+\frac12(1-\rho_{\mathrm m}^2)(D-2e_2)}{1-A^2}\Big), \quad D:=\nabla^\perp\Psi, & \text{in } S_t,\\[2mm] \Big(-1,\ \dfrac{v}{1-A},\ -\dfrac{v}{1-A}\Big) & \text{in } \Omega^-(z^-).\end{cases}$$

By construction, in S_t the Darcy field of the state is $\bar u+A\bar m=v$, and the quantities (3.5) are $d=D-2e_2$, $D=\nabla^\perp\Psi$ and $a=e_2+\frac{A}{\mu_{\mathrm m}}v$. This is the single-layer ansatz of [LM26, Secs. 4.2–4.6] in our variables: $v=-\nabla p-\bar\rho e_2$, with ∇p a sum of Cauchy integrals over the two boundaries as in [LM26, (4.8)]; the densities $f^\pm$ are fixed by the boundary motion through the Saffman–Taylor densities of $\Gamma^\pm$, as in [LM26, (4.7)] (and [LM26, (7.16)] for general density jumps); $F^\pm$ correspond to the Rankine–Hugoniot boundary data [LM26, (4.12)]; and the stream function Ψ, interpolated linearly across the strip between primitives of these data, plays the role of the potential g of [LM26, (4.13)] (for $\rho_{\mathrm m}=A$, $D=(Au+e_2)-2\nabla^\perp g$, Remark 5.8).

Lemma 4.2 (The linear equations)**.** *Assume that*

$$\int_{\mathbb T}M(\alpha,t)\,\mathrm d\alpha=0\qquad\text{for all } t\in(0,T_1]. \tag{4.12}$$

Then D is well defined and 2π-periodic in x_1, and $(\bar\rho,\bar u,\bar m)(\cdot,t)$ is bounded for each $t\in(0,T_1]$. If moreover $(\bar\rho,\bar u,\bar m)$ is bounded on $\mathbb T\times\mathbb R\times(0,T_1]$, uniformly in t (this holds under the hypotheses of Lemma 4.5; see the proof of Theorem 1.1), then it solves (3.2) in the sense of distributions on $\mathbb T\times\mathbb R\times[0,T_1)$ with initial datum ρ_0. Outside the closure of $\bigcup_t S_t\times\{t\}$ it takes values in $\mathsf K$.

Proof. Periodicity. $\Psi^\pm(\alpha+2\pi,t)=\Psi^\pm(\alpha,t)+\Phi^\pm(t)$ with $\Phi^\pm=\int_{\mathbb T}F^\pm$, and $\Phi^+=\Phi^-$ by (4.12). Since $X(\alpha+2\pi,\lambda,t)=X(\alpha,\lambda,t)+2\pi e_1$, we get $\Psi(x+2\pi e_1)=\Psi(x)+\Phi^+$, so $D=\nabla^\perp\Psi$ is periodic. Ψ is C^1 up to ∂S_t, hence $D(\cdot,t)$ is bounded for each t. A bound uniform in t needs more: since the strip has width $O(t)$, it requires $\int_0^\alpha M+C_\Psi=O(t)$, as in the hypotheses of Lemma 4.5.

The equations in the open regions. In each of the three open regions $\bar\rho$ is constant; v is divergence-free and curl-free there by Lemma 2.3(a), and $\operatorname{div}D=\operatorname{div}\nabla^\perp\Psi=0$. Hence $\operatorname{div}\bar m=0$, $\operatorname{div}\bar u=\operatorname{div}v-A\operatorname{div}\bar m=0$ and $\operatorname{curl}(\bar u+A\bar m)=\operatorname{curl}v=0$ in each region. Outside the strip, $\bar m=\bar\rho\bar u$ with $\bar\rho=\pm1$, i.e. the state is in $\mathsf K$.

Jump conditions. A piecewise C^1 bounded field satisfies (3.2) in $\mathcal D'$ if and only if it does so in each open region and satisfies the Rankine–Hugoniot conditions across the moving curves:

$$[\![\bar\rho]\!]V=[\![\bar m\cdot n]\!],\qquad [\![\bar u\cdot n]\!]=0,\qquad [\![(\bar u+A\bar m)\cdot\tau]\!]=-[\![\bar\rho]\!]\tau_2.$$

(The initial condition holds because the fields are bounded uniformly in t and $\bar\rho(\cdot,t)\to\rho_0$ in L^1_{loc}.) Consider Γ^+. The jump of v across Γ^+ is f^+ (the field $\mathcal C[z^-;f^-]$ is continuous there), and $f^+\cdot\tau^+=-[\![\rho]\!]^+\tau_2^+$, $f^+\cdot n^+=A[\![\rho]\!]^+V^+$. Since $\bar u+A\bar m=v$ on both sides, the third condition holds. Next, the flux of $\nabla^\perp\Psi$ through Γ^+ per unit α is $\nabla\Psi\cdot\partial_\alpha z^+=\partial_\alpha(\Psi\circ z^+)=\partial_\alpha\Psi^+=F^+$, so on the strip side

$$D\cdot n^+=\frac{F^+}{|\partial_\alpha z^+|},\qquad (D-2e_2)\cdot n^+=\frac{2}{1+\rho_{\mathrm m}}\big(v_S^+\cdot n^+-(1-A)V^+\big),$$

and therefore

$$(1-A^2)\,\bar m^S\cdot n^+=(\rho_{\mathrm m}-A)v_S^+\cdot n^++(1-\rho_{\mathrm m})\big(v_S^+\cdot n^+-(1-A)V^+\big)=(1-A)\big(v_S^+\cdot n^+-(1-\rho_{\mathrm m})V^+\big).$$

Above Γ^+, $\bar m\cdot n^+=v^{\text{above}}\cdot n^+/(1+A)=(v_S^+\cdot n^++A(1-\rho_{\mathrm m})V^+)/(1+A)$. Subtracting, $[\![\bar m\cdot n^+]\!]=(1-\rho_{\mathrm m})V^+=[\![\bar\rho]\!]V^+$. Moreover $\bar u^S\cdot n^+=v_S^+\cdot n^+-A\bar m^S\cdot n^+=(v_S^+\cdot n^++A(1-\rho_{\mathrm m})V^+)/(1+A)=\bar u^{\text{above}}\cdot n^+$. The computation on Γ^- is identical: there $(D-2e_2)\cdot n^-=\frac{2}{1-\rho_{\mathrm m}}((1+A)V^--v_S^-\cdot n^-)$, which gives $(1-A^2)\bar m^S\cdot n^-=(1+A)\big((1+\rho_{\mathrm m})V^--v_S^-\cdot n^-\big)$,

while below Γ^-, $\bar m\cdot n^- = -(v_S^-\cdot n^- - A(1+\rho_{\mathrm m})V^-)/(1-A)$; hence $[\![\bar m\cdot n^-]\!] = (1+\rho_{\mathrm m})V^-$ and $[\![\bar u\cdot n^-]\!]=0$. □

Thus the ansatz (4.11) solves the linear system exactly, whatever the curves $z^\pm$ are, provided (4.12) holds and the fields are bounded uniformly in t; the only remaining requirement for a subsolution is the open condition $D\cdot(D-2a)<0$ in S_t. The next lemma identifies (4.12) with the conservation of mass.

Lemma 4.3 (Mass identity)**.** *Let* $\mathcal{A}^\pm(t) := \int_{\mathbb T} z_2^\pm\,\partial_\alpha z_1^\pm\,\mathrm d\alpha$, *so that* $\frac{\mathrm d}{\mathrm dt}\mathcal{A}^\pm = \int_{\mathbb T} V^\pm|\partial_\alpha z^\pm|\,\mathrm d\alpha$, *and*

$$Q(t) := (1-\rho_{\mathrm m})\tfrac{\mathrm d}{\mathrm dt}\mathcal{A}^+(t) + (1+\rho_{\mathrm m})\tfrac{\mathrm d}{\mathrm dt}\mathcal{A}^-(t).$$

Then for $t\in(0,T_1]$

$$\int_{\mathbb T} M(\alpha,t)\,\mathrm d\alpha = -\frac{2\,Q(t)}{1-\rho_{\mathrm m}^2}.$$

Proof. The identity for $\frac{\mathrm d}{\mathrm dt}\mathcal{A}^\pm$ follows from an integration by parts: $\frac{\mathrm d}{\mathrm dt}\mathcal{A} = \int(\partial_t z_2\partial_\alpha z_1 - \partial_\alpha z_2\partial_t z_1) = \int \partial_t z\cdot(\partial_\alpha z)^\perp$. By Lemma 2.3(c), the constant parts $-[\![\rho]\!]^\pm e_2$ of the densities contribute $\mp\frac12\sum_\pm[\![\rho]\!]^\pm e_2 = \mp e_2$ to $v_{\pm\infty}$, while by (2.2) the normal parts contribute $\pm\frac{1}{4\pi}\sum_\pm[\![\rho]\!]^\pm(2\pi + A\frac{\mathrm d}{\mathrm dt}\mathcal{A}^\pm)e_2$. Hence

$$v_{\pm\infty} = \pm\frac{AQ}{4\pi}e_2.$$

Since v is divergence-free and periodic above Γ^+, the flux of v through Γ^+ from above equals $2\pi(v_{+\infty})_2 = \frac{AQ}{2}$; subtracting the jump $f^+\cdot n^+ = A(1-\rho_{\mathrm m})V^+$,

$$\int_{\mathbb T} v_S^+\cdot n^+|\partial_\alpha z^+|\,\mathrm d\alpha = \frac{AQ}{2} - A(1-\rho_{\mathrm m})\tfrac{\mathrm d}{\mathrm dt}\mathcal{A}^+, \qquad \int_{\mathbb T} v_S^-\cdot n^-|\partial_\alpha z^-|\,\mathrm d\alpha = -\frac{AQ}{2} + A(1+\rho_{\mathrm m})\tfrac{\mathrm d}{\mathrm dt}\mathcal{A}^-,$$

the second identity being obtained in the same way from below. Using $\int n_2^\pm|\partial_\alpha z^\pm| = 2\pi$ we get

$$\int_{\mathbb T} F^+ = \frac{2}{1+\rho_{\mathrm m}}\Big(\frac{AQ}{2} - (1-A\rho_{\mathrm m})\tfrac{\mathrm d}{\mathrm dt}\mathcal{A}^+\Big) + 4\pi,$$
$$\int_{\mathbb T} F^- = \frac{2}{1-\rho_{\mathrm m}}\Big((1-A\rho_{\mathrm m})\tfrac{\mathrm d}{\mathrm dt}\mathcal{A}^- + \frac{AQ}{2}\Big) + 4\pi,$$

and the difference is $\frac{2}{1-\rho_{\mathrm m}^2}\big(-\rho_{\mathrm m}AQ - (1-A\rho_{\mathrm m})Q\big) = -\frac{2Q}{1-\rho_{\mathrm m}^2}$. □

Remark 4.4. $Q=0$ means that $(1-\rho_{\mathrm m})\mathcal{A}^+ + (1+\rho_{\mathrm m})\mathcal{A}^-$ is constant, i.e. that the total mass $\int(\bar\rho-\rho_{\mathrm{ref}})$ over a period is conserved. In view of (4.3), the central interface Z separates, to first order in the width of the strip, the same amounts of heavy and light fluid as the strip itself. Heuristically, this is why Z must move, to leading order, with the normal velocity of the Muskat problem. Lemma 4.3 is, however, a global identity: the pointwise statement comes from the local flux balance $M_0=0$ (Lemma 5.1), and its necessity is the content of Theorem 1.2(a). In [LM26], too, the central curve starts with the normal velocity of the Muskat problem [LM26, Sec. 3.3.4 and Lemma 5.2]. The condition (4.12), Lemma 4.3 and the vertical correction h of Lemma 5.6 are specific to the periodic setting: in [LM26] the boundary values of the potential are primitives normalised at $\alpha=0$ [LM26, (4.13)], and the smallness of their transverse difference follows from uniform estimates for primitives on $\mathbb R$ [LM26, Prop. 7.2 and Lemmas 8.2, 8.6].

The following lemma, the counterpart for our stream function Ψ of the gradient formula [LM26, (4.11)] for the potential of [LM26], shows that the tangential component of D is controlled by the primitive of M, and its normal component by the boundary fluxes.

Lemma 4.5 (The state in the strip)**.** *Assume* (4.12)*, and that there are* $C_1 \geq 1$*,* $F_0 \in C^0(\mathbb{T})$ *and* $\delta \in C^0(\mathbb{T})$ *such that for* $t \in (0, T_1]$

$$\begin{aligned} &\|F^{\pm}(\cdot,t) - F_0\|_{C^0(\mathbb{T})} \leq C_1 t, \qquad \|M(\cdot,t)\|_{C^0(\mathbb{T})} \leq C_1 t, \\ &\sup_{\alpha\in[0,2\pi]} \Big| \int_0^\alpha M(\beta,t)\,\mathrm{d}\beta + C_\Psi(t) + 2t\kappa(\alpha)\delta(\alpha)\Big| \leq C_1 t^2. \end{aligned} \tag{4.13}$$

Then there is C *such that, for* $t \in (0, T_1]$ *and* $x = X(\alpha,\lambda,t) \in S_t$*,*

$$\Big| D(x,t) - \frac{F_0(\alpha)}{|\partial_\alpha z_0(\alpha)|}\, n_0(\alpha) - \delta(\alpha)\,\tau_0(\alpha)\Big| \leq Ct.$$

In particular, if $\|M(\cdot,t)\|_{C^0} \leq C_1 t^2$ *and* $C_\Psi \equiv 0$*, the conclusion holds with* $\delta \equiv 0$*.*

Proof. Differentiating (4.10) and using (4.2),

$$\begin{aligned} t\kappa\,\nabla\Psi\cdot n_0 &= \partial_\lambda\big(\Psi\circ X\big) = \tfrac12(\Psi^+ - \Psi^-) = \tfrac12\Big(\int_0^\alpha M(\beta,t)\,\mathrm{d}\beta + C_\Psi(t)\Big), \\ \nabla\Psi\cdot\partial_\alpha X &= \partial_\alpha\big(\Psi\circ X\big) = \tfrac12(1-\lambda)F^- + \tfrac12(1+\lambda)F^+. \end{aligned}$$

By (4.13), $\nabla\Psi\cdot n_0 = -\delta + O(t)$ (we use $\min\kappa > 0$) and the right-hand side of the second identity is $F_0 + O(t)$. Since $\partial_\alpha X = |\partial_\alpha z_0|\tau_0 + O(t)$ and $\nabla\Psi$ is bounded, $\nabla\Psi\cdot\tau_0 = F_0/|\partial_\alpha z_0| + O(t)$. As $D = \nabla^\perp\Psi = (\nabla\Psi\cdot\tau_0)\,n_0 - (\nabla\Psi\cdot n_0)\,\tau_0$, the claim follows. □

Remark 4.6. Lemma 4.5 isolates the two components of the relaxed flux D in the strip. The normal component is fixed by the boundary fluxes $F^\pm$, i.e. by the Rankine–Hugoniot conditions; the tangential component $D\cdot\tau_0$ is the transverse difference quotient $-(\Psi^+ - \Psi^-)/(2t\kappa)$, and it is of order one exactly when the primitive of the flux mismatch is of order t. This is the structure of the construction of [LM26], with their potential g in place of Ψ: the boundary motion determines the tangential derivatives of the values $G_\pm$ of g on the two boundaries [LM26, Sec. 3.3.1 and (4.12)], and the transverse derivative of g involves $t^{-1}\partial_\lambda G$, where $\partial_\lambda G$ is the derivative of g in the transverse variable of their strip coordinates [LM26, Sec. 3.3.4 and (4.11)]. For our ansatz, $M = O(t^2)$ together with $C_\Psi = O(t^2)$ forces $D\cdot\tau_0 = O(t)$; prescribing instead the $O(t)$ part of M produces any desired tilt δ (Lemma 5.6). The normalisation of [LM26] is a different one. It is imposed on g: the acceleration of the central curve cancels the $O(t)$ part of $\partial_\alpha G_+ - \partial_\alpha G_-$, so that $\partial_\lambda G = O(t^2)$ [LM26, Sec. 3.3.4, Lemma 5.3 and Prop. 7.2]. For $\rho_{\mathrm{m}} = A$, in our variables this gives the tilt $\delta_{\mathrm{LM}} = -(1-A^2)\gamma_0/(2+A^2)$, not $\delta = 0$, and on their subsolution $M = -2t\,\partial_\alpha(\kappa\delta_{\mathrm{LM}}) + O(t^2)$ (Remark 5.8). Thus the correspondence between M and $\partial_\alpha G_+ - \partial_\alpha G_-$ holds only at order t^0, where both vanish by the first order compatibility condition (Lemma 5.1 and [LM26, Lemma 5.2]).

5. Proof of Theorem 1.1

We keep the notation of Section 4. The proof consists in choosing the normal acceleration a, the vertical correction h and the constant C_Ψ so that the flux mismatch has a prescribed first order part, $M = -2t\,\partial_\alpha(\kappa\delta) + O(t^2)$ with $\int_{\mathbb{T}} M = 0$, for a tilt δ to be chosen; and then in checking the open condition $D\cdot(D - 2a) < 0$ in the strip by comparison with its limit as $t \to 0$.

The scheme is that of [LM26, Secs. 5–7]: a first order matching, which fixes the initial velocity of the central curve (Lemma 5.1; cf. [LM26, Lemmas 5.1–5.2]); a second order step, in which the normal acceleration of the central curve is found by inverting $I - 2A\mathcal{K}_0$ (Lemmas 5.5–5.6; cf. [LM26, Lemma 5.3 and Prop. 6.1]); and the hull condition at $t = 0$, followed by a perturbation argument and the h-principle (cf. [LM26, Sec. 7.4]). What differs is the target of the second order step, which we use to prescribe the tilt, the evaluation of the hull condition at $t = 0$ with its full tangential part (Lemma 5.2 and Remarks 5.3 and 5.8), the intermediate density ρ_{m}, which may differ from A, and the periodic setting. The time regularity of Lemma 5.4 and the regularity of G in Lemma 5.5 are only sketched (Appendix A.4); the proof of Theorem 1.1 given here is complete modulo these sketches.

5.1. **The limit $t \to 0$: velocity matching.** For the computation of the limits it is convenient to allow arbitrary normal velocities at $t = 0$.

Lemma 5.1. *Let $z^\pm(\cdot,t)$ be polynomials in t with coefficients in $C^{3,\gamma}$, with $z^\pm(\cdot,0) = z_0$ and $\partial_t z^\pm(\cdot,0)\cdot n_0 = W^\pm \in C^{3,\gamma}(\mathbb{T})$, such that the conclusions of Lemma 4.1 hold, and define $f^\pm, v, F^\pm, M$ by (4.4)–(4.8). Let*

$$\overline{W} := \tfrac{1}{2}\big[(1-\rho_{\mathrm{m}})W^+ + (1+\rho_{\mathrm{m}})W^-\big], \qquad P := 2\mathcal{K}_0[n_{0,2}] + 2A\mathcal{K}_0\overline{W}.$$

Then, uniformly on $\mathbb{T}$ as $t \to 0$, $v_S^\pm(\cdot,t) \to v_S^{(0)} := \mathcal{C}[z_0; f_0^+]^- + \mathcal{C}[z_0; f_0^-]^+$, where $f_0^\pm = [\![\rho]\!]^\pm(-e_2 + (n_{0,2} + AW^\pm)n_0)$, and

(5.1)
$$v_S^{(0)}\cdot n_0 = P + \tfrac{A}{2}\big[(1+\rho_{\mathrm{m}})W^- - (1-\rho_{\mathrm{m}})W^+\big], \qquad v_S^{(0)}\cdot\tau_0 = \mathcal{T}_0\big[2(n_{0,2}+A\overline{W})\big] - \rho_{\mathrm{m}}\tau_{0,2},$$

(5.2)
$$M(\cdot,t) \to M_0 := -\frac{4|\partial_\alpha z_0|}{1-\rho_{\mathrm{m}}^2}\Big[\big(I - 2A\mathcal{K}_0\big)\overline{W} - 2\mathcal{K}_0[n_{0,2}]\Big].$$

For the ansatz (4.1), i.e. $W^\pm = U_0 \pm c^\pm$, we have $\overline{W} = U_0$, $P = U_0$, $M_0 = 0$,

$$v_S^{(0)}\cdot n_0 = \mu_{\mathrm{m}}U_0 - A(1-\rho_{\mathrm{m}}^2)\kappa, \qquad v_S^{(0)}\cdot\tau_0 = -\mathcal{T}_0\sigma_0 - \rho_{\mathrm{m}}\tau_{0,2}, \tag{5.3}$$
$$F^\pm(\cdot,t) \to F_0 := |\partial_\alpha z_0|\big(-\sigma_0 - 2(1-A\rho_{\mathrm{m}})\kappa\big).$$

Proof. The convergence of $v_S^\pm$ follows from Lemma A.2: for instance $\mathcal{C}[z^+; f^+]^- \to \mathcal{C}[z_0; f_0^+]^-$, and $\mathcal{C}[z^-; f^-](z^+(\alpha,t)) \to \mathcal{C}[z_0; f_0^-]^+(\alpha)$ because $z^+(\alpha,t)$ lies above $\Gamma^-(t)$ and tends to $z_0(\alpha)$. By Lemma 2.3(b), $v_S^{(0)} = \langle\mathcal{C}[z_0; f_0^+ + f_0^-]\rangle - \frac{1}{2}f_0^+ + \frac{1}{2}f_0^-$, and $f_0^+ + f_0^- = -2e_2 + 2(n_{0,2} + A\overline{W})n_0$ because $[\![\rho]\!]^+ + [\![\rho]\!]^- = 2$. Since $\langle\mathcal{C}[z_0; e_2]\rangle = 0$ by Lemma 2.3(c), $\langle\mathcal{C}[z_0; f_0^+ + f_0^-]\rangle\cdot n_0 = P$ and $\langle\mathcal{C}[z_0; f_0^+ + f_0^-]\rangle\cdot\tau_0 = \mathcal{T}_0[2(n_{0,2} + A\overline{W})]$, while $f_0^\pm\cdot n_0 = A[\![\rho]\!]^\pm W^\pm$ and $f_0^\pm\cdot\tau_0 = -[\![\rho]\!]^\pm\tau_{0,2}$; this gives the normal part of (5.1) and $v_S^{(0)}\cdot\tau_0 = \mathcal{T}_0[2(n_{0,2} + A\overline{W})] - \rho_{\mathrm{m}}\tau_{0,2}$. Passing to the limit in (4.6)–(4.7),

$$\frac{M_0}{|\partial_\alpha z_0|} = \frac{2}{1-\rho_{\mathrm{m}}^2}\Big\{2v_S^{(0)}\cdot n_0 - (1-\rho_{\mathrm{m}})(1-A)W^+ - (1+\rho_{\mathrm{m}})(1+A)W^-\Big\} = \frac{4}{1-\rho_{\mathrm{m}}^2}\big(P - \overline{W}\big),$$

which is (5.2). For the ansatz, $(1-\rho_{\mathrm{m}})c^+ = (1+\rho_{\mathrm{m}})c^- = (1-\rho_{\mathrm{m}}^2)\kappa$ gives $\overline{W} = U_0$. Applying $\mathcal{K}_0$ to $(I - 2A\mathcal{K}_0)\sigma_0 = -2n_{0,2}$ and using $U_0 = -\mathcal{K}_0\sigma_0$ we get $(I - 2A\mathcal{K}_0)U_0 = 2\mathcal{K}_0[n_{0,2}]$, i.e. $P = U_0$ and $M_0 = 0$; moreover $2(n_{0,2} + AU_0) = -\sigma_0$ by Proposition 2.7, which gives the tangential part of (5.3). The normal part gives $v_S^{(0)}\cdot n_0 = U_0 + A\rho_{\mathrm{m}}U_0 - A(1-\rho_{\mathrm{m}}^2)\kappa$, and inserting this in (4.6),

$$\frac{F_0}{|\partial_\alpha z_0|} = \frac{2}{1+\rho_{\mathrm{m}}}\Big(A(1+\rho_{\mathrm{m}})U_0 - (1+\rho_{\mathrm{m}})\kappa\big[A(1-\rho_{\mathrm{m}}) + 1 - A\big]\Big) + 2n_{0,2} = -\sigma_0 - 2(1-A\rho_{\mathrm{m}})\kappa. \quad \square$$

The vanishing of M at order t^0 is the statement that the centre of mass Z of the strip starts with the normal velocity of the Muskat problem, which we built into the ansatz. This is the first order compatibility condition of [LM26, Lemma 5.2], where the central curve starts with the Muskat normal velocity and the boundary derivatives of their potential agree at $t = 0$ [LM26, (5.14)]; for $\rho_{\mathrm{m}} = A$ and $c = (1-A^2)\kappa$, the value $F_0/|\partial_\alpha z_0| = -\sigma_0 - 2c$ is the normal component of D on their single layer at $t = 0$ (Remark 5.8).

5.2. **The relaxation condition at $t = 0$.**

Lemma 5.2. *Let $a^{(0)} := e_2 + \frac{A}{\mu_{\mathrm{m}}}v_S^{(0)}$ and, for a function δ, $D_\delta^{(0)} := \frac{F_0}{|\partial_\alpha z_0|}n_0 + \delta\,\tau_0$. Then*

$$a^{(0)}\cdot n_0 = -\frac{\sigma_0}{2} - \frac{A^2(1-\rho_{\mathrm{m}}^2)\kappa}{\mu_{\mathrm{m}}}, \qquad a^{(0)}\cdot\tau_0 = -\frac{(1-A^2)\gamma_0}{2\mu_{\mathrm{m}}}, \tag{5.4}$$

and, with Φ *as in* (1.5),

$$D_\delta^{(0)} \cdot \big(D_\delta^{(0)} - 2a^{(0)}\big) = \frac{2(1-A^2)}{\mu_{\mathrm{m}}}\,\Phi(\kappa) + \big(\delta - a^{(0)}\cdot\tau_0\big)^2. \tag{5.5}$$

In particular, for the tilt $\delta_* := a^{(0)}\cdot\tau_0 = -\frac{(1-A^2)\gamma_0}{2\mu_{\mathrm{m}}}$, *the left-hand side equals* $\frac{2(1-A^2)}{\mu_{\mathrm{m}}}\Phi(\kappa)$, *which is negative under* (1.6)*; and* δ_* *is the optimal tilt.*

Proof. By (5.3), $a^{(0)}\cdot n_0 = n_{0,2} + AU_0 - \frac{A^2(1-\rho_{\mathrm{m}}^2)\kappa}{\mu_{\mathrm{m}}}$ with $n_{0,2} + AU_0 = -\sigma_0/2$; and $a^{(0)}\cdot\tau_0 = \tau_{0,2} - \frac{A}{\mu_{\mathrm{m}}}(\mathcal{T}_0\sigma_0 + \rho_{\mathrm{m}}\tau_{0,2}) = \frac{\tau_{0,2} - A\mathcal{T}_0\sigma_0}{\mu_{\mathrm{m}}}$, which equals $-\frac{(1-A^2)\gamma_0}{2\mu_{\mathrm{m}}}$ by (2.8). (For $\rho_{\mathrm{m}} = A$, where $a = (e_2 - A\nabla p)/(1+A^2)$ in the variables of [LM26] (Remark 5.8), the normal component is [LM26, (7.11)] combined with the computation in the proof of [LM26, Prop. 7.4], and the tangential component follows from the tangential part of [LM26, (7.11)] and from [LM26, (3.7b)], which is Proposition 2.10 in our normalisation.) Writing $D_n = -\sigma_0 - 2(1-A\rho_{\mathrm{m}})\kappa$, $a_n = a^{(0)}\cdot n_0$ and $a_\tau = a^{(0)}\cdot\tau_0$, we have

$$D_n - 2a_n = -2\kappa\Big[(1-A\rho_{\mathrm{m}}) - \frac{A^2(1-\rho_{\mathrm{m}}^2)}{1+A\rho_{\mathrm{m}}}\Big] = -\frac{2\kappa(1-A^2)}{\mu_{\mathrm{m}}},$$

so $D_n(D_n - 2a_n) = \frac{2(1-A^2)}{\mu_{\mathrm{m}}}\kappa\big(2(1-A\rho_{\mathrm{m}})\kappa + \sigma_0\big)$. Adding $\delta(\delta - 2a_\tau) = (\delta - a_\tau)^2 - a_\tau^2$ and $a_\tau^2 = \frac{(1-A^2)^2\gamma_0^2}{4\mu_{\mathrm{m}}^2}$ gives (5.5). □

Remark 5.3 (Three normalisations of the tilt). By (5.5), the tilt enters the relaxation inequality at $t=0$ only through $(\delta-\delta_*)^2$. We compare three choices.

(a) $\delta = 0$, which our construction gives when $M = O(t^2)$ and $C_\Psi \equiv 0$ (Lemma 4.5). The condition $D_0^{(0)}\cdot(D_0^{(0)} - 2a^{(0)}) < 0$ reads $0 < 2(1-A\rho_{\mathrm{m}})\kappa < -\sigma_0$ and involves σ_0 only. For $\rho_{\mathrm{m}} = A$ this is the range stated in [LM26] for their single layer, $\bar{c} < \frac{1}{2}$ in [LM26, (7.13)]. The untilted strip is, however, not the subsolution of [LM26]: it shares with their single layer only the first order geometry and this stated range, and it is not used in [LM26].

(b) $\delta = \delta_{\mathrm{LM}} := -\frac{(1-A^2)\gamma_0}{2+A^2} = \frac{2(1+A^2)}{2+A^2}\,\delta_*$ with $\rho_{\mathrm{m}} = A$, which is the tilt of the single layer of [LM26] (Remark 5.8). Since $\delta_{\mathrm{LM}} - \delta_* = \frac{A^2}{2+A^2}\,\delta_*$, the condition reads

$$\kappa\big(2(1-A^2)\kappa + \sigma_0\big) < \frac{(1-A^2)\gamma_0^2}{2(2+A^2)^2},$$

which is (1.8).

(c) $\delta = \delta_*$: the condition is $\Phi(\kappa) < 0$, i.e. (1.6). The tilt δ_* moves $D^{(0)}$ onto the line through the centre $a^{(0)}$ of the relaxation disc parallel to n_0, and by Theorem 1.2 no single strip of intermediate density ρ_{m} does better.

When $\gamma_0 \neq 0$, the ranges in (b) and (c) are strictly larger than the corresponding range in (a), and they are non-empty also where $\sigma_0 \geq 0$. For $A = 0$ we have $\delta_{\mathrm{LM}} = \delta_*$, so that (b) and (c) coincide; for $A \neq 0$ the coefficient of γ_0^2 in (c) is larger than in (b) by the factor computed in Remark 5.8.

5.3. **Time regularity and acceleration matching.** We only sketch the proof of the following lemma, in Appendix A.4, which also lists the points that are not proved in detail there. Fully proved analogues in the asymptotically flat setting are [LM26, Lemmas 8.4–8.5 and Prop. 7.1], except for the Lipschitz bound on $\partial_t F^\pm$: for the second time derivative, [LM26, Lemma 8.6 and Prop. 7.2] prove only the weaker bound on spatial primitives that their argument needs.

Lemma 5.4. *Let* $a \in C^{3,\gamma}(\mathbb{T})$ *and let* h *be a polynomial with* $h(0) = h'(0) = 0$. *There are* $T_1 > 0$ *and* C *such that* $F^\pm$ *and* M, *extended to* $t = 0$ *by their limits, belong to* $C^1([0,T_1]; C^0(\mathbb{T}))$, $\|\partial_t F^\pm(\cdot,t) - \partial_t F^\pm(\cdot,s)\|_{C^0} \leq C|t-s|$, *and*

$$\sup_{x = X(\alpha,\lambda,t)\in S_t} \big|v(x,t) - v_S^{(0)}(\alpha)\big| \leq Ct^\gamma \qquad (0 < t \leq T_1). \tag{5.6}$$

Lemma 5.5. *There is $G \in C^{3,\gamma}(\mathbb{T})$, depending only on $z_0, \kappa, \rho_{\mathrm{m}}$ and A, such that for every $a \in C^{3,\gamma}(\mathbb{T})$ and $h \equiv 0$,*

$$\partial_t M(\cdot,0) = G - \frac{4|\partial_\alpha z_0|}{1-\rho_{\mathrm{m}}^2}\big(I - 2A\mathcal{K}_0\big)a.$$

Proof. For fixed t the quantities $V^\pm$, $f^\pm$, v, $v_S^\pm$, $F^\pm$ and M are affine functions of the velocities $(\partial_t z^+, \partial_t z^-)$, with coefficients depending only on the positions $z^\pm(\cdot,t)$: $V^\pm$ is linear in $\partial_t z^\pm$, $f^\pm$ is affine in $V^\pm$, and v is linear in $f^\pm$. Write accordingly $M(t) = \mathcal{L}_t[\partial_t z^+, \partial_t z^-] + \mathcal{B}_t$. By the formulas of Appendix A, $\mathcal{L}_t$ and $\mathcal{B}_t$ are differentiable at $t = 0$, where the two curves merge (this is part of the sketch of Appendix A.4), and their derivatives only involve z_0 and $\partial_t z^\pm(\cdot,0) = (U_0 \pm c^\pm)n_0$, which do not depend on a (the term $\frac{t^2}{2}an_0$ in the positions does not contribute at $t = 0$). Hence $\partial_t M(\cdot,0) = \mathcal{L}_0[\partial_t^2 z^+, \partial_t^2 z^-](\cdot,0) + G$ with G independent of a. Since $\partial_t^2 z^\pm(\cdot,0) = an_0$ and, by (5.2), the linear part of the map $(W^+, W^-) \mapsto M_0$ is $-\frac{4|\partial_\alpha z_0|}{1-\rho_{\mathrm{m}}^2}(I - 2A\mathcal{K}_0)\overline{W}$, we obtain the formula. The regularity $G \in C^{3,\gamma}$ is sketched in Appendix A.4. □

The vanishing of M at order t^0 leaves no freedom (it is the Muskat equation for the velocity of the centre of mass), but at order t^1 the acceleration a is a free function, and by Proposition 2.6 we can make $\partial_t M(\cdot,0)$ equal to *any* function in $C^{3,\gamma}(\mathbb{T})$. This is the freedom used for the tilt. It is the second order step of [LM26]: there, too, the normal acceleration of the central curve enters the $O(t)$ part of the boundary data affinely, through $I - AP$, which is our $I - 2A\mathcal{K}_0$ [LM26, Lemma 5.3, (5.21) and (8.41)], and it is determined by inverting this operator [LM26, Prop. 6.1] (in the periodic setting the invertibility is due to [CCG11]; here Proposition 2.6). Lemmas 5.5 and 5.6 are this step with a nonzero target, which prescribes the tilt. [LM26] use two different normalisations of it, for the single layer [LM26, (5.16)] and for the multilayers [LM26, (8.41)]; in our variables they correspond to the tilts δ_{LM} and, for one intermediate strip, $2\delta_*$ (Remark 5.8).

Lemma 5.6 (Prescribed first order mismatch and mass correction)**.** *Let $\delta \in C^{4,\gamma}(\mathbb{T})$ and*

$$a := \big(I - 2A\mathcal{K}_0\big)^{-1}\Big[\frac{(1-\rho_{\mathrm{m}}^2)\big(G + 2\partial_\alpha(\kappa\delta)\big)}{4|\partial_\alpha z_0|}\Big] \in C^{3,\gamma}(\mathbb{T}), \tag{5.7}$$

so that $\partial_t M(\cdot,0) = -2\partial_\alpha(\kappa\delta)$ when $h \equiv 0$. Let $Q^{(0)}$ be the function Q of Lemma 4.3 computed with $h \equiv 0$, and $h(t) := -\frac{1}{4\pi}\int_0^t Q^{(0)}(s)\,\mathrm{d}s$. Then h is a polynomial with $|h(t)| \le Ct^3$, and the curves (4.1) with these a and h satisfy (4.12) and

$$\big\|M(\cdot,t) + 2t\,\partial_\alpha(\kappa\delta)\big\|_{C^0(\mathbb{T})} \le Ct^2 \qquad \text{for } t \in (0, T_1].$$

Proof. The choice (5.7) is possible by Proposition 2.6 with $k = 6$, $j = 3$, and Lemma 5.5 gives $\partial_t M(\cdot,0) = -2\partial_\alpha(\kappa\delta)$ for $h \equiv 0$. Let $M^{(0)}$ be the mismatch for $h \equiv 0$. By Lemmas 5.1 and 5.4, $M^{(0)}(\cdot,0) = 0$ and $\partial_t M^{(0)}$ is Lipschitz in time, so $\|M^{(0)}(\cdot,t) + 2t\partial_\alpha(\kappa\delta)\|_{C^0} \le Ct^2$. Since $\int_{\mathbb{T}} \partial_\alpha(\kappa\delta) = 0$, Lemma 4.3 gives $|Q^{(0)}(t)| \le Ct^2$. The areas $\mathcal{A}^\pm$ are polynomials in t, hence so are $Q^{(0)}$ and h, and $h(0) = h'(0) = 0$, $|h'(t)| \le Ct^2$, $|h(t)| \le Ct^3$. Adding $h(t)e_2$ to both curves adds $2\pi h$ to $\mathcal{A}^\pm$, so $Q = Q^{(0)} + 4\pi h' = 0$ and (4.12) holds by Lemma 4.3. Finally, Cauchy fields commute with translations, $\mathcal{C}[z + he_2; f](x + he_2) = \mathcal{C}[z; f](x)$, so the new $F^\pm$ are obtained from the old ones by adding $h'e_2$ to the velocities; by the affine dependence on the velocities and Lemma 2.3(b), $\|M - M^{(0)}\|_{C^0} \le C|h'(t)| \le Ct^2$.

The bounds $|h(t)| \le Ct^3$ and $\|M + 2t\partial_\alpha(\kappa\delta)\|_{C^0} \le Ct^2$ rest on Lemmas 5.4 and 5.5, whose proofs are only sketched (Appendix A.4); the identity $Q = Q^{(0)} + 4\pi h' = 0$, and hence (4.12), does not. □

5.4. **Conclusion.**

Proof of Theorem 1.1. Let $\delta := \delta_* = -\frac{(1-A^2)\gamma_0}{2\mu_{\mathrm{m}}} \in C^{4,\gamma}(\mathbb{T})$ be the optimal tilt of Lemma 5.2, choose a and h by Lemma 5.6, and set $C_\Psi(t) := -2t\kappa(0)\delta(0)$ in (4.9). Consider the curves (4.1) and the triple $(\bar\rho, \bar u, \bar m)$ of (4.11).

By Lemma 5.6, (4.12) holds and

$$\int_0^\alpha M(\beta,t)\,\mathrm{d}\beta + C_\Psi(t) + 2t\kappa(\alpha)\delta(\alpha) = \int_0^\alpha \big(M + 2t\partial_\beta(\kappa\delta)\big)(\beta,t)\,\mathrm{d}\beta = O(t^2)$$

uniformly in $\alpha \in [0,2\pi]$; by Lemma 5.4 and since $h'(0)=0$, $\|F^\pm(\cdot,t) - F_0\|_{C^0} \le Ct$ with F_0 given by (5.3). Thus (4.13) holds, and Lemma 4.5 gives $|D - D_\delta^{(0)}(\alpha)| \le Ct$ in S_t; in particular D is bounded uniformly in t. By (5.6), $|a - a^{(0)}(\alpha)| \le \frac{|A|}{\mu_\mathrm{m}} Ct^\gamma$ in S_t, where $a = e_2 + \frac{A}{\mu_\mathrm{m}} v$ is the vector (3.5) of the state in the strip. The field v is bounded uniformly in t by Lemma 2.3(b) and Lemma 4.1, so $(\bar\rho,\bar u,\bar m)$ is bounded on $\mathbb{T}\times\mathbb{R}\times(0,T_1]$; by Lemma 4.2 it therefore solves (3.2) with initial datum ρ_0, and it lies in K outside the strip. Since all quantities are bounded, Lemma 5.2 gives, in S_t,

$$D\cdot(D-2a) \le \frac{2(1-A^2)}{\mu_\mathrm{m}} \max_{\mathbb{T}} \Phi(\kappa) + C' t^\gamma < 0$$

for $0 < t \le T$, provided $T \le T_1$ and $C'T^\gamma < -\frac{2(1-A^2)}{\mu_\mathrm{m}} \max_{\mathbb{T}} \Phi(\kappa)$; note that $\max_{\mathbb{T}} \Phi(\kappa) < 0$ by (1.6) and continuity. Hence, with $\Omega_\mathrm{mix} := \{(x,t) : 0 < t < T,\ x \in S_t\}$, $(\bar\rho,\bar u,\bar m)$ is continuous on Ω_mix with values in $\mathcal{U}_A$ (Proposition 3.2), i.e. a subsolution in the sense of Definition 3.4. Theorem 3.5 yields infinitely many weak solutions with $\rho = \bar\rho$ outside Ω_mix, mixing in Ω_mix, and a sequence converging weakly-$*$ to $\bar\rho$. Since $\Omega_\mathrm{mix}(t) = S_t$ is the strip between $\Gamma^\pm(t)$ and Z has the form (1.7), all the assertions of Theorem 1.1 follow. □

Remark 5.7. The proof works with any tilt $\delta \in C^{4,\gamma}$ for which $\frac{2(1-A^2)}{\mu_\mathrm{m}}\Phi(\kappa) + (\delta-\delta_*)^2 < 0$ on $\mathbb{T}$, for instance with each of the three choices of Remark 5.3 under the corresponding condition on κ. In particular, if $0 < 2(1-A\rho_\mathrm{m})\kappa < -\sigma_0$, then $\delta = 0$ is admissible; for $\rho_\mathrm{m} = A$ this gives, in our periodic Hölder setting, single strips with the opening range stated in [LM26, (7.13)] for the curved single layer of [LM26], whose intermediate density is A ([LM26] work in the asymptotically flat setting). These strips are not the subsolutions of [LM26], whose tilt is δ_LM (Remark 5.8). The multilayers of [LM26, Sec. 7.5] with $N \ge 2$, whose $2N-1$ strips have the densities θ_j of [LM26, (7.14)], are not single strips (for $N=1$ their only intermediate strip has density A, the tilt $2\delta_*$ and the range $\bar c < \frac12$; Remark 5.8); for fully unstable data they reach any $0 < \bar c < 1$, and so open faster than our single strips wherever $|\gamma_0|$ is small.

Remark 5.8 (The single layer of [LM26] in our variables). Let $\rho_\mathrm{m} = A$ and $c = (1-A^2)\kappa$, so that the boundaries (4.1) move away from the central curve with the normal speeds $c/(1\mp A)$ of [LM26, (4.2)]. In the mixing zone of [LM26] the momentum is $m = Au - \nabla^\perp g - \frac12(Au+e_2)$, with a potential g [LM26, (4.6)], and Darcy's law [LM26, (4.10)] reads $(1+\frac{A^2}{2})u = -\nabla p + A\nabla^\perp g - \frac{A}{2}e_2$. In the variables (3.5) this gives

$$D = (Au+e_2) - 2\nabla^\perp g, \qquad a = \frac{e_2 - A\nabla p}{1+A^2}, \qquad D\cdot(D-2a) = \frac{4|\nabla g|^2 - |Au+e_2|^2}{1+A^2},$$

so that the exact hull condition $2|\nabla g| < |Au+e_2|$ of [LM26, (4.14)] is our condition $|D-a| < |a|$. In [LM26] the acceleration of the central curve is chosen so as to cancel the $O(t)$ part of $\partial_\alpha G_+ - \partial_\alpha G_-$, where $G_\pm$ are the values of g on the two boundaries [LM26, Sec. 3.3.4 and Lemma 5.3]. Then the transverse derivative $\partial_\lambda G$ of g in their strip coordinates is $O(t^2)$, and ∇g is tangential at leading order [LM26, Props. 7.2–7.3 and (7.9)], so that $D\cdot\tau_0 = (Au+e_2)\cdot\tau_0 + O(t)$; and the tangential part of [LM26, (4.10)], together with [LM26, (7.11)] and [LM26, (3.7b)] (our Proposition 2.10), gives $(Au+e_2)\cdot\tau_0 = \delta_\mathrm{LM} + O(t)$. Thus the single layer of [LM26] is tilted towards the centre of the relaxation disc, and exactly onto it when $A=0$; on their subsolution our mismatch is $M = -2t\,\partial_\alpha(\kappa\delta_\mathrm{LM}) + O(t^2)$. By [LM26, (7.9)–(7.10)], the normal component of D on their layer is $D\cdot n_0 = -\sigma_0 - 2c + O(t)$, the value $F_0/|\partial_\alpha z_0|$ of (5.3), and by [LM26, (7.11)] the vector a tends to $a^{(0)}$ of Lemma 5.2. Hence, by (5.5) with $\delta = \delta_\mathrm{LM}$, the unchanged single layer of [LM26] satisfies the strict hull condition at $t=0$ if and only if κ lies in the range (b) of Remark 5.3. This range is not stated in [LM26]. Up to their final estimate the opening c is free [LM26, (5.1)]; the restriction $\bar c < \frac12$ comes only

from the lower bound $|Au+e_2| \geq (Au+e_2)\cdot n_0 = -\frac{1}{2}\sigma_0 + O(t)$ used in [LM26, Sec. 7.4], which keeps only the normal component of $Au+e_2$, computed in [LM26, Prop. 7.4], and discards the tangential (vortex-sheet) part. In this sense the range (b) is latent in [LM26]: turning it into an existence statement in their setting requires only a modification of a few lines of [LM26, Sec. 7.4], with an opening c such that $c - c_\infty \in H^{k-1}$ and $\inf c > 0$ [LM26, Sec. 4.1 and (5.1)]. Compared with (b), the optimal tilt δ_* multiplies the coefficient of γ_0^2 by

$$\frac{(2+A^2)^2}{4(1+A^2)} = 1 + \frac{A^4}{4(1+A^2)} \in \Big[1, \frac{9}{8}\Big),$$

with no gain for $A = 0$. Finally, for one intermediate strip ($N = 1$), the normalisation of the multilayers of [LM26, Sec. 7.5] (see [LM26, (8.38) and (8.43)]) gives $D\cdot\tau_0 \to 2\delta_*$, the reflection of $\delta = 0$ about δ_*, whose margin in (5.5) is the same as that of $\delta = 0$.

6. Necessary conditions: proof of Theorem 1.2

We now show that the range (1.6) cannot be improved by subsolutions with a single intermediate density. This is a statement about the type of subsolution, not about the geometry of the mixing zone: the multilayer subsolutions of [LM26, Sec. 7.5], whose mixing zone is one band containing several intermediate densities, are not single-strip subsolutions, and for fully unstable data they reach any $0 < \bar{c} < 1$; for $\bar{c}$ close to 1 this exceeds the single-strip bound wherever $|\gamma_0|$ is small (if $\rho_{\mathrm{m}} = A$, $\sigma_0 < 0$ and $\gamma_0 = 0$, Theorem 1.2 gives $\kappa \leq \kappa_{\max} = -\sigma_0/(2(1-A^2))$, i.e. $\bar{c} \leq \frac{1}{2}$ when $(1-A^2)\kappa = -\bar{c}\,\sigma_0$). We first fix the class of subsolutions; it contains the subsolutions of Section 4.

Definition 6.1. Let $z_0 \in \mathcal{Z}^{3,\gamma}$, $\rho_{\mathrm{m}} \in (-1,1)$ and $T > 0$. A *single-strip subsolution with intermediate density* ρ_{m} *emanating from* z_0 is a subsolution $(\bar{\rho}, \bar{u}, \bar{m})$ on $[0,T]$ with initial datum $\rho_0 = \mathbf{1}_{\Omega^+(z_0)} - \mathbf{1}_{\Omega^-(z_0)}$ (Definition 3.4), together with curves $z^\pm(\cdot,t)$ and a map X, such that:

(S1) $t \mapsto z^\pm(\cdot,t) - (\cdot,0)$ belongs to $C^0([0,T]; C^{1,\gamma}(\mathbb{T})) \cap C^1([0,T]; C^{0,\gamma}(\mathbb{T}))$, $\delta(z^\pm(\cdot,t))$ is bounded below, $z^\pm(\cdot,0) = z_0$, and for $t > 0$ the curve $\Gamma^+(t)$ lies in $\Omega^+(z^-(\cdot,t))$; we let $S_t := \Omega^-(z^+(\cdot,t)) \cap \Omega^+(z^-(\cdot,t))$;

(S2) $X(\cdot,\cdot,t) : \mathbb{T}\times[-1,1] \to \overline{S_t}$ is C^1, $X(\cdot,\pm1,t) = z^\pm(\cdot,t)$, and $|\partial_\lambda X| \leq C_X t$;

(S3) $\bar{\rho} = 1$ in $\Omega^+(z^+(\cdot,t))$, $\bar{\rho} = \rho_{\mathrm{m}}$ in S_t, $\bar{\rho} = -1$ in $\Omega^-(z^-(\cdot,t))$, and $\Omega_{\mathrm{mix}} = \{(x,t) : 0 < t < T,\ x \in S_t\}$;

(S4) for every $t \in (0,T]$ the restrictions of $(\bar{u},\bar{m})(\cdot,t)$ to each of the three regions extend continuously to their closures;

(S5) $\bar{u}(x,t) \to 0$ as $|x_2| \to \infty$, uniformly in t.

The *initial normal velocities* of the strip are $W^\pm := \partial_t z^\pm(\cdot,0)\cdot n_0 \in C^{0,\gamma}(\mathbb{T})$.

The subsolutions constructed in Section 4 satisfy (S1)–(S5), with X given by (4.2).

In the asymptotically flat setting of [LM26] one can define an analogous class, which contains the single-layer subsolutions of [LM26] (with $\rho_{\mathrm{m}} = A$). We expect the proof below to apply to it after adapting Steps 1 and 4, which use periodic tools: Liouville's theorem on the cylinder and the periodic far-field limits of Cauchy integrals (Lemma 2.3(c)) in Step 1, and the injectivity of $I - 2A\mathcal{K}_0$ on $C^{0,\gamma}(\mathbb{T})$ (Proposition 2.6) in Step 4. In [LM26] the Saffman–Taylor densities do not decay at infinity, the Cauchy integrals are taken in the principal value sense at infinity [LM26, Sec. 3.3.2], and $I - AP$ is inverted on $H^j(\mathbb{R})$ [LM26, Prop. 6.1]; we have not carried out this adaptation. For the single-layer subsolutions actually constructed in [LM26], the conclusions of Theorem 1.2 are immediate: their central curve $z = \frac{1}{2}[(1-A)z_+ + (1+A)z_-]$ is the centre of mass of the strip and starts with the Muskat normal velocity by construction [LM26, Sec. 3.3.4 and Lemma 5.2]; and, for every opening c with which their single layer is a subsolution, letting $t \to 0$ in their hull condition [LM26, (4.14)] gives (1.8) with $\leq$ in place of $<$, hence $\Phi(\kappa) \leq 0$ (see the discussion after Theorem 1.2). For their stated range $0 < \bar{c} < \frac{1}{2}$ [LM26, (7.13)], with $(1-A^2)\kappa = c = -\bar{c}\,\sigma_0$, this is simply $2(1-A^2)\kappa + \sigma_0 = (1-2\bar{c})\sigma_0 < 0$, which gives $\Phi(\kappa) < 0$.

The content of Theorem 1.2 concerns the other single strips: other boundary motions, other tilts, other intermediate densities and, more generally, other states in the strip.

The proof is built from ingredients of [LM26], which are used there constructively. The boundary motion determines the pressure, hence the Darcy field [LM26, Sec. 3.3.2] (Step 1); the Rankine–Hugoniot conditions determine the boundary data of the relaxed flux [LM26, Sec. 3.3.1 and (4.12)] (Step 2); on a strip of width $O(t)$ the transverse derivative involves t^{-1} times the difference of the boundary data [LM26, Sec. 3.3.4] (Step 3); equal leading boundary data correspond to the Muskat velocity of the central curve [LM26, Lemma 5.2], through the invertibility of $I - AP$ [LM26, Prop. 6.1] (Step 4); and Step 5 evaluates at $t = 0$ the hull condition, which for $\rho_{\mathrm{m}} = A$ is [LM26, (4.14)], the term in γ_0 coming from [LM26, (3.7b)] through Proposition 2.10 and the computation of Lemma 5.2. For the flat interface, (a) and (b) are contained in [LM26, Sec. 3.2], within its ansatz $\bar u = 0$, $\bar m$ constant and vertical, with an arbitrary intermediate density θ_{mix}: the mass balance $(1-\theta_{\mathrm{mix}})h_+ = (1+\theta_{\mathrm{mix}})h_-$ puts the centre of mass at rest, which is (a) with $U_0 = 0$, and the hull condition $|\gamma| < \frac12$ there reads $0 < \kappa < 1/(1 - A\theta_{\mathrm{mix}}) = \kappa_{\max}$, which is (b) with $\sigma_0 = -2$ and $\gamma_0 = 0$.

We use Lemma 5.1 in the following form, in which the curves are only assumed to satisfy (S1).

Lemma 6.2. *Let $z_0 \in \mathcal{Z}^{3,\gamma}$, let $z^\pm$ satisfy (S1), with $W^\pm := \partial_t z^\pm(\cdot,0)\cdot n_0 \in C^{0,\gamma}(\mathbb{T})$, and define $f^\pm, v_S^\pm, F^\pm, M$ by (4.4)–(4.8). Then, uniformly on $\mathbb{T}$ as $t \to 0$, $v_S^\pm(\cdot,t) \to v_S^{(0)}$, $F^\pm(\cdot,t) \to F_0^\pm$ and $M(\cdot,t) \to M_0 = F_0^+ - F_0^-$, where $v_S^{(0)}$ and M_0 are given by (5.1)–(5.2), and $F_0^\pm$ are obtained from (4.6)–(4.7) by replacing $z^\pm$, $n^\pm$, $V^\pm$ and $v_S^\pm$ with z_0, n_0, $W^\pm$ and $v_S^{(0)}$. If moreover $W^\pm = U_0 \pm (1 \pm \rho_{\mathrm{m}})\kappa$ with $\kappa \in C^{0,\gamma}(\mathbb{T})$, then (5.3) holds.*

Proof. By (S1), $z^\pm(\cdot,t) \to z_0$ in $C^{1,\gamma}$, with $\delta(z^\pm(\cdot,t))$ bounded below, and $\partial_t z^\pm(\cdot,t) \to \partial_t z^\pm(\cdot,0)$ in $C^{0,\gamma}$. Hence $n^\pm$, $|\partial_\alpha z^\pm|$ and $V^\pm$ converge in $C^{0,\gamma}$ to n_0, $|\partial_\alpha z_0|$ and $W^\pm$, and $f^\pm$ converges in $C^{0,\gamma}$ to $f_0^\pm$. The proof of Lemma 5.1 uses only these convergences, Lemma A.2 (each of the curves $\Gamma^\pm(t)$ lies on the appropriate side of the other by (S1)), Lemma 2.3(b)–(c) and Proposition 2.7; it does not use the polynomial dependence on t or the $C^{3,\gamma}$ regularity assumed there. It therefore applies without change. □

Proof of Theorem 1.2. Let $n^\pm$, $V^\pm = \partial_t z^\pm \cdot n^\pm$, $f^\pm$ and $v_S^\pm$ be defined from $z^\pm$ as in Section 4, and let $v := \bar u + A\bar m$ be the Darcy field of the subsolution. The identities of Steps 1–3 are obtained by testing the equations with functions of the form $\psi(t)\chi(x)$, so they hold for a.e. $t \in (0,T)$; in Steps 4 and 5 we pass to the limit $t \to 0$ along a sequence $t_k \to 0$ of such times.

Step 1: the Darcy field. By (3.2), $\operatorname{curl} v = -\partial_1\bar\rho$ and $\operatorname{div} v = \operatorname{div}\bar u + A\operatorname{div}\bar m = -A\partial_t\bar\rho$ in $\mathcal{D}'$. By (S1) and (S3), $\partial_t\bar\rho = -\sum_\pm [\![\rho]\!]^\pm V^\pm \mathcal{H}^1\llcorner\Gamma^\pm(t)$, so for a.e. t the field $v(\cdot,t)$ has the same distributional curl and divergence as $W := \mathcal{C}[z^+;f^+] + \mathcal{C}[z^-;f^-]$ (Lemma 2.3(b) and the computation of the jumps of $f^\pm$ in the proof of Lemma 4.2). Hence $w := v - W$ is a bounded, periodic field with $\operatorname{div} w = \operatorname{curl} w = 0$ in $\mathcal{D}'(\mathbb{R}^2)$; by Weyl's lemma $w^\flat$ is entire, and by Liouville's theorem on the cylinder it is constant. Outside the strip $\bar m = \pm\bar u$, so (S5) gives $v \to 0$ as $|x_2| \to \infty$, whereas $W \to \pm\frac{AQ}{4\pi}e_2$ (see the proof of Lemma 4.3). Comparing the limits at $\pm\infty$, the constant is 0 (and $AQ = 0$). Thus $v(\cdot,t) = W$ for a.e. t: the Darcy field of any single-strip subsolution is the one of our ansatz.

Step 2: boundary fluxes. By (S4), the conservation law $\partial_t\bar\rho + \operatorname{div}\bar m = 0$ and $\operatorname{div}\bar u = 0$ imply, for a.e. t, the Rankine–Hugoniot conditions $[\![\bar\rho]\!]V^\pm = [\![\bar m\cdot n^\pm]\!]$ and $[\![\bar u\cdot n^\pm]\!] = 0$ pointwise on $\Gamma^\pm(t)$: for a.e. t, $\operatorname{div}\bar m(\cdot,t) = -\partial_t\bar\rho(\cdot,t) = \sum_\pm [\![\rho]\!]^\pm V^\pm \mathcal{H}^1\llcorner\Gamma^\pm(t)$ by Step 1, while the Gauss–Green formula in each region, which applies by (S4), identifies $\operatorname{div}\bar m(\cdot,t)$ with $\sum_\pm [\![\bar m\cdot n^\pm]\!]\mathcal{H}^1\llcorner\Gamma^\pm(t)$; similarly for $\bar u$, with $\operatorname{div}\bar u = 0$. Inside the strip, $\operatorname{div}\bar m = 0$ and $\operatorname{div} v = 0$, so the field $D := \frac{2[(1-A^2)\bar m - (\rho_{\mathrm{m}} - A)v]}{1-\rho_{\mathrm{m}}^2} + 2e_2$ of (3.5) is divergence-free in S_t and continuous on $\overline{S_t}$. Reading the computation in the proof of Lemma 4.2 backwards, the Rankine–Hugoniot conditions determine the normal traces of D from inside the strip: for a.e. t,

$$D\cdot n^\pm\,|\partial_\alpha z^\pm| = F^\pm \qquad \text{on } \Gamma^\pm(t), \tag{6.1}$$

with $F^{\pm}$ given by (4.6)–(4.7). Since $(\bar\rho,\bar u,\bar m)\in\mathcal{U}_A$ in S_t, D lies in the disc $\{|D-a|<|a|\}$, so $|D|\le 2|a|\le C$, with C independent of t because $v=\bar u+A\bar m$ is bounded (Definition 3.4).

Step 3: flux balance. Let t be such that (6.1) holds. On the periodic lift of $\overline{S_t}$, a simply connected strip, write $D=\nabla^{\perp}\Psi$ with $\Psi\in C^1$. By (6.1), $\partial_\alpha\big(\Psi\circ z^{\pm}\big)=F^{\pm}$, hence

$$\Psi(z^+(\alpha,t))-\Psi(z^-(\alpha,t))=\Psi(z^+(0,t))-\Psi(z^-(0,t))+\int_0^{\alpha}M(\beta,t)\,\mathrm{d}\beta.$$

By (S2) the left-hand side, and the first two terms on the right, are bounded by $\sup|D|\cdot 2C_Xt\le Ct$. Thus $\big|\int_{\alpha_1}^{\alpha_2}M(\beta,t)\,\mathrm{d}\beta\big|\le Ct$ for all α_1,α_2, for a.e. t.

Step 4: velocity of the centre of mass. By Lemma 6.2, $M(\cdot,t)\to M_0$ uniformly as $t\to 0$, with M_0 given by (5.2). Applying Step 3 along a sequence $t_k\to 0$ gives $\int_{\alpha_1}^{\alpha_2}M_0=0$ for all α_1,α_2, so $M_0\equiv 0$, i.e. $(I-2A\mathcal{K}_0)\overline{W}=2\mathcal{K}_0[n_{0,2}]$. Since U_0 solves the same equation (proof of Lemma 5.1) and $I-2A\mathcal{K}_0$ is injective on $C^{0,\gamma}(\mathbb{T})$ (Proposition 2.6), $\overline{W}=U_0$. This is (a): writing $\kappa=\frac12(W^+-W^-)$, the identity $(1-\rho_{\mathrm m})(W^+-U_0)+(1+\rho_{\mathrm m})(W^--U_0)=0$ gives $W^{\pm}=U_0\pm(1\pm\rho_{\mathrm m})\kappa$.

Finally, $\kappa\ge 0$. For some $\varepsilon>0$ the map $Y(\beta,s):=z_0(\beta)+s\,n_0(\beta)$ is a C^1 diffeomorphism from $\mathbb{R}\times(-\varepsilon,\varepsilon)$ onto a neighbourhood of $z_0(\mathbb{R})$, with $DY(\beta,0)^{-1}w=(w\cdot\tau_0/|\partial_\beta z_0|,\,w\cdot n_0)$. By (S1), $z^{\pm}(\cdot,t)\to z_0$ in C^1 and $z^{\pm}(\cdot,t)=z_0+t\,\partial_t z^{\pm}(\cdot,0)+o(t)$ uniformly. Hence, for small t, $(\beta^{\pm},s^{\pm})(\cdot,t):=Y^{-1}\circ z^{\pm}(\cdot,t)$ satisfy $\partial_\alpha\beta^{\pm}\to 1$, $\beta^{\pm}(\alpha,t)=\alpha+O(t)$ and $s^{\pm}(\alpha,t)=tW^{\pm}(\alpha)+o(t)$ uniformly. So $\beta^{\pm}(\cdot,t)$ is a diffeomorphism, $\Gamma^{\pm}(t)=\{Y(\beta,\eta^{\pm}(\beta,t))\}$ with $\eta^{\pm}(\cdot,t):=s^{\pm}(\cdot,t)\circ\beta^{\pm}(\cdot,t)^{-1}$, and, by the uniform continuity of $W^{\pm}$, $\eta^{\pm}(\cdot,t)=tW^{\pm}+o(t)$ uniformly: the tangential parts of $\partial_t z^{\pm}(\cdot,0)$ only reparametrise the curves. Near $z_0(\mathbb{R})$ the region $\Omega^+(z^-(\cdot,t))$ is $\{Y(\beta,s):s>\eta^-(\beta,t)\}$ (tubular neighbourhood argument, as for Lemma 4.1), so $\Gamma^+(t)\subset\Omega^+(z^-(\cdot,t))$ gives $\eta^+>\eta^-$. Dividing by t and letting $t\to 0$, $W^+\ge W^-$, i.e. $\kappa\ge 0$.

Step 5: the relaxation inequality. Fix α, and $t\in(0,T)$ such that (6.1) holds. The trace of D at $z^+(\alpha,t)$ from inside the strip lies in the closed disc centred at $a_S^+:=e_2+\frac{A}{\mu_{\mathrm m}}v_S^+(\alpha,t)$ with radius $|a_S^+|$, and by (6.1) its component along n^+ is $F^+/|\partial_\alpha z^+|$. Therefore

$$\Big(\frac{F^+}{|\partial_\alpha z^+|}-a_S^+\cdot n^+\Big)^2\le|a_S^+|^2.$$

Let $t=t_k\to 0$. By Lemma 6.2 and part (a), the computation of Lemma 5.1 applies with $W^{\pm}=U_0\pm(1\pm\rho_{\mathrm m})\kappa$, so $F^+/|\partial_\alpha z^+|\to-\sigma_0-2(1-A\rho_{\mathrm m})\kappa$, and $a_S^+\to a^{(0)}$, whose components are given by (5.4). Writing D_n, a_n, a_τ as in the proof of Lemma 5.2, the limit inequality reads $D_n(D_n-2a_n)-a_\tau^2\le 0$, and the computation there shows that the left-hand side equals $\frac{2(1-A^2)}{\mu_{\mathrm m}}\Phi(\kappa)$. Hence $\Phi(\kappa)\le 0$, which is (b). Since $\Phi(\kappa)>0$ for $\kappa>\kappa_{\max}$, we get $\kappa\le\kappa_{\max}$, and $\kappa_{\max}=0$ on $\Pi(z_0)$. □

Remark 6.3. The proof shows where the constraints come from, and it reads the mechanism of the construction of [LM26] as a necessity argument. Part (a) is a consequence of mass conservation in a strip of width $O(t)$ (Step 3): as for the potential of [LM26], whose transverse derivative involves $t^{-1}\partial_\lambda G$ [LM26, Sec. 3.3.4], the boundary fluxes must agree at leading order, and this forces the Muskat velocity of the centre of mass, as in the first order compatibility condition [LM26, Lemma 5.2]. Part (b) only uses the relaxation inequality on the *boundary* of the strip, where the normal component of the relaxed flux is fixed by the Rankine–Hugoniot conditions, as in [LM26, (4.12)]; Steps 2 and 5 use only the closed disc (see Section 8.3). The tangential component is free in (b), and minimising over it gives $\Phi(\kappa)\le 0$; the minimiser is the optimal tilt δ_* of Lemma 5.2, with which Theorem 1.1 realises, for $z_0\in\mathcal{Z}^{6,\gamma}$, every $\kappa\in C^{4,\gamma}(\mathbb{T})$ with $0<\kappa<\kappa_{\max}$.

The gap between Theorems 1.1 and 1.2 is larger than the boundary case $\Phi(\kappa)=0$. Theorem 1.1 requires $z_0\in\mathcal{Z}^{6,\gamma}$, $\kappa\in C^{4,\gamma}(\mathbb{T})$ and $\min\kappa>0$, whereas Theorem 1.2 applies to $z_0\in\mathcal{Z}^{3,\gamma}$ and yields $\kappa\in C^{0,\gamma}(\mathbb{T})$ with $\kappa\ge 0$. Openings $0\le\kappa<\kappa_{\max}$ that vanish at some points are not covered by Theorem 1.1, even if $\gamma_0\neq 0$ at these points, so that $\Phi(\kappa)<0$ holds everywhere; nor are openings or interfaces of lower regularity (see Section 8).

The proof of Theorem 1.2 uses Lemma A.2, hence the uniformity of the Plemelj–Privalov constants over the curves $z^{\pm}(\cdot,t)$ (point (a) of Appendix A.4), but not the sketched Lemmas 5.4 and 5.5.

7. Consequences: partially stable data, equal viscosities and viscous fingering

7.1. **Pinch-free data.**

Proof of Corollary 1.3. If $\Pi(z_0)=\emptyset$, then $\kappa_{\max}>0$ on $\mathbb{T}$ and $\kappa_{\max}\in C^{4,\gamma}(\mathbb{T})$ (end of Section 2), so $\kappa:=\theta\kappa_{\max}$, $\theta\in(0,1)$, satisfies (1.6), and Theorem 1.1 applies. Conversely, if a single-strip subsolution with opening $\kappa>0$ everywhere exists, Theorem 1.2 gives $\kappa\le\kappa_{\max}$, hence $\kappa_{\max}>0$ everywhere, i.e. $\Pi(z_0)=\emptyset$. On $\{\sigma_0\ge 0\}$ the condition $2(1-A\rho_{\mathrm{m}})\kappa<-\sigma_0$ of the untilted strip (Remark 5.3(a); for $\rho_{\mathrm{m}}=A$ it is the range stated in [LM26]) fails for every $\kappa>0$, so there the admissible opening comes entirely from the term $(1-A^2)\gamma_0^2/(8\mu_{\mathrm{m}})$ in Φ, i.e. from the vortex sheet. □

For $\rho_{\mathrm{m}}=A$ the existence half of Corollary 1.3 does not need the optimal tilt, and it is latent in [LM26]. With the tilt $\delta_{\mathrm{LM}}=-(1-A^2)\gamma_0/(2+A^2)$ of the single layer of [LM26] (Remarks 5.3(b) and 5.8), the admissible openings are those of (1.8), that is, $0<\kappa<\kappa_{\mathrm{LM}}$ with

$$\kappa_{\mathrm{LM}}:=\frac{-\sigma_0+\sqrt{\sigma_0^2+4\delta_{\mathrm{LM}}^2}}{4(1-A^2)},$$

the positive root of the quadratic in (1.8). It vanishes exactly on $\Pi(z_0)$. Hence, if $\Pi(z_0)=\emptyset$, then $\kappa_{\mathrm{LM}}\in C^{4,\gamma}(\mathbb{T})$ is positive, and Remark 5.7 applies with $\delta=\delta_{\mathrm{LM}}$ and $\kappa=\theta\kappa_{\mathrm{LM}}$, $\theta\in(0,1)$. In the asymptotically flat setting of [LM26] the same conclusion follows from their unmodified single layer, once the full modulus $|Au+e_2|$ is kept in the final estimate of [LM26, Sec. 7.4] and the opening satisfies $c-c_\infty\in H^{k-1}$ and $\inf c>0$ (Section 1.5); [LM26] do not state this. What Corollary 1.3 adds to [LM26] is the 'only if' half (Theorem 1.2); the sharp bound $\kappa_{\max}\ge\kappa_{\mathrm{LM}}$, whose range improves the coefficient of γ_0^2 in (1.8) by the factor $(2+A^2)^2/(4(1+A^2))\in[1,\frac{9}{8})$ (Remark 5.8); an arbitrary intermediate density ρ_{m}; and the periodic setting.

By (5.4) (for $\rho_{\mathrm{m}}=A$, cf. [LM26, (7.11), Prop. 7.4 and (3.7b)]), as $\kappa\to 0$ the relaxation disc $\{|D-a|<|a|\}$ degenerates only where $\sigma_0=\gamma_0=0$; this is the trace on the interface of the pinch singularity of [Men22] (Remark 3.3(b)). On the rest of $\Pi(z_0)$, where $\sigma_0>0$ and $\gamma_0=0$, the disc has radius at least $\frac{1}{2}\sigma_0$, but for every $\kappa>0$ the normal component $D_n=-\sigma_0-2(1-A\rho_{\mathrm{m}})\kappa$ of D forced at leading order by the Rankine–Hugoniot conditions lies outside the projection of the disc onto n_0: there $a^{(0)}\cdot\tau_0=0$ and $D_n-2a^{(0)}\cdot n_0=-2(1-A^2)\kappa/\mu_{\mathrm{m}}<0$ (proof of Lemma 5.2).

As in the asymptotically flat setting of [LM26], where $\sigma^{(0)}+2\in H^{k-1}$ [LM26, Prop. 6.3], so that $\sigma^{(0)}\to-2$ at infinity, our data, which have the heavy fluid on top ($\rho_0=\mathbf{1}_{\Omega^+(z_0)}-\mathbf{1}_{\Omega^-(z_0)}$), are never fully stable: $\sigma_0\ge 0$ on all of $\mathbb{T}$ is impossible, because $\int_{\mathbb{T}}\sigma_0\,\mathrm{d}s=-4\pi$ (Proposition 2.7). This identity is the periodic counterpart of the normalisation of [LM26, Prop. 6.3], and it also follows by integrating $\sigma_0=-2(n_{0,2}+AU_0)$, which is [LM26, (3.8)] in our normalisation, over a period, since $\int_{\mathbb{T}}U_0\,\mathrm{d}s=0$ (Lemma 2.5(b)). With the heavy fluid below, the integral would be $+4\pi$, and fully stable data exist, for instance the flat interface.

7.2. **Equal viscosities.**

Proof of Corollary 1.4. For $A=0$, Example 2.12 gives $\sigma_0=-2n_{0,2}$ and $\gamma_0=-2\tau_{0,2}$ (the formulas of [LM26, Secs. 1.2 and 3.1.3]), so $\Pi(z_0)=\{\tau_0=-e_1\}$ and, since $n_{0,2}^2+\tau_{0,2}^2=1$,

$$\kappa_{\max}=\frac{2n_{0,2}+\sqrt{4n_{0,2}^2+4\tau_{0,2}^2}}{4}=\frac{1+n_{0,2}}{2}.$$

The result follows from Corollary 1.3 and Theorem 1.1; the regularity of $\kappa_{\max}$ is clear. □

For $A = 0$ the untilted strip ($\delta = 0$) has the range $0 < \kappa < n_{0,2}$, which is also the range stated in [LM26] ($\bar{c} < \frac{1}{2}$, with $c = \kappa$) and only allows mixing where the interface is a graph. The single layer of [LM26] is, however, tilted: for $A = 0$ we have $\delta_{\mathrm{LM}} = \delta_* = \tau_{0,2}$, so that at leading order it is exactly the optimally tilted strip. Moreover, the range of Corollary 1.4 can be read off from [LM26] directly: for $A = 0$ we have $|Au + e_2| = 1$, and [LM26, (7.9)] gives $2|\nabla g| = |2\kappa - n_{0,2}| + O(t)$ in their mixing zone, so that at $t = 0$ their exact hull condition [LM26, (4.14)] reads $|2\kappa - n_{0,2}| < 1$, which for $\kappa > 0$ means $\kappa < \frac{1}{2}(1 + n_{0,2})$. Turning this into an existence statement in their setting requires the modification of [LM26, Sec. 7.4] described in Section 1.5.

The range $0 < \kappa < \frac{1}{2}(1+n_{0,2})$ coincides with $0 < \kappa < n_{0,2}$ at horizontal points with the heavy fluid on top ($n_{0,2} = 1$). There the total opening speed 2κ of a single strip is less than 2, as for the flat interface, whose single strip with intermediate density $A = 0$ has fronts approaching $\pm t$ [LM26, Sec. 3.2]; this is half of the maximal mixing zone $|x_2| < 2t$ of [Ott99, Sze12] (Example 3.6 with $\theta \to 1$). Elsewhere the range is strictly larger. On the overturned part ($n_{0,2} < 0$) the tangential component of gravity drives the heavy fluid along the interface, the interface carries a vortex sheet $\gamma_0 = -2\tau_{0,2} \neq 0$, and single strips of normal width up to $(1+n_{0,2})t$ are admissible. These are single-strip bounds: for graphs, the multilayers of [LM26, Sec. 7.5] give mixing zones of width $4\bar{c}\, n_{0,2} t$ for every $\bar{c} < 1$, which exceeds $(1 + n_{0,2})t$ for $\bar{c}$ close to 1 wherever $n_{0,2} > \frac{1}{3}$. Within single-strip subsolutions, Theorem 1.2 forces the linear opening rate to vanish at the points where the interface is horizontal with the heavy fluid below, and only there; if there is no such point, every opening $\kappa \in C^{4,\gamma}(\mathbb{T})$ with $0 < \kappa < \frac{1}{2}(1 + n_{0,2})$ is attained (Corollary 1.4). We prove neither that mixing is impossible near such points nor existence for data that have them.

For $A = 0$, see also [FS18] for stable non-horizontal data and [CFM22] for turned interfaces, where the mixing zone is opened slightly inside the stable regime and, reportedly, the horizontal stable points ('fully stable points') are identified as an obstruction to mixing; we have not checked this against the full text of [CFM22], and we have not compared Corollary 1.4 with their precise admissible ranges. The mixing zones of [CFM22] for bubbles are localised, and closed bubbles lie outside our horizontally periodic class. Heuristically, for $A = 0$ the obstruction of Theorem 1.2 also concerns them: a closed C^1 curve has a point where the normal pointing into the heavy fluid is $-e_2$, and there a single strip should not open at a positive linear rate. We have not proved a closed-curve analogue of Theorem 1.2, and for $A \neq 0$ the corresponding point need not belong to the analogue of $\Pi(z_0)$, since σ_0 and γ_0 are nonlocal. In any case this obstruction is pointwise: $\kappa_{\max} = \frac{1}{2}(1+n_{0,2}) > 0$ on the rest of the Rayleigh–Taylor stable part, so Theorem 1.2 by itself does not force a mixing zone to be localised.

7.3. **Viscous fingering driven by injection.** The original experiments of Saffman and Taylor [ST58] concern a less viscous fluid injected into a more viscous one, without gravity. Consider the problem with gravity constant $G \geq 0$ and a uniform velocity Ve_2 at infinity:

(7.1)
$$\partial_t \rho + \operatorname{div}(\rho u) = 0, \qquad \operatorname{div} u = 0, \qquad (1+A\rho)u = -\nabla p - G\rho\, e_2, \qquad u(x,t) \to Ve_2 \ \text{ as } |x_2| \to \infty.$$

A weak solution of (7.1) on $\mathbb{R}^2 \times (0,T')$ with initial datum ρ_0 is a pair $(\rho, u) \in L^\infty(\mathbb{R}^2 \times (0,T'))^{1+2}$ with $\rho^2 = 1$ a.e. which satisfies (3.1) with $\mathbb{T} \times \mathbb{R}$ replaced by $\mathbb{R}^2$ and $\rho\, \partial_1 \varphi$ replaced by $G\rho\, \partial_1 \varphi$, and such that $\operatorname{ess\,sup}_{|x_2|>L} |u(x,t) - Ve_2| \to 0$ as $L \to \infty$ for a.e. t. Darcy's law is thus imposed in curl form; since $\mathbb{R}^2$ is simply connected, a pressure can then be recovered distributionally, as in [LM26, Sec. 7.4].

Corollary 7.1. *Let $G \geq 0$, $A \in (-1,1)$ and $V \in \mathbb{R}$ with $G_{\mathrm{eff}} := G + AV > 0$. Let $z_0 \in \mathcal{Z}^{6,\gamma}$, $\rho_{\mathrm{m}} \in (-1,1)$ and $\kappa \in C^{4,\gamma}(\mathbb{T})$ with $0 < \kappa < \kappa_{\max}$, where $\kappa_{\max}$ and $\Pi(z_0)$ are computed from z_0 and A as in (1.5) (they do not depend on G and V). Let T, Z and $z^\pm$ be given by Theorem 1.1. Then there are infinitely many weak solutions of (7.1) on $\mathbb{R}^2 \times (0, T/G_{\mathrm{eff}})$ with initial datum $\mathbf{1}_{\Omega^+(z_0)} - \mathbf{1}_{\Omega^-(z_0)}$ which are pure outside, and mixing inside, the strip between the curves*

$$z^\pm\big(\cdot, G_{\mathrm{eff}}\, t\big) + Vt\, e_2.$$

In particular, for $G = 0$ and $AV > 0$ (the displacing fluid is the less viscous one), mixing solutions exist for all pinch-free data.

Proof. Let $(\tilde\rho, \tilde u)$ be one of the solutions given by Theorem 1.1, and set

$$\rho(x,t) := \tilde\rho\big(x - Vte_2,\, G_{\mathrm{eff}}t\big), \qquad u(x,t) := Ve_2 + G_{\mathrm{eff}}\,\tilde u\big(x - Vte_2,\, G_{\mathrm{eff}}t\big).$$

For $\varphi \in C_c^\infty(\mathbb{R}^2 \times [0, T/G_{\mathrm{eff}}))$ let $\psi(y,s) := \varphi\big(y + \frac{Vs}{G_{\mathrm{eff}}}e_2, \frac{s}{G_{\mathrm{eff}}}\big)$, so that $\psi \in C_c^\infty(\mathbb{R}^2 \times [0,T))$, $\psi(\cdot,0) = \varphi(\cdot,0)$, $G_{\mathrm{eff}}\,\partial_s\psi = \partial_t\varphi + V\partial_2\varphi$ and $\nabla_y\psi = \nabla_x\varphi$ at corresponding points. Since $\rho u = V\rho e_2 + G_{\mathrm{eff}}\tilde\rho\tilde u$, $(1+A\rho)u + G\rho e_2 = Ve_2 + G_{\mathrm{eff}}\big((1+A\tilde\rho)\tilde u + \tilde\rho e_2\big)$, and constant fields integrate to zero against derivatives of test functions, the change of variables $(y,s) = (x - Vte_2, G_{\mathrm{eff}}t)$, with $\mathrm{d}x\,\mathrm{d}t = G_{\mathrm{eff}}^{-1}\,\mathrm{d}y\,\mathrm{d}s$, turns the three identities of the weak formulation for (ρ,u) and φ into those of (3.1), on $\mathbb{R}^2$, for $(\tilde\rho,\tilde u)$ and ψ. Outside the mixing zone, $\tilde u$ coincides a.e. with $\bar u = v/(1 \pm A)$ (Theorem 3.5 and (4.11)), and $v \to \pm\frac{AQ}{4\pi}e_2$ as $x_2 \to \pm\infty$, uniformly in x_1 (Lemma 2.3(c) and the proof of Lemma 4.3), where $Q = 0$ because (4.12) holds (Lemmas 4.3 and 5.6). Hence $\tilde u \to 0$ and $u \to Ve_2$. Finally, the affine map $(x,t) \mapsto (x - Vte_2, G_{\mathrm{eff}}t)$ maps open sets to open sets and null sets to null sets, so the solutions are pure outside, and mixing inside, the preimage of the mixing zone of Theorem 1.1, which is the strip of the statement. □

The physical opening speed of the strip is $G_{\mathrm{eff}}\kappa$: $\kappa_{\max}$ and $\Pi(z_0)$ refer to the rescaled time $G_{\mathrm{eff}}t$. Corollary 7.1 transfers only existence statements (Theorem 1.1, the 'if' part of Corollary 1.3, and Corollary 1.4); we do not transfer Theorem 1.2, which would require replacing condition (S5) of Definition 6.1 by $\bar u \to Ve_2$ and checking that the change of variables preserves single-strip subsolutions. The hypothesis $G \geq 0$ is not used in the proof: $G + AV > 0$ suffices, which also covers a stabilising density contrast ($G < 0$) overcome by injection.

Thus injection enters only through the effective gravity $G_{\mathrm{eff}} = G + AV$: in the frame moving with the injection, and after rescaling time by G_{eff}, the problem is the one studied above. The condition $G + AV > 0$ is the classical Saffman–Taylor criterion for the flat interface [ST58] (see also [Hom87]): a flat interface translating with velocity Ve_2 has the Saffman–Taylor function $-2(G + AV)$, as given by $\sigma = -2(Gn_2 + AU)$, the analogue with gravity constant G of the identity $\sigma = -2(n_2 + AU)$ of Proposition 2.7, i.e. of [LM26, (3.8)]. The rescaling of time is the normalisation of the gravity constant in [LM26, Sec. 1.1]. For $G = 0$ the instability is purely viscous: in the laboratory frame the Saffman–Taylor function and the vortex-sheet strength of the data are $AV\sigma_0$ and $AV\gamma_0$, with σ_0 and γ_0 as in (1.4).

The same change of frame applies to any mixing theorem for the normalised problem (1.1), in particular to [LM26, Thm. 1.1] and to the multilayers of [LM26, Sec. 7.5]: for every fully unstable datum they give injection-driven mixing solutions, also for fluids of equal density ($G = 0$, $AV > 0$). Full instability does not depend on the frame, since the Saffman–Taylor function in the laboratory frame is $G_{\mathrm{eff}}\sigma_0$. What Corollary 7.1 adds is only the range of Theorem 1.1 and the pinch-free data of Corollary 1.3. For an optimal bound on the growth rate of the mixing zone in viscous fingering see [Ott97].

7.4. **An example.** Let $z_0(\alpha) = (\alpha - 1.5\sin\alpha,\ 0.8\sin\alpha)$. Then $\partial_\alpha z_0 = (1 - 1.5\cos\alpha,\ 0.8\cos\alpha) \neq 0$, and z_0 is injective modulo $2\pi e_1$: if $z_{0,2}(\alpha) = z_{0,2}(\beta)$ with $\beta \not\equiv \alpha$ then $\beta \equiv \pi - \alpha \pmod{2\pi}$, and $z_{0,1}(\alpha) - z_{0,1}(\pi - \alpha) = 2\alpha - \pi$ lies in $2\pi\mathbb{Z}$ only if $\alpha \equiv \frac{\pi}{2} \pmod{\pi}$, in which case $\beta \equiv \alpha$. Hence $z_0 \in \mathcal{Z}^{k,\gamma}$ for every k. The interface is overturned: $n_{0,2} < 0$ for $\cos\alpha > 2/3$. The tangent is horizontal only where $\cos\alpha = 0$, and there $\tau_0 = e_1$. By Corollary 1.4, for $A = 0$ the data are pinch-free although Rayleigh–Taylor stable on the set $\{\cos\alpha \geq 2/3\}$, which makes up 26.8% of the parameter interval and 15.4% of the arclength of one period, and mixing solutions exist with any opening $\kappa \in C^{4,\gamma}(\mathbb{T})$ with $0 < \kappa < \frac{1}{2}(1 + n_{0,2})$; the minimum of $\frac{1}{2}(1 + n_{0,2})$ is 0.2350. On the stable set the range stated in [LM26] is empty, whereas for $A = 0$ the range (1.8) of the single layer of [LM26] with its exact hull condition is that of Corollary 1.4.

For $A \neq 0$ we computed σ_0 and γ_0 numerically, solving the second kind equations (1.4) with the trapezoidal rule on 512 nodes (the kernel of $\mathcal{K}_0$ is smooth). As a consistency check of this discretisation, identity (2.8) (the left relation of [LM26, (3.7b)] in our normalisation), with $\mathcal{T}_0$ evaluated by the alternating-point rule, holds with maximal absolute residual $\leq 2 \cdot 10^{-13}$. For

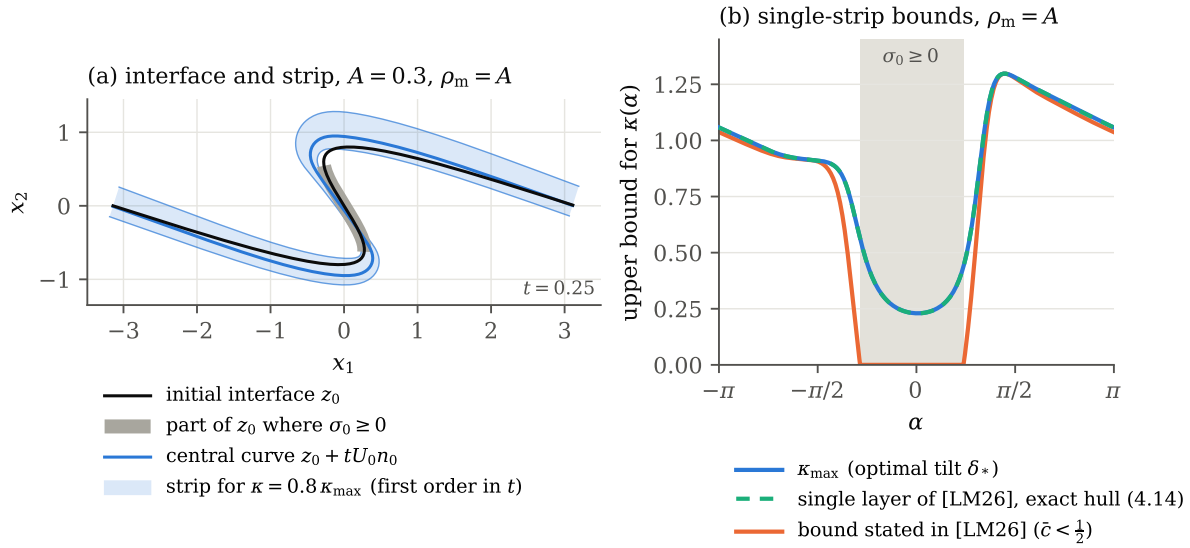


FIGURE 1. Overturned interface $z_0(\alpha) = (\alpha - 1.5\sin\alpha, 0.8\sin\alpha)$ with $A = 0.3$ and $\rho_m = A$. (a) The initial interface z_0, with its Saffman–Taylor stable part $\{\sigma_0 \geq 0\}$ marked in grey, and the strip between $z_0 + t(U_0 \pm (1 \pm \rho_m)\kappa)n_0$ for $\kappa = 0.8\,\kappa_{\max}$, i.e. the first order part in t of the mixing zone of Theorem 1.1, with its central curve $z_0 + tU_0 n_0$. The strip extends along the whole moving interface, including the part where $\sigma_0 \geq 0$; since it is centred on the moving central curve, it does not contain all of z_0. The time $t = 0.25$ is illustrative: we have not checked that it is smaller than the existence time T of Theorem 1.1. (b) Single-strip upper bounds for the opening function, as functions of α: $\kappa_{\max}$ of Theorem 1.1, obtained with the optimal tilt δ_*; the bound κ_{LM} of the single layer of [LM26] with its exact hull condition [LM26, (4.14)], i.e. of the range (1.8) (dashed), which differs from $\kappa_{\max}$ by less than 0.2% for $A = 0.3$; and the bound $\max(-\sigma_0, 0)/(2(1 - A^2))$ stated in [LM26] ($\bar{c} < \frac{1}{2}$), which is also that of the untilted strip $\delta = 0$ and vanishes on the shaded set $\{\sigma_0 \geq 0\}$.

$A = 0.3$, -0.6 and 0.9 we find $\sigma_0 \geq 0$ on about 26%, 24% and 16% of the parameter interval (15%, 14% and 9% of the arclength), with $\min|\gamma_0| = 1.74$, 1.88 and 2.33 on that set. Thus the data are pinch-free and partially Saffman–Taylor stable; for $A \neq 0$ this is a numerical observation without validated error bounds (the values above do not change, to the digits given, with 256 or 1024 nodes). With $\rho_m = A$, the minimum of $\kappa_{\max}$ over $\mathbb{T}$ is 0.2297, 0.2265 and 0.2959, against 0.2294, 0.2224 and 0.2778 for the range (1.8) of the single layer of [LM26] with its exact hull condition, while the range stated in [LM26] is empty on the stable set. The existence conclusions rely on Theorem 1.1, whose proof is complete modulo the sketches of Appendix A.4. Figure 1 shows the case $A = 0.3$. The scripts are in the `checks/` folder of the source repository.

8. Remarks and open problems

8.1. **Continuous profiles with tilt (formal).** Single strips do not exhaust the relaxation: the multilayer subsolutions of [LM26, Sec. 7.5], whose mixing zone contains several intermediate densities, obtain larger opening rates [LM26, Sec. 7], for fully unstable data and wherever $|\gamma_0|$ is small (Section 1.5); as [LM26, Sec. 7.5] note, they are analogous to [CFM22, Sec. 6.2] and [FS18, Sec. 5], where $A = 0$. Consider a mixing zone of width $O(t)$ around Z in which the density has a continuous profile $\bar{\rho} = R(\alpha, \xi)$, $\xi = (x - Z)\cdot n_0/t$, increasing in ξ from -1 at $\xi_-(\alpha)$ to 1 at $\xi_+(\alpha)$ and normalised so that Z is the centre of mass. Repeating the computations of Section 5 at leading order, under the assumptions that Z moves with the Muskat velocity U_0 and that the traces outside the zone are those of the Muskat problem,

gives, with $\mathcal{P}(\xi) := -\int_{\xi_-}^{\xi} \eta\, \partial_\eta R \,\mathrm{d}\eta \geq 0$,

$$D \cdot n_0 = -\sigma_0 - \frac{2(1-AR)\,\mathcal{P}}{1-R^2}, \qquad a \cdot n_0 = -\frac{\sigma_0}{2} - \frac{A^2\mathcal{P}}{1+AR}, \qquad a \cdot \tau_0 = -\frac{(1-A^2)\gamma_0}{2(1+AR)},$$

where for the last identity we used that, at leading order, the tangential component of the Darcy field is $v \cdot \tau_0 = -\mathcal{T}_0\sigma_0 - \tau_{0,2}R$ across the zone, together with (2.8), that is, [LM26, (3.7b)]. Without tilt ($D \cdot \tau_0 = 0$), or with the reflected tilt $D \cdot \tau_0 = 2a \cdot \tau_0$, which gives the same value of $D \cdot (D - 2a)$, the condition $D \cdot (D - 2a) < 0$ is $(1 - AR)\mathcal{P} < -\frac{\sigma_0}{2}(1 - R^2)$. Equality gives $\mathcal{P} = -\frac{\sigma_0}{2}F_1(R)$, with $F_1(\rho) = \frac{1-\rho^2}{1-A\rho}$ as in Example 3.6, hence $\xi = \frac{\sigma_0}{2}F_1'(R)$; since $F_1'(\pm 1) = \mp\frac{2}{1\mp A}$, this extremal profile has fronts at $\xi_+ = -\sigma_0/(1-A)$ and $\xi_- = \sigma_0/(1+A)$. It lies on the boundary circle of the hull and is not itself admissible. Its fronts are the supremum of the opening speeds of the multilayer subsolutions of [LM26, Sec. 7.5]: for each N (with $2N$ interfaces) the outer boundaries of their mixing zone move away from the central curve with the normal speeds $c/(1 \mp A)$, where $c = -\bar{c}\,\sigma_0$ and $\bar{c} < 1 - \frac{1}{2N}$ [LM26, (7.13)–(7.15)], so these speeds are approached as $\bar{c} \to 1$ (hence $N \to \infty$) but never attained. Luna-Velasco and Mengual also identify the profile: they interpret [LM26, (7.15)] as the Rankine–Hugoniot discretisation of a frozen scalar flux proportional to F_1, and they state that, as $N \to \infty$, the step profiles converge to the viscosity-dependent Otto profile with time scaled by $\bar{c}$ [LM26, end of Sec. 7.5]. The extremal profile above is that profile at $\bar{c} = 1$; its flux F_1 is that of the flat profile of [Men22, Sec. 2], and for the flat interface the profiles with $\bar{c} < 1$ are those of Example 3.6 with $\theta = \bar{c}$. The inequality without tilt is a continuum version of the per-layer hull condition [LM26, (7.18)–(7.19)]. In fact the strips of the multilayers of [LM26, Sec. 7.5] carry the reflected tilt (unlike the single layer of [LM26, Secs. 4–7.4], whose tilt is δ_{LM}; Remark 5.8): by [LM26, (7.18)], $m - \theta_j u$ is normal to the interface at leading order in the strip of density θ_j, and since $D - 2a = \frac{2(1-A^2)}{(1-\rho^2)\mu}(m - \rho u)$, this means $D \cdot \tau_0 = 2a \cdot \tau_0$. This is why the margin in [LM26, (7.19)] contains no term in γ_0.

With the optimal tilt $D \cdot \tau_0 = a \cdot \tau_0$ the condition becomes $(D \cdot n_0 - a \cdot n_0)^2 < |a|^2$ pointwise in the zone, which enlarges the admissible profiles wherever $\gamma_0 \neq 0$. By the computation of Lemma 5.2, with $\rho_{\mathrm{m}} = R$ and $\kappa = \mathcal{P}/(1 - R^2)$, it is the single-strip condition $\Phi(\kappa) < 0$ applied pointwise. Compared with [LM26], only this tilt is new in this subsection, and it is formal. We expect that multilayer subsolutions with a tilt in every layer approximate these larger profiles as the number of layers tends to infinity, by combining the tilt of Section 5 with the multilayer construction of [LM26, Sec. 7.5 and Lemma 8.8]. Each layer carries its own potential [LM26, (7.17) and (8.38)], so one tilt per layer is available. Given the Darcy field, each internal interface imposes a single Rankine–Hugoniot relation, so the tilts enter only the total flux mismatch, and, as in [LM26, Lemma 8.8], one acceleration per order should suffice.

8.2. **The role of the vortex sheet.** Lemma 5.2, together with Theorems 1.1 and 1.2, shows that within single strips the admissible opening is governed by $|a|$, whose components at $t = 0$ are $-\frac{1}{2}\sigma_0 - \frac{A^2(1-\rho_{\mathrm{m}}^2)\kappa}{\mu_{\mathrm{m}}}$ and $-\frac{(1-A^2)\gamma_0}{2\mu_{\mathrm{m}}}$: the vortex sheet enters through the relaxation condition, not as an additional assumption. This is not a mechanism absent from [LM26]. Their exact hull condition [LM26, (4.14)] involves the full modulus $|Au + e_2|$, whose tangential part carries γ_0, and their single layer is already tilted (Remark 5.8); the vortex-sheet contribution is discarded only in their final estimate [LM26, Sec. 7.4], so it is latent in their construction. The qualitative statement that a pinch singularity in the relaxation prevents the two fluids from mixing wherever there is neither Rayleigh–Taylor nor vorticity at the interface is due to Mengual [Men22] (see Remark 3.3(b) for its relation with $\Pi(z_0)$). What we add is the explicit range for $A \neq 0$, its sharpness within single strips, and the small gain of the optimal tilt; for $A = 0$, see [FS18] for stable non-horizontal data and [CFM22] for turned interfaces (we have not compared our $A = 0$ statements with their precise admissible ranges).

The analogy with the vortex-sheet driven mixing zones for the Euler equations [MS23] is only partial: vortex sheets in Darcy flow have no Kelvin–Helmholtz instability, and where $\sigma_0 > 0$ the interface is Saffman–Taylor stable, so that mixing zones there reflect the non-uniqueness

of weak solutions rather than an instability. For $A = 0$ the 'vortex sheet' is tangential gravity, $\gamma_0 = -2\tau_{0,2}$ and $|Au + e_2| = 1$: on a Rayleigh–Taylor stable but inclined part of the interface the heavy fluid slides down along the interface, and the relaxation allows this shear to produce a mixing zone of positive width; calling this vorticity-assisted is a choice of name. For $A \neq 0$, $\gamma_0 = -2(I + 2A\mathcal{K}_0)^{-1}[\tau_{0,2}]$ is nonlocal, and this physical reading is only heuristic. Whether such mixing zones are selected by a physical criterion, for instance Otto's relaxation [Ott99] or maximal dissipation, is an interesting question; for $A = 0$ and the flat interface the maximal mixing rate is selected [Ott99, Sze12], and for $|A| < 1$ the flat construction of [Men22] preserves the connection with Otto's relaxation.

8.3. **The target set.** In the proof of Theorem 1.1 we only used that $\mathcal{U}_A$ consists of laminates (Proposition 3.2). By Section 3, $\mathcal{U}_A$ is the set O of [Men22], written (O) in [LM26]: with the function $g(\rho, u, m) := (1 - A\rho)|m - \rho u|^2 + (1 - \rho^2)(m - \rho u) \cdot (Au + e_2)$ of [LM26, (A.1)] (see [Men22, Lemmas 4.1–4.2], and [Sze12, Prop. 2.3] for $A = 0$), a direct computation gives $D \cdot (D - 2a) = \frac{4(1-A^2)}{(1-\rho^2)^2(1+A\rho)}\, g$ for $|\rho| < 1$. The closure $\overline{O}$ is the lamination convex hull K^{lc} [Men22], and, by [LM26, Thm. 2.4], it is also the Λ-convex hull K^{Λ} and the weak relaxation of (1.1) in the sense of [LM26, Sec. 2.1]. The proof of Theorem 1.2 uses only the closed disc $|D - a| \leq |a|$ (Steps 2 and 5), which is the slice $\{\rho = \rho_{\mathrm{m}}\}$ of $\overline{O}$. Hence Theorem 1.2 holds verbatim for single-strip subsolutions with values in $\overline{O}$ instead of $\mathcal{U}_A$. Moreover, weak limits of the triples $(\rho, u, \rho u)$ of solutions take values in $\overline{O}$ [LM26, Lemma A.4 and Thm. 2.4], so no larger target set is compatible with weak limits of solutions, and the range (1.6) cannot be enlarged by changing the target set. This concerns single strips of constant intermediate density only: the multilayers of [LM26, Sec. 7.5] and, formally, the continuous profiles of Section 8.1 achieve larger openings by changing the structure of the subsolution, not the target set.

8.4. **Open problems.**

(1) *Pinch points.* When $\Pi(z_0) \neq \emptyset$, Theorem 1.2 forces $\kappa = 0$ on $\Pi(z_0)$: there a single strip cannot open at a positive linear rate. So no single strip with everywhere positive linear opening exists; one must let the opening vanish on $\Pi(z_0)$, or use mixing zones that do not surround the whole interface, such as the localised zones of [CFM22] for $A = 0$. Extending Theorem 1.1 to opening functions vanishing on $\Pi(z_0)$ requires weighted versions of the estimates of Lemma 4.5, since the margin in (5.5) degenerates there.

(2) *Asymptotically flat interfaces.* In our periodic setting the strip carries a global constraint, $\int_{\mathbb{T}} M = 0$ (Lemma 4.3), which we meet with the vertical correction h; in exchange, the control of the stream function reduces to sup-norm bounds on M (Lemma 4.5). In the asymptotically flat setting of [LM26] there is no such constraint: the boundary values of the potential are primitives normalised at $\alpha = 0$ [LM26, (4.13)], and the transverse derivative is controlled by uniform bounds on $\mathbb{R}$ for the primitive of $\partial_\alpha G_+ - \partial_\alpha G_-$ [LM26, Prop. 7.2], which rest on primitive estimates for the pressure terms [LM26, Lemmas 8.2 and 8.6] and on an L^1 bound for the geometric term. As far as we can see, transferring the tilt to $\mathbb{R}$ would require (a) a nonzero target, involving $\partial_\alpha(\kappa\delta)$, in the second order equations [LM26, (5.15)–(5.16)], to be solved in the appropriate spaces $H^j(\mathbb{R})$ by [LM26, Prop. 6.1]; (b) a relative normalisation of the two boundary values of the potential, playing the role of our C_Ψ, in place of $G_\pm(t, 0) = 0$; and (c) redoing [LM26, Prop. 7.2] for the $O(t^2)$ remainder. We have not carried this out.

(3) *The latent range and tilted multilayers in the setting of* [LM26]. For asymptotically flat interfaces it remains to write out the modification of [LM26, Sec. 7.4] that turns the range (1.8) of their unchanged single layer into an existence statement, with an opening c such that $c - c_\infty \in H^{k-1}$ and $\inf c > 0$ [LM26, Sec. 4.1 and (5.1)]; to transfer the optimal tilt as in (2); and to construct the tilted multilayers of Section 8.1, by combining these with [LM26, Sec. 7.5 and Lemma 8.8].

(4) *Longer times and lower regularity.* The time expansion only gives a short existence time, and we assumed $z_0 \in C^{6,\gamma}$, far from the H^5 of [LM26] and the $C^{3,\alpha}$ of [FS18] for $A = 0$.

Appendix A. Periodic Cauchy integrals

It is convenient to work with complex densities. For $z \in \mathcal{Z}^{1,\gamma}$ and $G \in C^{0,\gamma}(\mathbb{T};\mathbb{C})$ set

$$\mathfrak{c}[z;G](x) := \frac{1}{4\pi i}\int_{\mathbb{T}} G(\beta)\,\partial_\beta z(\beta)\cot\frac{z(\beta)-x}{2}\,\mathrm{d}\beta, \qquad x \in \mathbb{C}\setminus\Gamma,$$

so that $\mathcal{C}[z;f]^\flat = \mathfrak{c}[z;f^\flat]$.

A.1. **Reduction to closed curves.** With $w = e^{ix}$ and $\omega = e^{i\zeta}$, a direct computation gives

$$\tfrac{1}{2}\cot\tfrac{\zeta-x}{2}\,\mathrm{d}\zeta = \frac{\mathrm{d}\omega}{\omega-w} - \frac{i}{2}\,\mathrm{d}\zeta.$$

The map $x \mapsto e^{ix}$ sends the cylinder $\mathbb{T}\times\mathbb{R}$ to $\mathbb{C}\setminus\{0\}$, Γ to a closed Jordan curve γ of class $C^{k,\gamma}$ winding once around the origin, $\Omega^+(z)$ to the punctured interior of γ and $\Omega^-(z)$ to its exterior. Hence $\mathfrak{c}[z;G]$ is the classical Cauchy integral $\frac{1}{2\pi i}\int_\gamma \frac{\tilde G(\omega)\,\mathrm{d}\omega}{\omega-w}$ of the transported density $\tilde G$, minus the constant $\frac{1}{4\pi}\int_{\mathbb{T}} G\partial_\beta z\,\mathrm{d}\beta$. The classical Plemelj–Privalov theorem for closed $C^{1,\gamma}$ curves [Mus53, Chapter 2], [Kre14, Chapter 7] then gives Lemma 2.3(b): $\mathfrak{c}[z;G]$ extends to a $C^{0,\gamma}$ function on each closed region, with norm bounded by $C\|G\|_{C^{0,\gamma}}$, and its boundary values are given by the Plemelj formulas $\mathfrak{c}^\pm = \pm\frac{1}{2}G + \mathrm{p.v.}\,\mathfrak{c}$. The value at $w=0$ corresponds to the limit $x_2 \to +\infty$. Two points are not covered by this reduction, and we do not prove them here: the dependence of C only on γ, $\delta(z)$ and $\|z\|_{\mathcal{Z}^{1,\gamma}}$ (the cited references treat a fixed curve), and the Hölder bound on the unbounded region $\Omega^-(z)$, on which the map $x \mapsto e^{ix}$ is not uniformly Lipschitz (see the end of Section A.4).

A.2. **Derivative identities.**

Lemma A.1. *Let $z \in \mathcal{Z}^{2,\gamma}$ and $G \in C^{1,\gamma}(\mathbb{T};\mathbb{C})$. Then, in $\mathbb{C}\setminus\Gamma$,*

$$\partial_x\,\mathfrak{c}[z;G] = \mathfrak{c}\Big[z;\frac{\partial_\beta G}{\partial_\beta z}\Big]. \tag{A.1}$$

If $z = z(\beta,t)$ and $G = G(\beta,t)$ are C^1 in t with values in $\mathcal{Z}^{2,\gamma}$ and $C^{1,\gamma}$, then

$$\partial_t\,\mathfrak{c}[z;G](x) = \mathfrak{c}\Big[z;\partial_t G - \frac{\partial_\beta G}{\partial_\beta z}\,\partial_t z\Big](x), \qquad x \notin \Gamma(t). \tag{A.2}$$

Proof. Since $\partial_\beta\cot\frac{z(\beta)-x}{2} = -\frac{1}{2}\csc^2\frac{z(\beta)-x}{2}\,\partial_\beta z$, we have $\partial_x\cot\frac{z(\beta)-x}{2} = -\partial_\beta\cot\frac{z(\beta)-x}{2}/\partial_\beta z$ and $\partial_t\cot\frac{z(\beta)-x}{2} = \partial_\beta\cot\frac{z(\beta)-x}{2}\cdot\partial_t z/\partial_\beta z$. Integrating by parts in β (there are no boundary terms by periodicity),

$$\partial_x\mathfrak{c} = -\frac{1}{4\pi i}\int G\,\partial_\beta\cot\,\mathrm{d}\beta = \frac{1}{4\pi i}\int \partial_\beta G\cot\,\mathrm{d}\beta,$$

which is (A.1), and

$$\partial_t\mathfrak{c} = \frac{1}{4\pi i}\int\Big[(\partial_t G\,\partial_\beta z + G\,\partial_\beta\partial_t z)\cot - \partial_\beta(G\partial_t z)\cot\Big]\,\mathrm{d}\beta = \frac{1}{4\pi i}\int\big(\partial_t G\,\partial_\beta z - \partial_\beta G\,\partial_t z\big)\cot\,\mathrm{d}\beta,$$

which is (A.2). □

Iterating (A.1) and using the Plemelj–Privalov theorem gives Lemma 2.3(d): if $z \in \mathcal{Z}^{j+1,\gamma}$ and $G \in C^{j,\gamma}$ then all derivatives of order $\le j$ of $\mathfrak{c}[z;G]$ are Cauchy integrals of $C^{0,\gamma}$ densities, hence $C^{0,\gamma}$ up to Γ. If $x = x(t)$ is a C^1 path in $\Omega^\pm(z(t))$, then since $\mathfrak{c}$ is holomorphic,

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\mathfrak{c}[z;G]\big(x(t)\big) = \mathfrak{c}\Big[z;\partial_t G - \frac{\partial_\beta G}{\partial_\beta z}\partial_t z\Big](x) + \mathfrak{c}\Big[z;\frac{\partial_\beta G}{\partial_\beta z}\Big](x)\,\dot x, \tag{A.3}$$

and we use the same formula for the boundary values $t \mapsto \mathfrak{c}[z;G]^\pm(\alpha)$ with $x(t) = z(\alpha,t)$. This boundary-value version on moving curves requires the derivatives to converge up to $\Gamma(t)$ uniformly in t, which we do not prove here. The identities (A.1)–(A.3) are the periodic versions of the differentiation identities [LM26, (5.11), (8.5), (8.24)–(8.25)] on $\mathbb{R}$.

A.3. **Continuity.**

Lemma A.2. *Let $z_k \to z$ in $\mathcal{Z}^{1,\gamma}$ and $G_k \to G$ in $C^{0,\gamma}(\mathbb{T};\mathbb{C})$. Then $\mathfrak{c}[z_k;G_k]^\pm \to \mathfrak{c}[z;G]^\pm$ uniformly on $\mathbb{T}$, and if $x_k \in \overline{\Omega^+(z_k)}$ with $x_k \to z(\alpha)$, then $\mathfrak{c}[z_k;G_k](x_k) \to \mathfrak{c}[z;G]^+(\alpha)$; similarly from below.*

Proof. By the Plemelj–Privalov theorem the functions $\mathfrak{c}_k := \mathfrak{c}[z_k;G_k]$ are uniformly $C^{0,\gamma}$ on $\overline{\Omega^+(z_k)}$, with a constant L independent of k (this uses the uniformity of the constant in Lemma 2.3(b) over the family $\{z_k\}$, see Section A.1). Given $\varepsilon > 0$ let $y = z(\alpha) + \varepsilon n(\alpha)$; for k large $y \in \Omega^+(z_k)$ at distance $\geq \varepsilon/2$ from Γ_k, so $\mathfrak{c}_k(y) \to \mathfrak{c}[z;G](y)$ by dominated convergence. Hence

$$\limsup_k |\mathfrak{c}_k(x_k) - \mathfrak{c}[z;G]^+(\alpha)| \leq \limsup_k \left(L|x_k - y|^\gamma + L\varepsilon^\gamma\right) \leq 2L\varepsilon^\gamma,$$

and ε is arbitrary. Uniformity in α follows from the same argument and the equicontinuity. □

A.4. **Sketch of the proof of Lemma 5.4 and of the regularity of G in Lemma 5.5.** We sketch the argument; the points that are not proved in detail are listed at the end of this section. By Lemma 4.1, the curves $z^\pm(\cdot,t)$ are polynomials in t with coefficients in $C^{3,\gamma}$, with $\delta(z^\pm) \geq \frac{1}{2}\delta(z_0)$ on $[0,T_1]$. The normals $n^\pm$, the lengths $|\partial_\alpha z^\pm|$ and the velocities $V^\pm = \partial_t z^\pm \cdot n^\pm$ are therefore smooth in t with values in $C^{2,\gamma}$, and so are the complex densities $G^\pm := (f^\pm)^\flat$ and their time derivatives.

By (4.5), $(v_S^+)^\flat = \mathfrak{c}[z^+;G^+]^- + \mathfrak{c}[z^-;G^-] \circ z^+$, where $z^+(\alpha,t) \in \Omega^+(z^-(\cdot,t))$ for $t > 0$, and similarly for v_S^-. Applying (A.3) once, $\partial_t v_S^\pm$ is a sum of boundary values and evaluations on the other curve of Cauchy integrals with densities

$$G_1^\pm = \partial_t G^\pm - \frac{\partial_\beta G^\pm}{\partial_\beta z^\pm}\partial_t z^\pm, \qquad G_2^\pm = \frac{\partial_\beta G^\pm}{\partial_\beta z^\pm},$$

multiplied by $\partial_t z^\pm$; these densities are bounded in $C^{1,\gamma}$ uniformly in t. Applying (A.3) once more, $\partial_t^2 v_S^\pm$ involves Cauchy integrals of densities bounded in $C^{0,\gamma}$, evaluated on $\Gamma^\pm(t)$ from the side of the strip, and these are bounded uniformly in $t \in (0,T_1]$ by the Plemelj–Privalov theorem (with constants uniform in t, point (a) below). Since $F^\pm$ are polynomials in $v_S^\pm \cdot n^\pm$, $V^\pm$, $n_2^\pm$ and $|\partial_\alpha z^\pm|$, we obtain that $\partial_t F^\pm$ and $\partial_t^2 F^\pm$ are bounded in $C^0(\mathbb{T})$ on $(0,T_1]$. By Lemma A.2 applied to the densities appearing in $F^\pm$ and $\partial_t F^\pm$, these functions extend continuously to $t = 0$, which gives the first part of Lemma 5.4.

For (5.6), let $x = X(\alpha,\lambda,t)$, so that $|x - z^\pm(\alpha,t)| \leq 2t\kappa(\alpha)$. The uniform Hölder bounds on $\overline{\Omega^-(z^+)}$ and $\overline{\Omega^+(z^-)}$ (points (a) and (c) below) give $|v(x,t) - v_S^+(\alpha,t)| \leq Ct^\gamma$, and the Lipschitz bound in time just proved gives $|v_S^+(\alpha,t) - v_S^{(0)}(\alpha)| \leq Ct$.

Finally, $G = \partial_t M(\cdot,0)$ for $a = 0$, $h = 0$ is obtained by letting $t \to 0$ in the formula for $\partial_t M$. In the limit all Cauchy integrals become boundary values on $z_0 \in \mathcal{Z}^{6,\gamma}$, and the densities involve $n_0 \in C^{5,\gamma}$, $\partial_t z^\pm(\cdot,0) = (U_0 \pm c^\pm)n_0 \in C^{4,\gamma}$, their first α-derivatives, and $\partial_t^2 z^\pm(\cdot,0) = 0$. Thus $G^\pm(\cdot,0) \in C^{4,\gamma}$, $G_1^\pm(\cdot,0), G_2^\pm(\cdot,0) \in C^{3,\gamma}$, and by Lemma 2.3(d) the boundary values are in $C^{3,\gamma}$. The remaining factors ($\partial_t z^\pm(\cdot,0)$, n_0, and the time derivatives at $t = 0$ of $n^\pm$, $V^\pm$ and $|\partial_\alpha z^\pm|$, which involve $\partial_\alpha \partial_t z^\pm(\cdot,0)$) belong to $C^{3,\gamma}$. Hence $G \in C^{3,\gamma}(\mathbb{T})$.

What is missing. The argument above is a sketch. It uses three facts that we do not prove: (a) the Plemelj–Privalov bounds of Lemma 2.3(b),(d) with constants that are uniform over the families of curves $z^\pm(\cdot,t)$, $t \in [0,T_1]$ (the classical references treat a fixed curve), which are also used in Lemma A.2; (b) the boundary-value version of (A.3) on the moving curves $z^\pm(\cdot,t)$, up to $t = 0$, where the two curves merge (the same point underlies the differentiability of $\mathcal{L}_t$ and $\mathcal{B}_t$ at $t = 0$ in the proof of Lemma 5.5); (c) Hölder bounds on the unbounded lower region $\Omega^-(z^+(\cdot,t))$, used for (5.6), on which the map $x \mapsto e^{ix}$ of Section A.1 is not uniformly Lipschitz. Fully proved analogues of these estimates in the asymptotically flat setting, with L^∞ bounds for Sobolev densities, are [LM26, Lemmas 8.4–8.5].

References

[Amb04] D. M. Ambrose, *Well-posedness of two-phase Hele-Shaw flow without surface tension*, European J. Appl. Math. **15** (2004), no. 5, 597–607.

[ACF21] V. Arnaiz, Á. Castro, D. Faraco, *Semiclassical estimates for pseudodifferential operators and the Muskat problem in the unstable regime*, Comm. Partial Differential Equations **46** (2021), no. 1, 135–164.

[CCF21] Á. Castro, D. Córdoba, D. Faraco, *Mixing solutions for the Muskat problem*, Invent. Math. **226** (2021), 251–348.

[CFM19] Á. Castro, D. Faraco, F. Mengual, *Degraded mixing solutions for the Muskat problem*, Calc. Var. Partial Differential Equations **58** (2019), Paper No. 58.

[CFM22] Á. Castro, D. Faraco, F. Mengual, *Localized mixing zone for Muskat bubbles and turned interfaces*, Ann. PDE **8** (2022).

[CCG11] A. Córdoba, D. Córdoba, F. Gancedo, *Interface evolution: the Hele-Shaw and Muskat problems*, Ann. of Math. (2) **173** (2011), no. 1, 477–542.

[CFG11] D. Córdoba, D. Faraco, F. Gancedo, *Lack of uniqueness for weak solutions of the incompressible porous media equation*, Arch. Ration. Mech. Anal. **200** (2011), no. 3, 725–746.

[CG07] D. Córdoba, F. Gancedo, *Contour dynamics of incompressible 3-D fluids in a porous medium with different densities*, Comm. Math. Phys. **273** (2007), no. 2, 445–471.

[DLS09] C. De Lellis, L. Székelyhidi Jr., *The Euler equations as a differential inclusion*, Ann. of Math. (2) **170** (2009), no. 3, 1417–1436.

[FS18] C. Förster, L. Székelyhidi Jr., *Piecewise constant subsolutions for the Muskat problem*, Comm. Math. Phys. **363** (2018), no. 3, 1051–1080.

[Gan17] F. Gancedo, *A survey for the Muskat problem and a new estimate*, SeMA J. **74** (2017), no. 1, 21–35.

[GBL20] R. Granero-Belinchón, O. Lazar, *Growth in the Muskat problem*, Math. Model. Nat. Phenom. **15** (2020).

[GKS21] B. Gebhard, J. J. Kolumbán, L. Székelyhidi Jr., *A new approach to the Rayleigh–Taylor instability*, Arch. Ration. Mech. Anal. **241** (2021), no. 3, 1243–1280.

[Hom87] G. M. Homsy, *Viscous fingering in porous media*, Annu. Rev. Fluid Mech. **19** (1987), 271–311.

[KMS03] B. Kirchheim, S. Müller, V. Šverák, *Studying nonlinear PDE by geometry in matrix space*, in: Geometric Analysis and Nonlinear Partial Differential Equations, Springer, Berlin, 2003, 347–395.

[Kov63] E. J. Koval, *A method for predicting the performance of unstable miscible displacement in heterogeneous media*, Soc. Petrol. Eng. J. **3** (1963), no. 2, 145–154.

[Kre14] R. Kress, *Linear Integral Equations*, 3rd ed., Applied Mathematical Sciences **82**, Springer, New York, 2014.

[LM26] P. Luna-Velasco, F. Mengual, *Mixing zone dynamics for the Saffman–Taylor instability with viscosity jump*, preprint, arXiv:2610.34504 (2026). Announced as *Mixing solutions for unstable Muskat data with viscosity jump*, Seminario EDPs y Mecánica de Fluidos, ICMAT, Madrid, 12 November 2026.

[Mat19] B.-V. Matioc, *The Muskat problem in two dimensions: equivalence of formulations, well-posedness, and regularity results*, Anal. PDE **12** (2019), no. 2, 281–332.

[Men22] F. Mengual, *H-principle for the 2-dimensional incompressible porous media equation with viscosity jump*, Anal. PDE **15** (2022), no. 2, 429–476.

[MO05] G. Menon, F. Otto, *Dynamic scaling in miscible viscous fingering*, Comm. Math. Phys. **257** (2005), no. 2, 303–317.

[MS23] F. Mengual, L. Székelyhidi Jr., *Dissipative Euler flows for vortex sheet initial data without distinguished sign*, Comm. Pure Appl. Math. **76** (2023), no. 1, 163–221.

[Mus34] M. Muskat, *Two fluid systems in porous media. The encroachment of water into an oil sand*, Physics **5** (1934), 250–264.

[Mus53] N. I. Muskhelishvili, *Singular Integral Equations*, P. Noordhoff, Groningen, 1953.

[NP20] H. Q. Nguyen, B. Pausader, *A paradifferential approach for well-posedness of the Muskat problem*, Arch. Ration. Mech. Anal. **237** (2020), no. 1, 35–100.

[NS21] F. Noisette, L. Székelyhidi Jr., *Mixing solutions for the Muskat problem with variable speed*, J. Evol. Equ. **21** (2021).

[Ott97] F. Otto, *Viscous fingering: an optimal bound on the growth rate of the mixing zone*, SIAM J. Appl. Math. **57** (1997), 982–990.

[Ott99] F. Otto, *Evolution of microstructure in unstable porous media flow: a relaxational approach*, Comm. Pure Appl. Math. **52** (1999), no. 7, 873–915.

[SCH04] M. Siegel, R. E. Caflisch, S. Howison, *Global existence, singular solutions, and ill-posedness for the Muskat problem*, Comm. Pure Appl. Math. **57** (2004), no. 10, 1374–1411.

[Shv11] R. Shvydkoy, *Convex integration for a class of active scalar equations*, J. Amer. Math. Soc. **24** (2011), no. 4, 1159–1174.

[ST58] P. G. Saffman, G. Taylor, *The penetration of a fluid into a porous medium or Hele-Shaw cell containing a more viscous liquid*, Proc. Roy. Soc. London Ser. A **245** (1958), 312–329.

[Sze12] L. Székelyhidi Jr., *Relaxation of the incompressible porous media equation*, Ann. Sci. Éc. Norm. Supér. (4) **45** (2012), no. 3, 491–509.